Local Music Organizer

Detailed User Guide

Advanced collection analysis and maintenance - created by a collector for collectors.

For serious local/offline Music.app libraries and physically available audio files.

macOS 14 or later | Apple Silicon and Intel | Local/offline workflow

About Local Music Organizer

Local Music Organizer (LMO) is a detailed, advanced macOS toolkit created by a music collector for music collectors who maintain serious offline Music.app libraries and large collections of local audio files. It is designed for the work that becomes difficult once a collection grows beyond ordinary browsing: understanding exactly which Music.app record points to which physical file, seeing where a track is used, separating playlist duplication from library-record duplication, finding missing or suspicious links, reviewing artwork and metadata, comparing physical versions, inspecting audio quality, searching disks, converting difficult formats, and playing local files through a live spectrum analyzer.

LMO deliberately distinguishes four things that are easy to confuse in a large collection: the Music.app library record, playlist membership, the physical audio file, and LMO's own saved Library Index. That separation is what makes the tools useful for careful maintenance: a scan can describe a problem without changing it, and an action can be limited to the exact record, playlist entry, tag, or file the collector has reviewed. The guiding principle is simple: inspect deeply, understand the evidence, and modify only through an explicit control whose scope is clear.

What LMO does not do
LMO does not repair, rebuild, optimize, or patch the Music.app database. It does not edit Music.app service databases, private Music system files, or other internal Music.app support files. It is not a subscription/streaming manager: it does not operate on Apple Music subscription streams, online-only/cloud-only audio as source files, or any other streaming source. It does not download or convert a stream into a local source file. Network/NAS workflows have not been validated for this release and are outside the supported scope of this manual.

Supported working model
The documented workflow is a local Music.app library plus physical audio files available through normal macOS file paths. A non-local Music.app record may still be visible as a record with no local path in a diagnostic count, but physical-file tools require an offline file that exists on disk.

How to read the control descriptions
Read-only = inspect/filter/report/open only. LMO index = only LMO's own saved working index changes. Music.app metadata = a supported field on a specific Music.app record can change after an explicit action. Playlist entry = playlist membership can change without deleting the audio file. Physical file/output = a local file may be renamed, moved, cleaned, or newly created only by the explicitly labeled operation.

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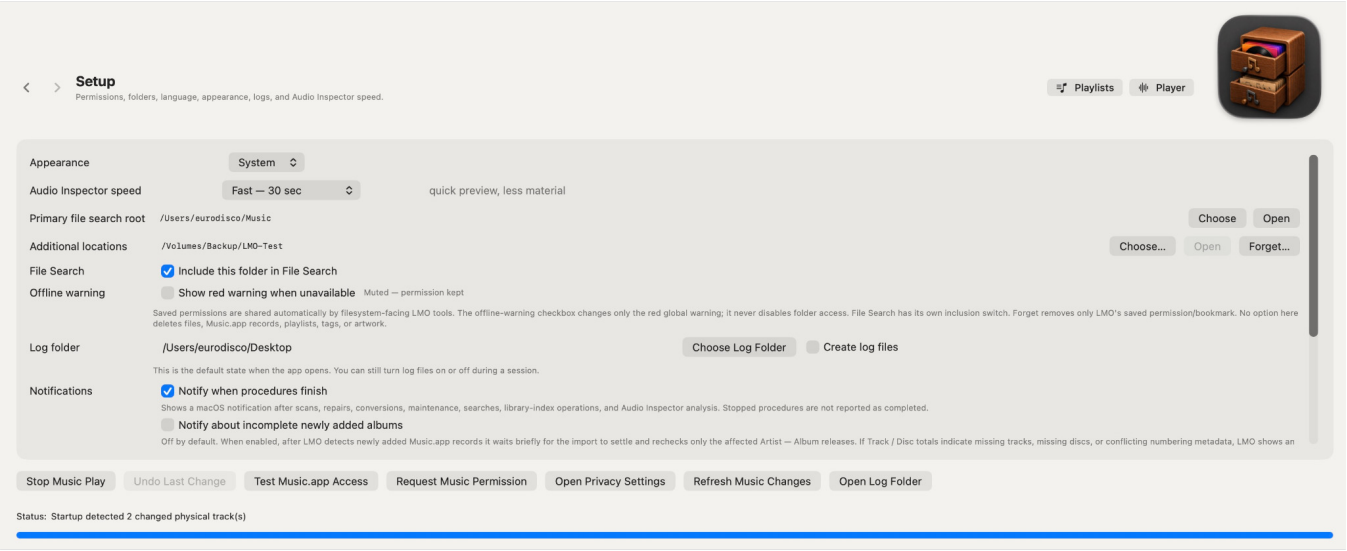
1. Setup

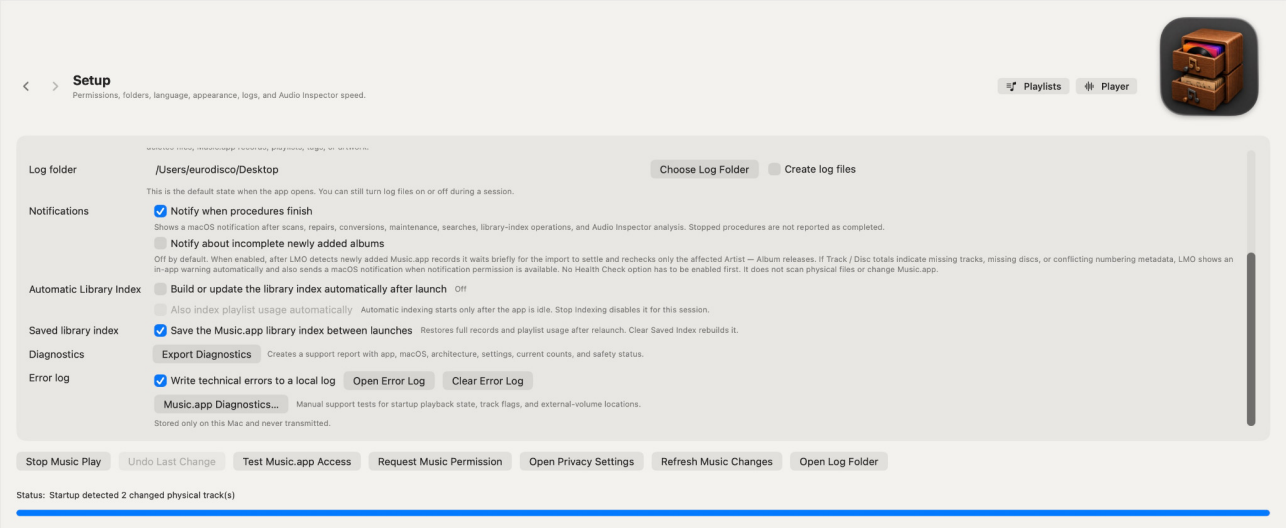


Two controls you will see in every main section
At the top right beside the LMO app icon, **Playlists** opens the Music.app playlist/folder navigator and **Player** opens or restores the Live Spectrum Player. These are navigation/playback shortcuts; they do not start a scan.

Setup controls LMO preferences, local/external folder access, File Search inclusion, offline-warning behavior, logging, notifications, automatic/saved Library Index behavior, diagnostics, and Audio Inspector speed. Changing a Setup option does not by itself delete a Music.app record or a physical audio file.

1.1 Appearance, analysis depth, and folders

Appearance LMO setting	Selects System, Light, or Dark. This changes only LMO's interface appearance.
Audio Inspector speed LMO setting	Selects the analysis-depth preset used by Audio Inspector. The Setup screen shows the active preset and approximate analysis window; the source audio file is not modified.
Primary file search root LMO setting	Main folder used by physical-file tools and File Search. Music.app record scans still use each record's actual saved location.
Choose / Open Folder permission / navigation	**Choose** selects or re-authorizes the primary folder through macOS. **Open** opens that folder in Finder. Neither control moves, imports, renames, or deletes audio.
Additional locations LMO setting / saved access	Shows another local or external folder whose permission LMO remembers for filesystem-facing tools.
Choose... / Open / Forget... Folder permission / navigation	**Choose...** authorizes an additional folder; **Open** reveals it in Finder; **Forget...** removes only LMO's saved permission/bookmark. It does not delete the folder, audio, Music.app records, playlists, tags, or artwork.
Include this folder in File Search LMO setting	Controls only whether the additional location participates in File Search. It does not disable the saved folder permission or filter Music.app library scans.
Show red warning when unavailable LMO warning setting	Controls only the red global warning for a saved location. Turning it off keeps the permission. An unavailable external location is treated as temporarily offline, not as proof its files were deleted.



1.2 Logs, notifications, automatic indexing, and saved index

Log folder Read-only display	Shows where operation logs are written when logging is enabled.
Choose Log Folder / Create log files LMO setting	**Choose Log Folder** authorizes the destination. **Create log files** enables detailed scan logs for tools that support the common logging option; it does not start a scan.
Notify when procedures finish LMO notification setting	Reports completion of supported scans, repairs, conversions, maintenance, searches, Library Index operations, and Audio Inspector analysis when macOS notifications are available. A stopped procedure is not reported as completed.
Notify about incomplete newly added albums Optional read-only watcher - OFF by default	When enabled, LMO checks newly detected Music.app records after an import and evaluates only the affected Artist - Album releases. If Track/Disc totals indicate missing tracks, missing discs, or conflicting numbering, LMO shows an in-app warning and can also use a macOS notification. The check is separate from Health Check and does not change Music.app metadata or physical files. Re-importing the same album is treated as a new import event and can warn again.
Automatic Library Index LMO setting / index	Controls the separate idle-time Library Index maintenance after launch. It is not the same feature as the optional incomplete-new-album watcher.
Build or update the library index automatically after launch LMO setting / index	Allows LMO to build/update its Library Index after the app has been idle and no conflicting operation is running.
Also index playlist usage automatically LMO setting / index	When automatic indexing is enabled, also refreshes visible playlist membership so relationship-based tools are ready.
Save the Music.app library index between launches LMO setting / index	Persists LMO's Full Library Records and available playlist-usage index in LMO's own SQLite storage for restoration after relaunch. Turning this off affects only LMO's saved index, not Music.app data.

Important distinction

Automatic Library Index is background index maintenance. **Notify about incomplete newly added albums** is a separate opt-in import warning. The warning does not require the Health Check album-completeness checkbox to be enabled first.

Export Diagnostics Support log only	Creates a support report containing app/macOS information, settings, current counts, index state, and safety status. It is separate from scan-specific logs.
Write technical errors to a local log LMO setting / log	Enables LMO's technical error log. The log is stored locally and is not transmitted automatically.
Open Error Log Navigation only	Opens the local technical error log.
Clear Error Log LMO log only	Clears LMO's technical error log. It does not clear Music.app data or scan results.
Music.app Diagnostics... Read-only support test	Appears while technical error logging is enabled. Opens explicit read-only support tests for startup state and track flags.

1.3 Setup toolbar and status areas

Stop Music Play Playback only	Stops playback LMO has started in Music.app or through supported local playback paths.
Undo Last Change Changes depend on last action	Reverses the most recent supported LMO change when an undo record exists. It is disabled when nothing supported can be undone.
Test Music.app Access Read-only support test	Performs a small access test so you can see whether macOS currently allows LMO to communicate with Music.app.
Request Music Permission macOS permission	Triggers the macOS automation permission flow for Music.app where available. This is separate from folder access for local/external files. LMO never receives an Apple Account password.
Open Privacy Settings Navigation only	Opens the relevant macOS Privacy & Security settings so Music.app automation permission can be reviewed.
Refresh Music Changes LMO index	Performs LMO's incremental reconciliation against Music.app. It is intended for small changes and does not automatically rebuild the entire index.
Open Log Folder Navigation only	Opens the selected log folder in Finder.
Status line and progress bar Read-only display	Show the current operation/index state and progress. If a saved external location is unavailable and its warning is enabled, the global status can show a red offline message. Muting that warning hides only the message; LMO keeps the saved folder permission.
Permissions card Read-only display	Explains the current Music.app automation status and which control to use if macOS blocks it. Local/external folder access is a separate macOS permission and can be requested through a folder picker when a filesystem-facing tool needs it.
Default playlists card Read-only display	Shows names LMO will use for support/recovery-oriented playlist workflows, such as the review playlist name.
Audio Inspector speed card Read-only display	Repeats the currently selected inspection speed and its practical meaning.

1.4 Music.app Diagnostics window

Capture Music.app Startup State Read-only support test	Explicitly launches/observes Music.app without sending Play, then records several short player-state snapshots for support. A large library that is still loading is recorded as not ready instead of blocking indefinitely.
Audit Music.app Track Flags Read-only support test	Reads Music.app track flags in bounded ranges and reports non-default playback-related values and storage-volume totals. It does not rewrite those flags.
Stop Stops diagnostic only	Requests cancellation of the active diagnostic.
Close Navigation only	Closes the diagnostics window when no diagnostic is running.

1.5 Setup: support, permissions and saved state

These controls operate on support information, LMO preferences or explicitly authorized locations. They do not themselves alter a Music.app audio record.

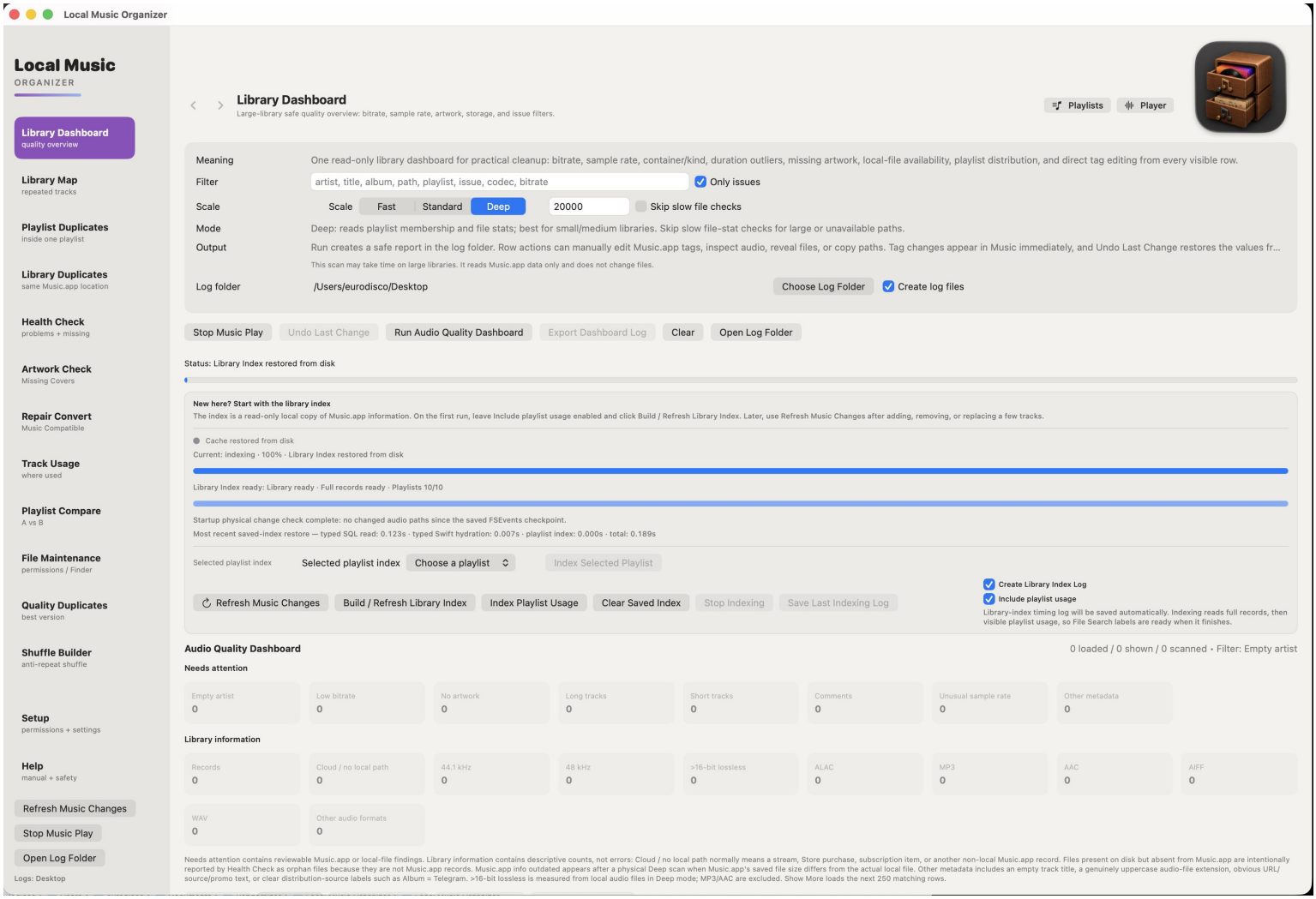
License Information...	Opens the license information window. It is information about LMO terms, not a music-library maintenance operation.
Open Full License Files	Inside Help > Open Source Licenses, opens the full bundled third-party license files. This does not edit Music.app, playlists, or audio files.
Open License Information...	The Setup button opens License Information. Full license texts are available through Help > Open Source Licenses; neither action changes the music collection.
Music.app Diagnostics...	Shown when technical error logging is enabled. Use its explicit startup-state and track-flag tests; opening the window does not issue a Play command.
Capture Music.app Startup State	Records short Music.app startup/player-state observations for troubleshooting. Music.app may still be loading; not-ready is a diagnostic state, not a failed track.
Audit Music.app Track Flags	Reads playback-related flags in bounded record ranges without rewriting them. The detailed result is a support observation, not a change recommendation.
Add Folder... / Warn when offline	Add Folder... authorizes another filesystem location. The per-location Warn when offline checkbox shows or suppresses its availability warning; switching the warning off does not remove permission, delete a folder or make an offline file available.
Saved folders vs File Search inclusion	A remembered permission and its Include this folder in File Search checkbox are separate. Forget removes the bookmark only; it never deletes a folder.
Technical error log vs operation log	The error log, Export Diagnostics, Library Index timing log and per-operation logs are different outputs. Enable the log required for the procedure before reproducing it.

When an external disk is offline, do not treat the missing path alone as proof that the tracks were deleted. Reconnect the same disk before deciding to relink or import.

Music / Media Library access on first use

macOS may ask LMO for access to your music and media library. This allows local Music.app metadata and file-location indexing through iTunesLibrary; it does not require an Apple Music subscription. This is separate from Music.app Automation permission and from folder access granted through macOS file pickers. If denied, certain index/path checks may be unavailable; no audio files are deleted.

2. Library Index



Library Index is the foundation of library-wide LMO work
Without the required saved index stage, most library-wide tools are unavailable, incomplete, or must wait for data. Once Full Library Records are built and the needed Playlist Usage is indexed, LMO can answer large-library questions quickly from its own saved SQLite rows instead of repeatedly reading the entire Music.app library. The index can be extended, refreshed incrementally, resumed after a stop, refreshed for one selected playlist, rebuilt when necessary, and saved between app launches.

The Library Index is the foundation for almost every library-wide LMO tool. It is LMO's saved working copy of Music.app information - not a repair of Music.app itself. Building it once converts repeated expensive library reads into fast local queries. Why this matters Without the required index stage, a library-wide tool may be unavailable, incomplete, or forced to wait for data. With the Full Library Records and the needed Playlist Usage saved, LMO can answer most library-wide questions from its own indexed rows instead of repeatedly reading the entire Music.app library.

2.1 Index controls

Build / Refresh Library Index **LMO index**	Reads Music.app records in bounded batches and builds/updates LMO's Full Library Records index. When Include playlist usage is enabled, visible playlist membership is indexed after full records are ready. Building the index is read-only with respect to Music.app tracks, playlists, and audio files.
Resume Library Index **LMO index**	The same main button changes to Resume when a previous full-record build was stopped after completed batches. Resume continues from the next safe record instead of throwing away completed work.
Include playlist usage **LMO index**	Adds visible playlist membership to the same indexing run. This is important for Track Usage, Playlist Compare, playlist labels, Library Map, and duplicate-safety relationships.

Index Playlist Usage LMO index	Builds or refreshes visible playlist membership independently after full records are ready. It is useful when the record index is good but playlist coverage is missing or stale.
Queue Playlist Usage / Playlist Usage Queued LMO index	During an active full-record build, the playlist-usage action can be queued so it begins after full records finish.
Selected playlist index LMO index	Picker used to choose one playlist whose saved membership should be indexed or refreshed without rereading every visible playlist.
Index Selected Playlist LMO index	Reads and saves only the chosen playlist when it is not yet indexed.
Refresh Selected Playlist LMO index	Rereads only the chosen playlist and replaces that playlist's indexed membership with the newly verified complete snapshot.
Refresh Music Changes LMO index	Checks for additions, removals, identity changes, and playlist count changes after small Music.app edits. It is the normal incremental refresh; it does not start a full-record rebuild.
Stop Indexing LMO index	Finishes the current small Music.app batch, saves completed progress, and leaves the index resumable. It also disables automatic indexing for the current LMO session.
Clear Saved Index LMO index	Deletes LMO's saved Library Index so the next build starts from the beginning. Use this only when a clean rebuild is actually needed; it does not clear Music.app.
Create Library Index Log Support log only	When enabled before indexing, automatically writes the completed indexing timing/row-count report to the selected log folder.
Save Last Indexing Log Support log only	Saves the most recent completed/ended index performance report without rerunning Library Index.

2.2 Index stages and meaning

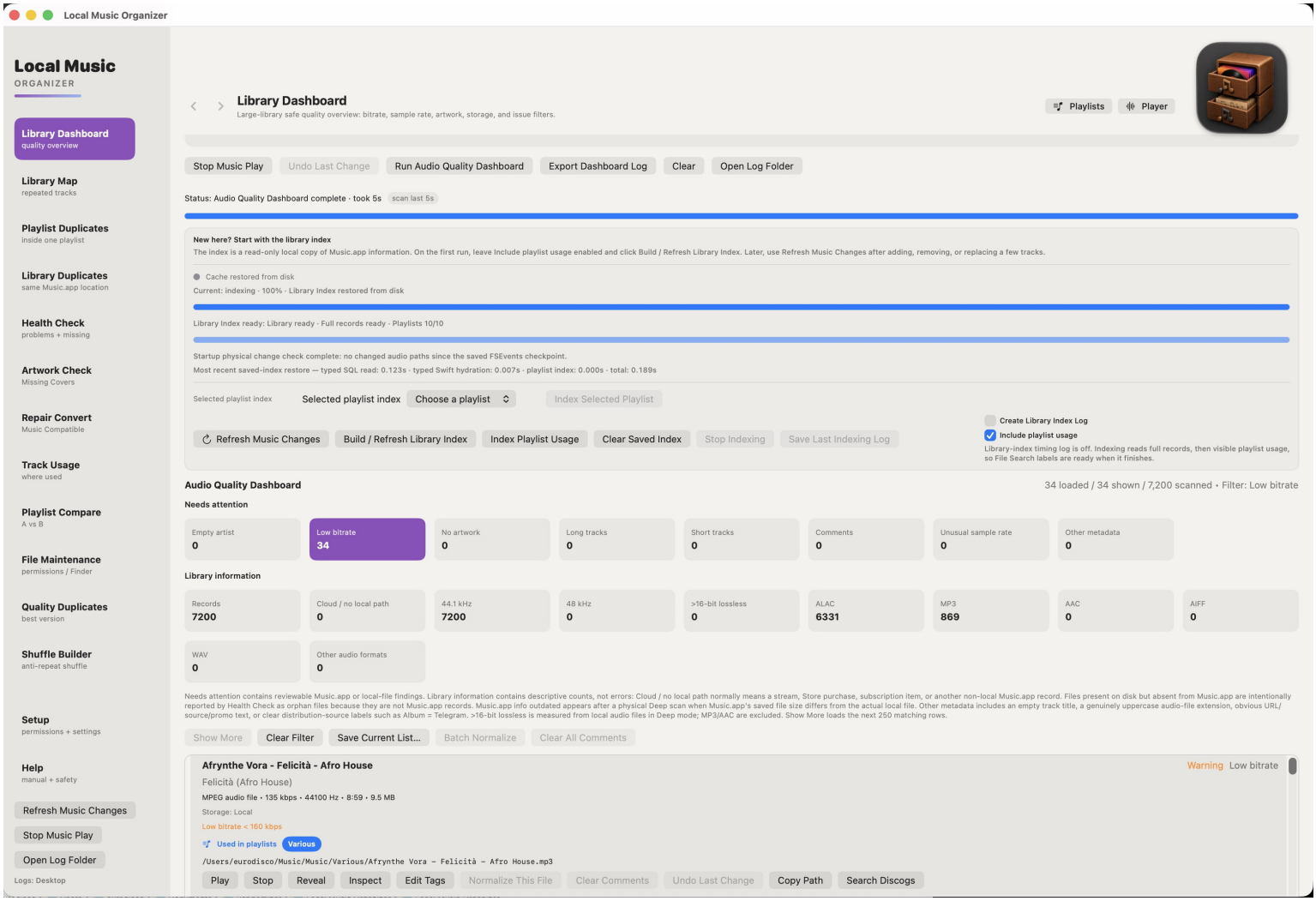
Full Library Records Read-only indexed data	The indexed Music.app record set used by Dashboard, whole-library Health Check, Library Duplicates, quality review, and library-wide search. Library Duplicates deliberately waits for complete full-record coverage.
Playlist Usage Read-only indexed data	Saved membership of visible playlists. It answers “where is this track used?” and protects playlist-used records from being incorrectly treated as unused.
Selected Playlist Read-only indexed data	A targeted playlist snapshot. It lets one changed playlist be refreshed without rebuilding all playlist usage.
Overall progress Read-only display	Shows progress across the required index stages. A full-record stage can be ready before every playlist has been indexed; LMO labels partial coverage rather than pretending it is complete.
Saved index restore LMO index	When Save the Music.app library index between launches is enabled, LMO reloads its own typed SQLite rows after relaunch. This is why a large library can become usable much faster on later launches.

Safe rule for large libraries Let one long index operation finish or stop it cleanly before starting another. After a small Music.app change, prefer Refresh Music Changes; after one playlist change, prefer Refresh Selected Playlist. A full rebuild is not the default answer to every change.

Large imports: keep the saved index protected

Refresh Music Changes handles ordinary small additions and removals. If a large import exceeds the safe targeted-reader limit, use Build / Refresh Library Index. LMO does not fall back to thousands of AppleScript file aliases. If identity verification fails, the previous saved SQLite index is retained but marked stale until verified again. Do not use Clear Saved Index just to clear this warning.

3. Library Dashboard



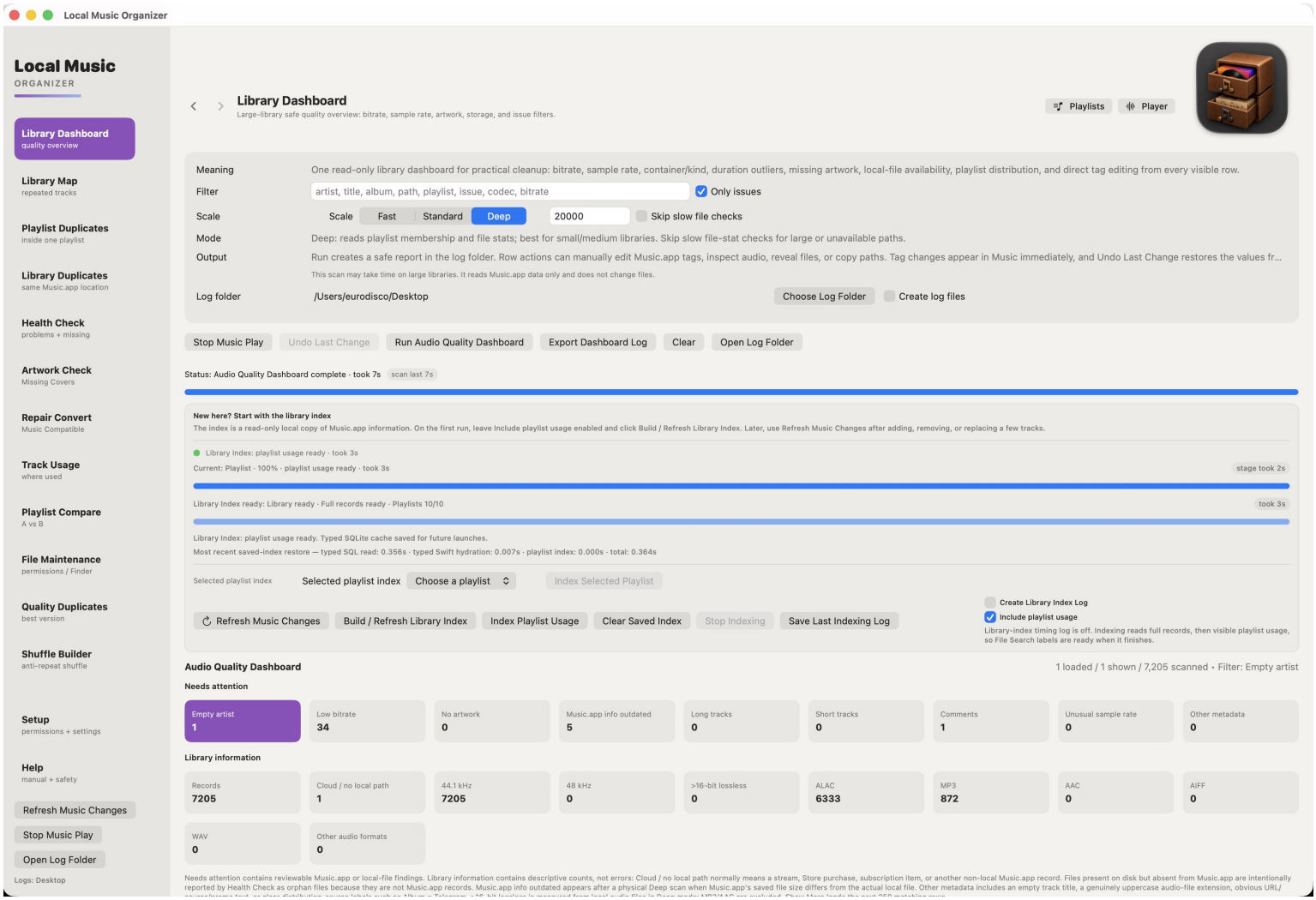
Library Dashboard is the main read-only overview of the indexed Music.app library. It summarizes both items that deserve review and descriptive facts about the collection. A high counter is a review signal, not proof that every matching track is wrong.

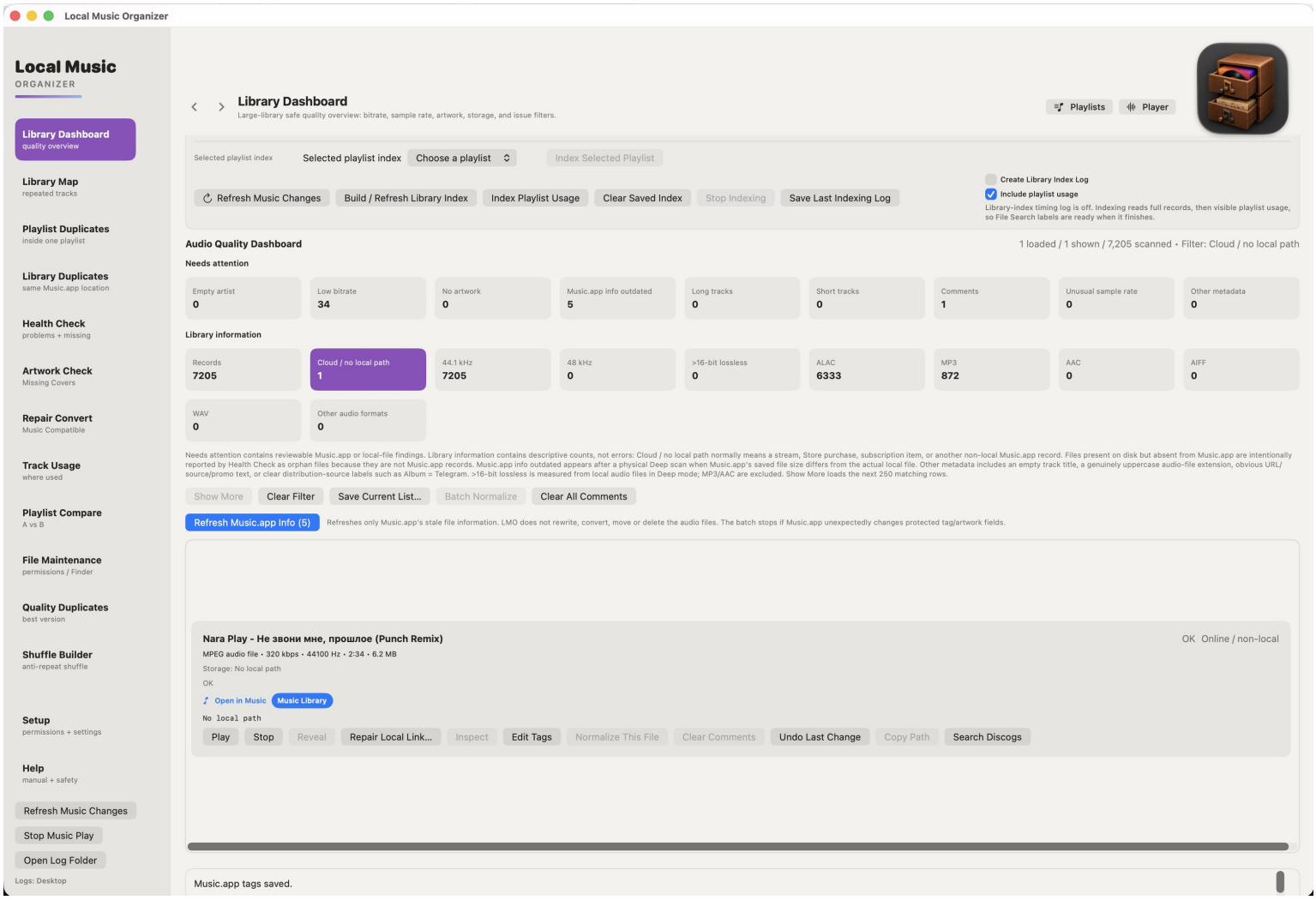
3.1 Dashboard controls

Meaning **Read-only display**	Explains what the Dashboard evaluates and how the chosen scan depth changes the source of technical facts.
Filter **Read-only filter**	Searches loaded/indexed Dashboard rows by artist, title, album, path, and related fields.
Only issues **Read-only filter**	Limits results to rows classified as reviewable issues rather than descriptive library information.
Scale - Fast **Read-only scan**	Uses indexed Music.app record information for fast review and skips playlist membership and slow physical-file checks.
Scale - Standard **Read-only scan**	Adds lightweight local-file checks when enabled while still avoiding playlist membership. It is the normal middle ground.
Scale - Deep **Read-only scan**	Uses playlist membership and physical-file facts where available, including real local sample rate/bit depth/file size. It is the most complete and can be slower.
Number field beside Scale **LMO scan setting**	The numeric field immediately to the right of Fast / Standard / Deep limits how many indexed records the Dashboard considers for that run. When the field is empty its placeholder is "Max records"; when a number is present, the number itself is what the user sees.
Skip slow file checks **LMO scan setting**	Visible checkbox that avoids slow physical-file checks and favors the saved/indexed data path. It is useful when external storage is slow or temporarily unavailable. It does not mean that NAS/network workflows are validated.

Log folder / Create log files **Support log only**	Controls the destination and creation of the Dashboard operation log. This is separate from the Library Index timing log.
Run Audio Quality Dashboard **Read-only scan**	Runs the Dashboard using the current scale/filter settings. The scan itself is read-only.
Export Dashboard Log **Support log only**	Exports the currently loaded Dashboard report when a scan exists.
Clear **LMO view only**	Clears the Dashboard result state in LMO. It does not delete library records or files.

3.2 Dashboard counters and filters





How to read the Dashboard example

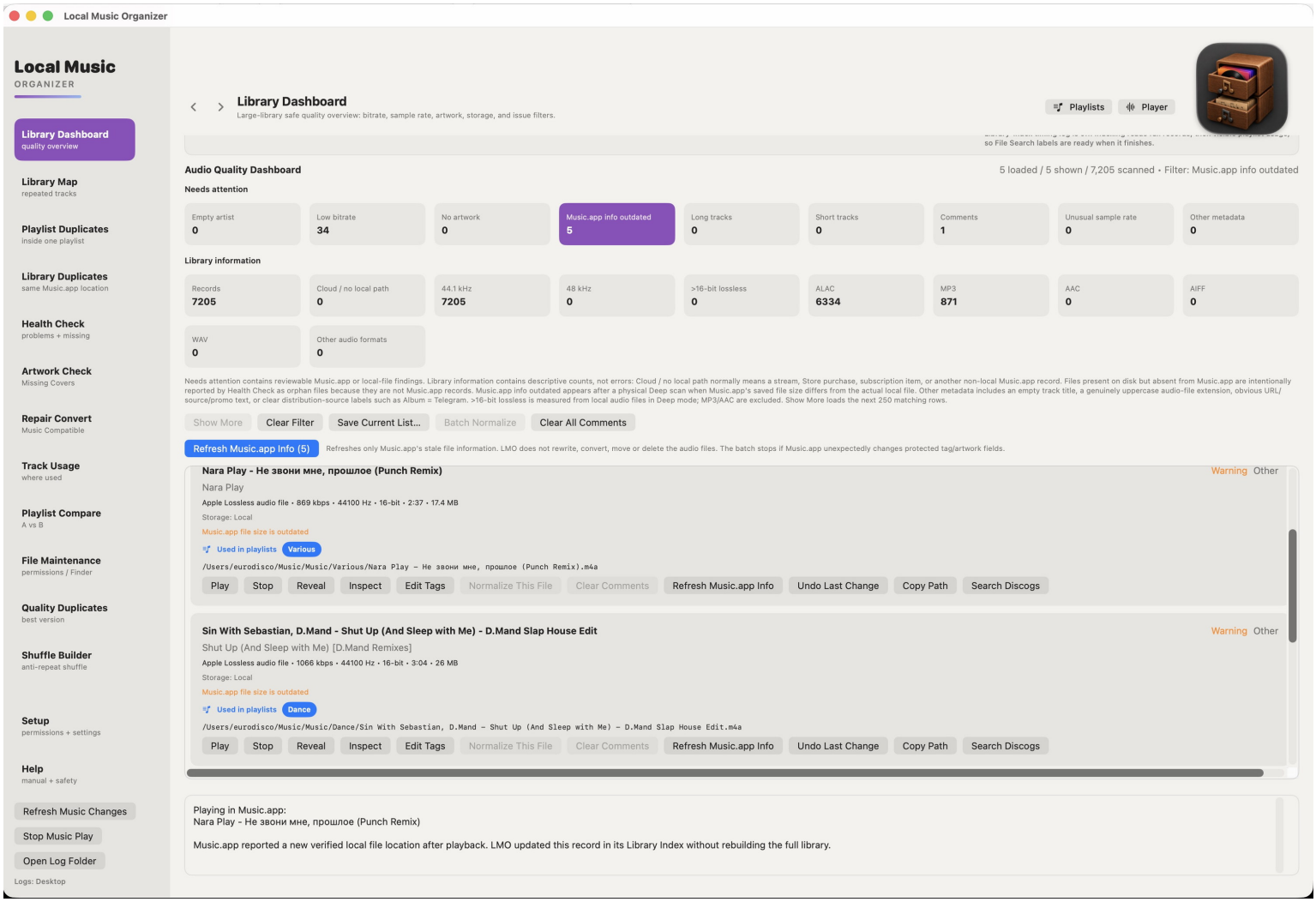
The screenshot above is an example of the **Dashboard counters and result list**. The small cards are filters and collection facts: a high count means "review these items", not "LMO has proved they are wrong". The selected counter changes which rows are shown below.

Typical workflow

Choose Fast, Standard, or Deep as needed, run the Dashboard, then use the counters to focus the result list. Review an individual row before using Play, Reveal, Inspect, Edit Tags, Repair Local Link, Refresh Music.app Info, or another explicit action. **Show More** only loads more matching rows; it does not rerun the scan.

Important interpretation

Comments are a review category, not automatically a defect. **Music.app info outdated** means saved Music.app technical information differs from the current local file and can be refreshed explicitly. The detailed counter definitions continue on the next pages.



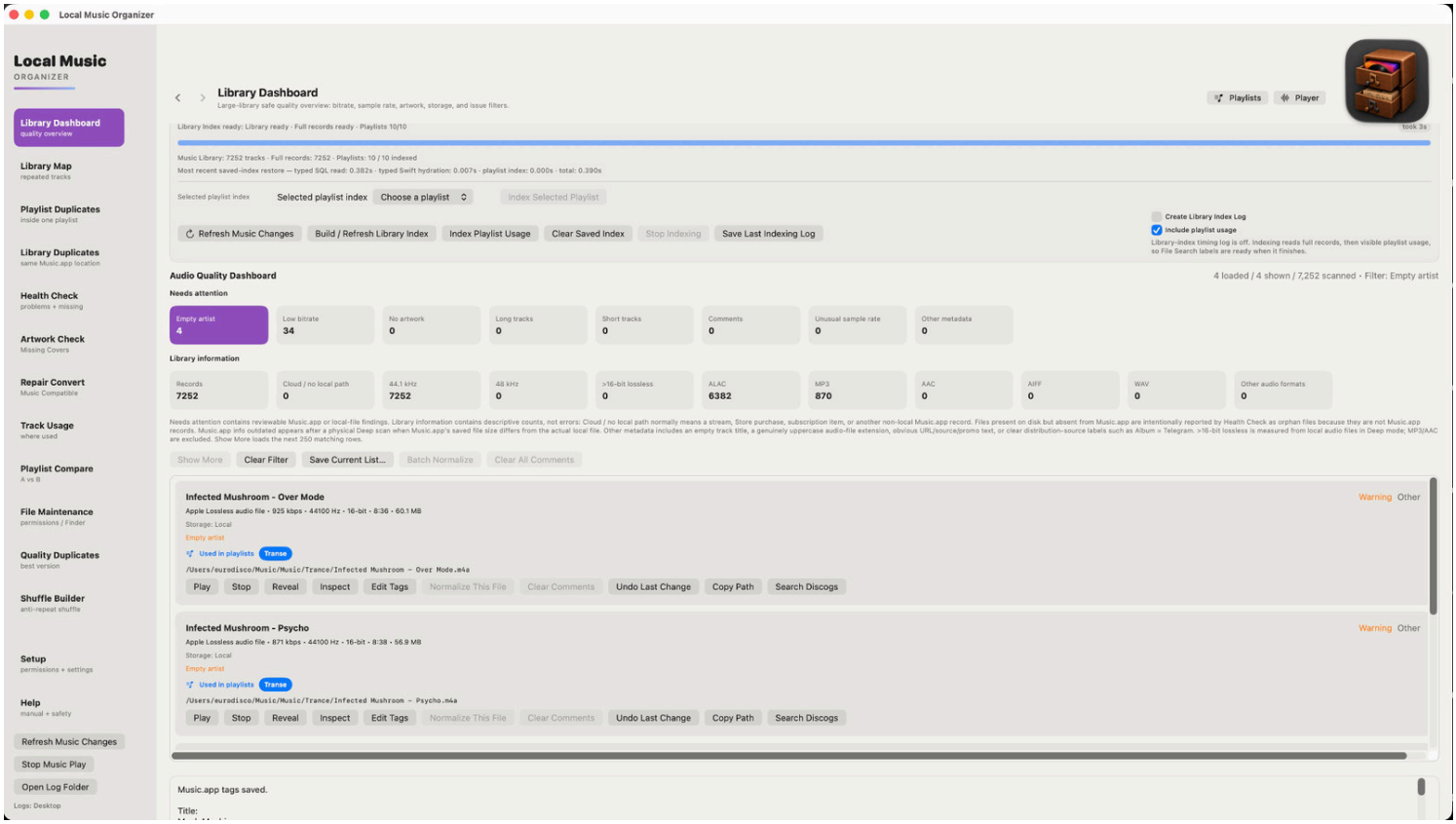
Comments are a review category, not a defect
A non-empty Comments field often contains harmless imported text: encoder/ripper signatures, DJ/remixer notes, release-group notes, source information, website/shop URLs, forum or tracker references, promo/advertising text, catalog/archive notes, or the owner's own notes. LMO shows the count so these strings can be reviewed; it does not assume that every comment should be removed.

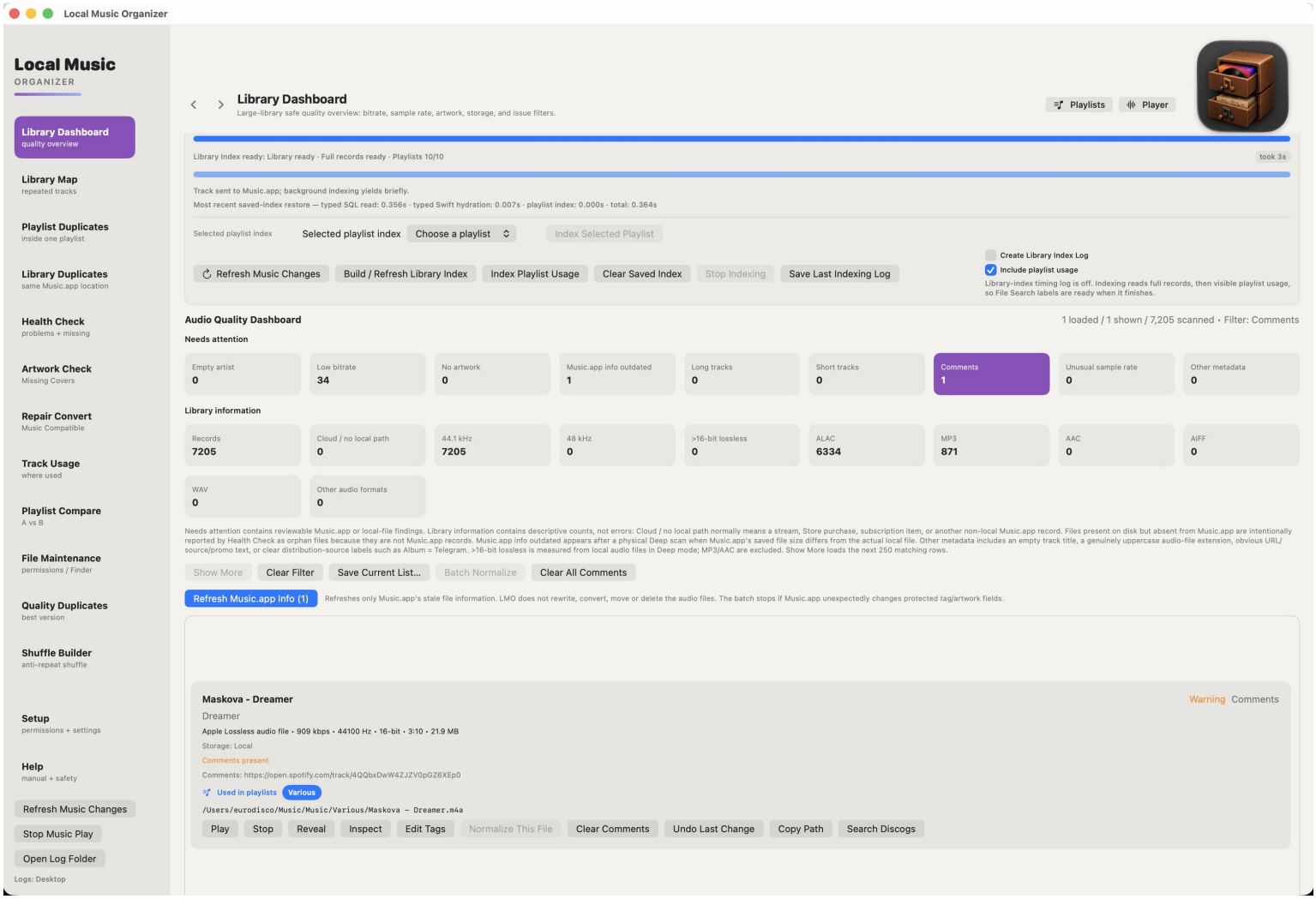
Needs attention **Read-only display**	Heading for review-oriented counters. These are categories to inspect, not automatic repair commands.
Empty Artist **Read-only filter**	Filters rows whose Music.app artist field is empty.
Low bitrate **Read-only filter**	Filters low-bitrate rows according to Dashboard criteria.
No artwork **Read-only filter**	Filters rows with no artwork count available in the indexed/verified data.
Music.app info outdated **Read-only filter**	Appears after Deep physical comparison when Music.app's saved file information, especially file size, differs from the actual local file.
Long tracks **Read-only filter**	Filters unusually long durations for review.
Short tracks **Read-only filter**	Filters unusually short durations for review.

Comments **Read-only filter**	Filters tracks whose Music.app Comments field is not empty. A comment is not automatically a problem. Comments are simply free-text metadata attached to a Music.app record and are often inherited from imported tags or added by software or people. Typical examples are an encoder/ripper signature or version, a DJ/remixer or release-group note, a source/rip note, a website or shop URL, forum/tracker references, advertising or promo text, catalog/archive notes, or the owner's own notes. The counter is therefore a review category: read the text before deciding whether it is useful or unwanted.
Unusual sample rate **Read-only filter**	Filters sample rates outside the common categories used by Dashboard.
Other metadata **Read-only filter**	Filters additional review conditions such as an empty title, obvious source/promo text, or other clearly suspicious metadata patterns.
Library information **Read-only display**	Heading for descriptive collection counters. These counts are not errors.
Records **Read-only filter**	Selects all scanned records.
Missing files **Read-only filter**	Filters Music.app records whose local path cannot be found. Non-local cloud/streaming records are a separate concept and are not repaired by LMO.
44.1 kHz / 48 kHz **Read-only filter**	Filters common sample-rate categories.
>16-bit lossless **Read-only filter**	Filters local lossless/PCM files measured above 16-bit in Deep mode. MP3/AAC are excluded from this bit-depth classification.
ALAC / MP3 / AAC / AIFF / WAV / Other audio formats **Read-only filter**	Filters the corresponding format families. These are descriptive format counts.

3.3 Result-list controls and track actions

Example - Empty Artist filter and result list





Show More **Read-only view**	Loads the next page/batch of matching Dashboard rows. It does not rerun the full scan.
Clear Filter **Read-only view**	Clears the search/focus filters so the broader Dashboard result set is visible again.
Save Current List **Support/report output**	Writes the currently focused Dashboard list to a report file.
Batch Normalize **Creates conversion queue**	Available for the supported 48 kHz or high-bit-depth focus categories. It queues the matching files into the explicit normalization/conversion workflow rather than silently rewriting them.
Clear All Comments **Changes Music.app metadata**	Explicitly clears only the Music.app Comments field for the Dashboard comment rows covered by the action. LMO first verifies the current live value, keeps supported undo data for the most recent reversible action, and synchronizes the result to its own saved index. It does not rename the file, rewrite audio samples, remove artwork, or clear unrelated tags such as Title, Artist, Album, Genre, Composer, or Year.
Refresh Music.app Info (N) **Changes Music.app technical info**	Targets rows whose Music.app file information is stale and asks Music.app to reread the same local file. LMO verifies protected metadata/artwork and does not convert or replace the audio file.
Play **Playback only**	Plays the matching record/file using the mode provided by that result card.
Stop **Playback only**	Stops current Music.app/local playback.
Reveal **Navigation only**	Reveals the local file in Finder when a local path exists.
Repair Local Link... **Changes Music.app file link**	For a broken local link, lets you choose the replacement file for the same existing Music.app Persistent ID. It is a targeted relink, not a new library import and not a database repair.

3.4 Dashboard: exact counter actions and complete list export

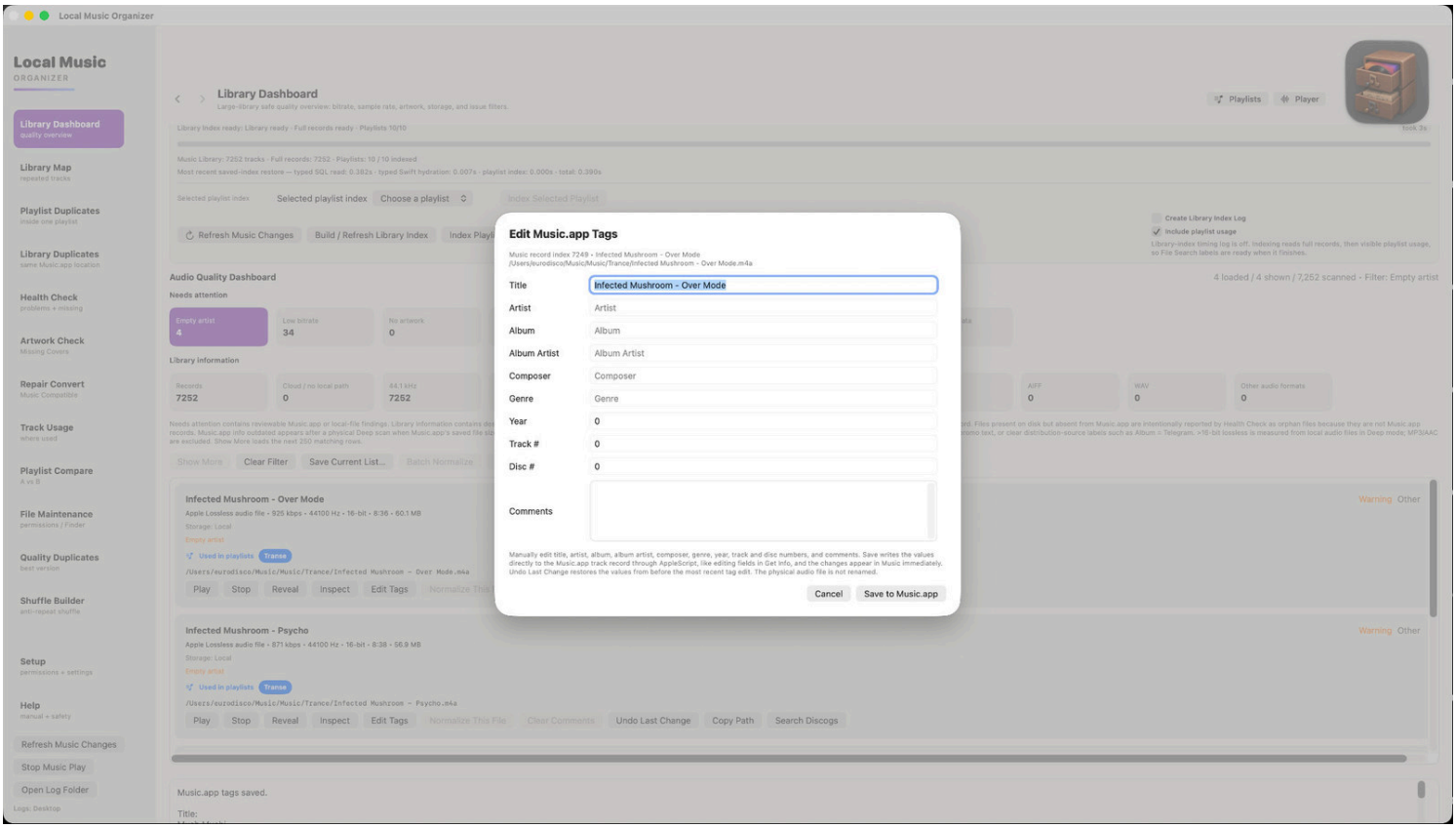
The Dashboard counters serve as result filters. A secondary click on a counter provides a separate text export; selecting a counter or exporting a list does not modify Music.app records.

Click a Dashboard counter	Focuses the current scan on that category and resets its search/list pagination. It is a result filter, not a scan and not a repair command.
Secondary-click a counter	Opens the context menu for that counter. The menu contains Save This List As... when a Dashboard scan has loaded.
Save This List As...	Opens a standard macOS Save dialog and writes a plain-text report of the selected category, including the current search criteria. In SQL mode it retrieves the complete matching set, not only the first 50 displayed cards.
Save Current List	The regular Dashboard export control saves the presently focused list. This is a distinct entry point from the right-click Save This List As... action.
Category vs actual correction	A count for Comments, Missing Files, Low bitrate or another category identifies candidates for review. The export does not invoke Clear All Comments, Repair Local Link or another modifying command.
When disabled	A counter cannot open the per-category export until a Dashboard scan has been loaded. A zero-result focus cannot produce a nonempty export.

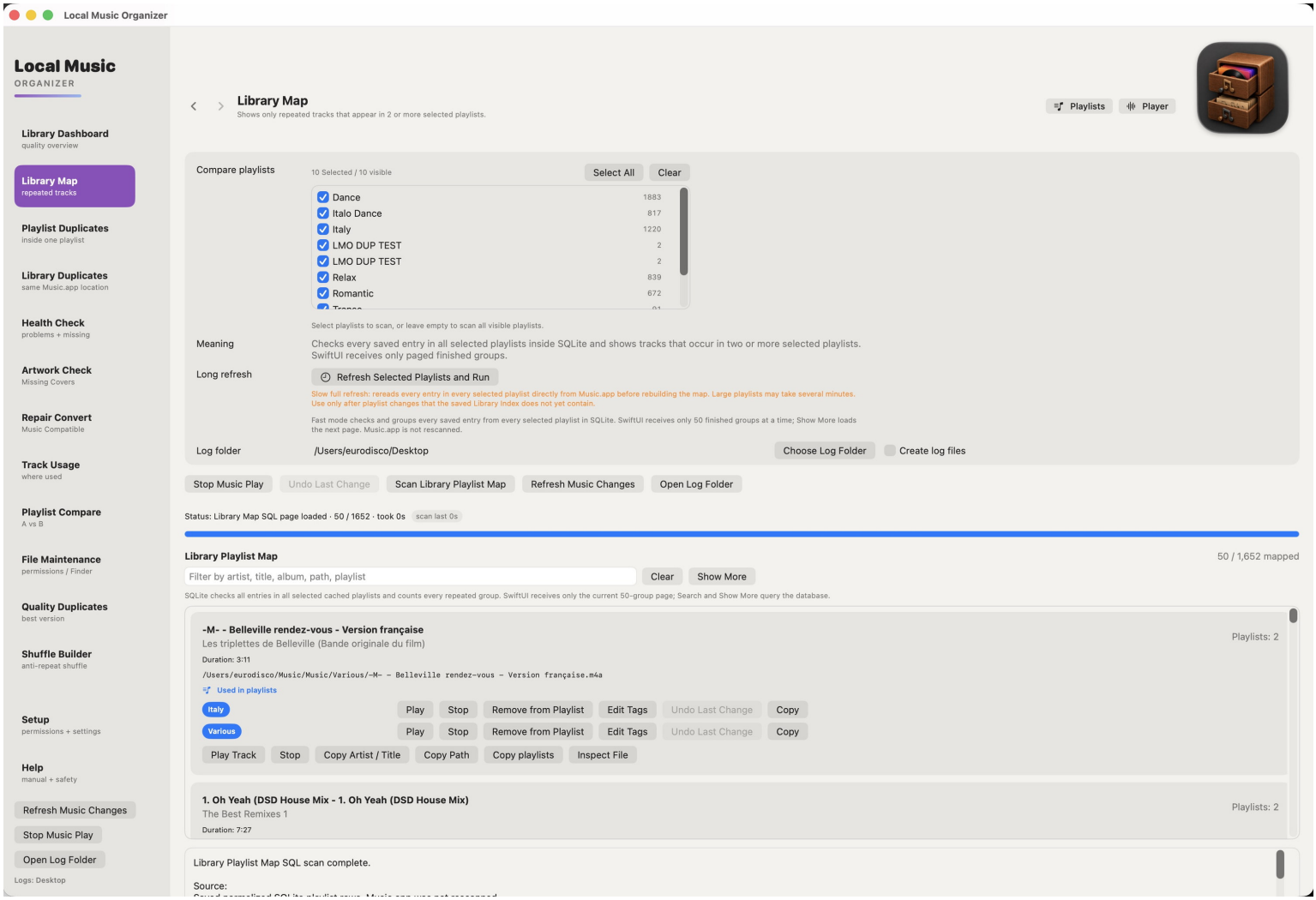
When checking large libraries, do not assume Show More must load every card before an SQL-backed Save This List As... export. The export retrieves all rows matching the chosen category and search.

Inspect **Read-only analysis**	Opens the exact local file in Audio Inspector.
Edit Tags **Changes only after save**	Opens the reviewed Music.app metadata editor for the exact record. Nothing changes until Save to Music is used.
Normalize This File **Physical/output file operation**	Queues/starts the explicit normalization path for that file using the selected safe conversion logic.
Clear Comments **Changes Music.app metadata**	Clears only the Comments field of this exact reviewed Music.app record after live verification. This is intended for unwanted free-text such as encoder/DJ signatures, URLs, promo text, advertising, or other imported notes. The physical audio file is not renamed or rewritten and other Music.app metadata fields are left alone.
Refresh Music.app Info **Changes Music.app technical info**	Refreshes technical file information for this exact Music.app record after verification.
Undo Last Change **Changes depend on last action**	Restores the most recent supported reversible LMO edit.
Copy Path **Clipboard only**	Copies the local file path to the clipboard.
Search Discogs **Navigation only**	Opens a Discogs web search based on the visible artist/title/album. This is a browser lookup, not a metadata write.

Example - Edit Music.app Tags for the reviewed Empty Artist record



4. Library Map



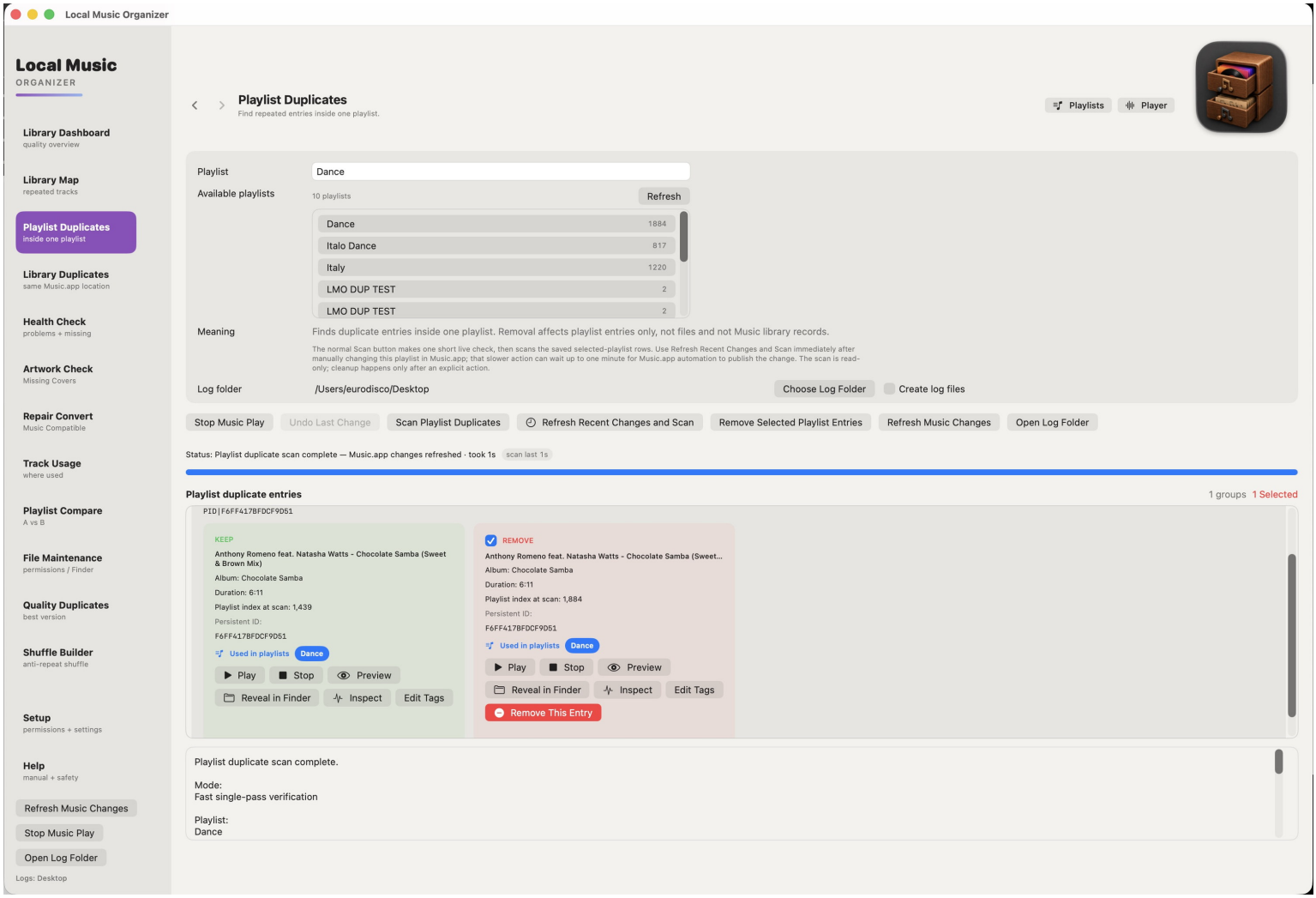
Library Map shows exact repeated track usage across selected playlists. It is a relationship map: its purpose is to answer which selected playlists contain the same track, not to diagnose duplicate entries inside one playlist.

4.1 Library Map controls and result actions

Compare playlists **Read-only selection**	The checklist/list of playlists included in the map. Only the selected playlist snapshots are grouped.
Select All / Clear **Read-only selection**	Selects or clears playlist scope for the map.
Meaning **Read-only display**	Explains that Library Map compares exact saved playlist memberships.
Scan Library Playlist Map **Read-only scan**	Fast path: groups the saved selected-playlist rows directly in SQLite. Music.app is not rescanned.
Refresh Selected Playlists and Run **LMO index then read-only scan**	Slow path: rereads every entry in the selected playlists from Music.app, saves complete snapshots, then runs the map. Use when the selected playlists changed after the saved index.
Filter **Read-only filter**	Filters map groups by artist/title/path/playlist information.
Show More **Read-only view**	Loads additional matching groups from the saved map query.
Playlist chips/buttons **Navigation only**	Each playlist chip identifies a destination containing the track and can reveal/open that exact playlist context.

Play **Playback only**	Plays the exact selected track from the indicated playlist/context.
Stop **Playback only**	Stops playback.
Remove From Playlist **Changes playlist entry**	Removes only the chosen playlist occurrence. The Music.app library record and physical file remain.
Edit Tags **Changes only after save**	Opens the exact Music.app record in the metadata editor.
Undo Last Change **Changes depend on last action**	Restores the most recent supported playlist/tag operation.
Copy **Clipboard only**	Copies the row information requested by the action.
Play Library Record **Playback only**	Plays the underlying Music.app library record represented by the group.
Copy Artist + Title / Copy Path / Copy Playlists **Clipboard only**	Copies the corresponding text without changing the library.
Inspect File **Read-only analysis**	Opens the local file in Audio Inspector.

5. Playlist Duplicates



What Playlist Duplicates is for

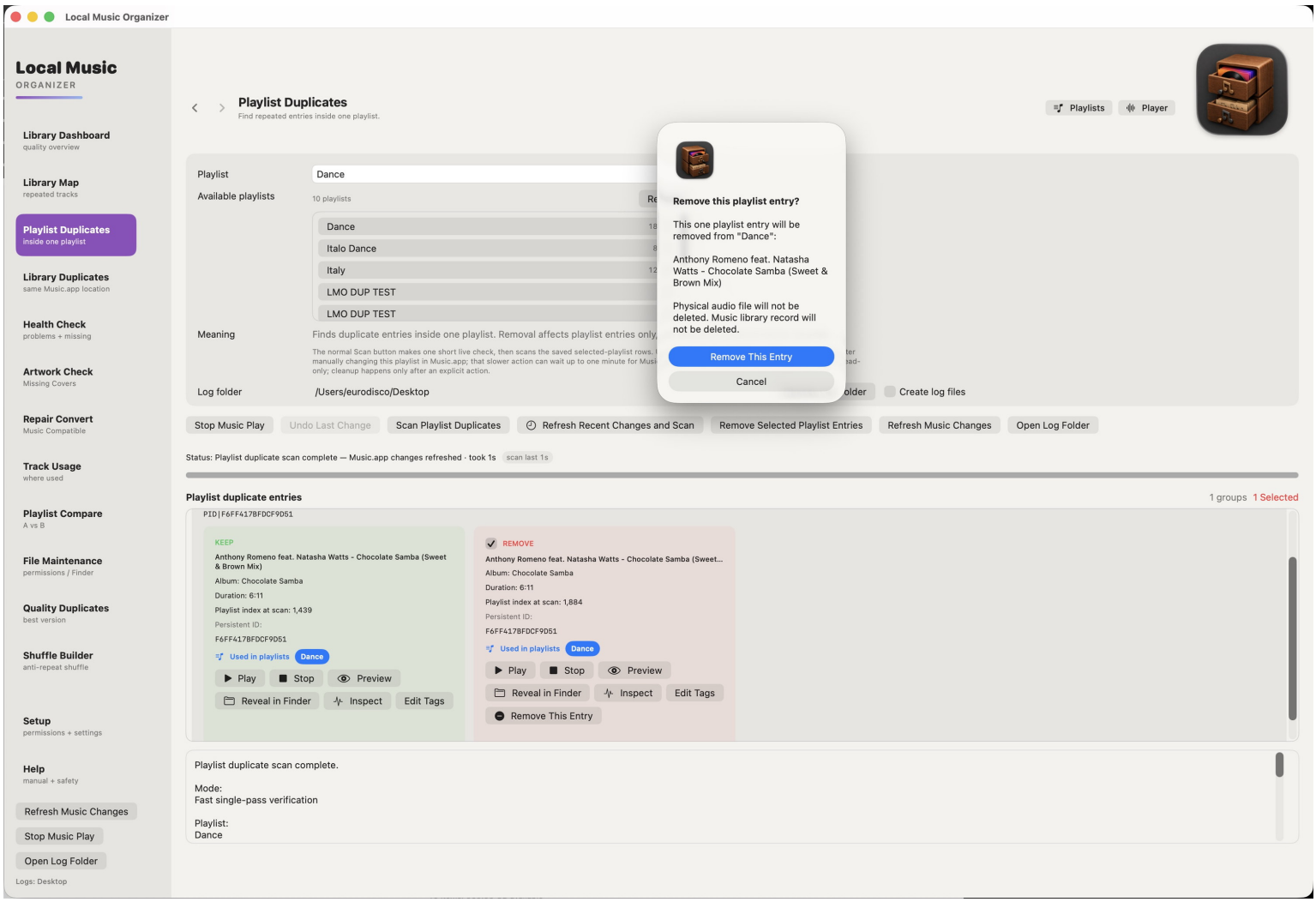
Use this tool when one Music.app playlist contains the same entry more than once. It answers a narrow question: **which occurrences inside this one playlist are duplicates?** It does not search the entire Music.app library for duplicate records and it does not compare several playlists with each other.

Typical workflow

1. Choose the playlist. 2. Run **Scan Playlist Duplicates**. If the playlist was just edited in Music.app, use **Refresh Recent Changes and Scan**. 3. Compare the KEEP and REMOVE occurrences with Play/Preview/Reveal/Inspect if needed. 4. Select only the duplicate playlist entries you want to remove.

Safety

Removing a duplicate here removes only that playlist occurrence. The Music.app library record and the physical audio file remain. Detailed controls and result-card actions continue on the next two pages.



Playlist Duplicates finds repeated entries inside one selected Music.app playlist. It deliberately separates “which occurrence to keep” from “which occurrence to remove.” Removing an entry here does not delete the audio file or the Music.app library record.

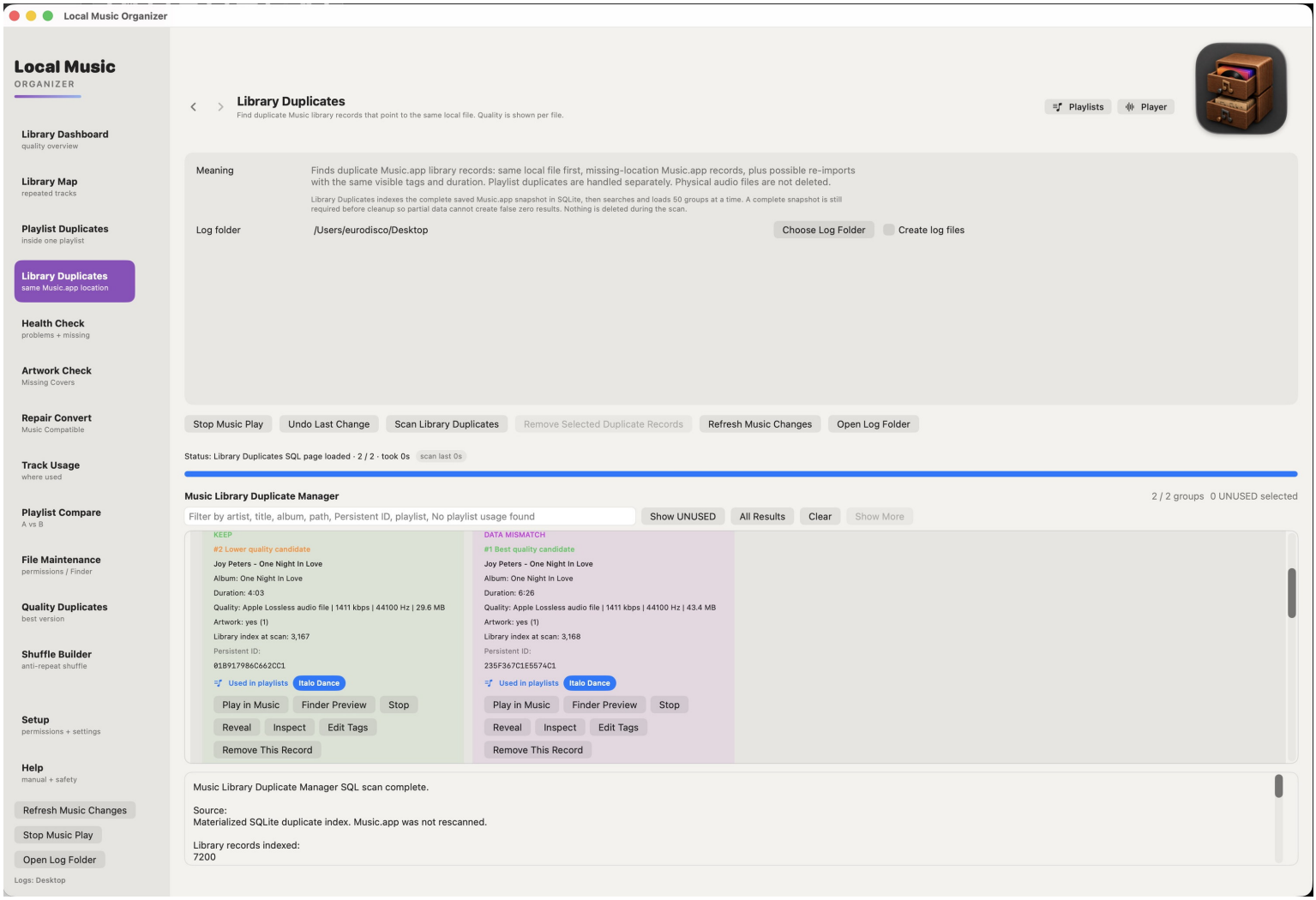
5.1 Playlist selection and scan controls

Playlist **Read-only selection**	Shows the selected source playlist.
Available playlists **Read-only selection**	List of visible playlists that can be selected as the scan scope.
Scan Playlist Duplicates **Read-only scan**	Runs the normal duplicate-entry scan for the selected playlist using current/saved playlist data as designed. The scan is read-only.
Refresh Recent **Changes and Scan / LMO index then read-only scan**	Slower synchronization scan intended immediately after manual changes in that Music.app playlist. It refreshes the selected playlist state before scanning.
Remove Selected Playlist Entries **Changes playlist entries**	Removes only the checked duplicate occurrences from the selected playlist after explicit review.
KEEP **Read-only label**	Marks the occurrence LMO presents as the retained entry in the duplicate group. It is a review label, not a physical-file operation.
REMOVE checkbox **Read-only selection**	Selects a duplicate playlist occurrence for removal. Checking it alone does not change anything.
Play **Playback only**	Plays the exact occurrence/record represented by the card.
Preview **Playback/preview only**	Uses local/Finder-style preview for the exact physical file so two similar entries can be compared independently.

Reveal in Finder **Navigation only**	Reveals the physical file path.
Inspect **Read-only analysis**	Opens the exact file in Audio Inspector.
Edit Tags **Changes only after save**	Opens the exact Music.app record in the metadata editor.
Remove This Entry **Changes playlist entry**	Removes only this reviewed occurrence from the selected playlist.

Confirmation meaning The confirmation dialog is intentionally specific about scope: the operation removes a playlist entry. It does not delete the physical audio file. Use Undo Last Change promptly if a supported playlist-entry removal was made in error.

6. Library Duplicates



Library Duplicates examines duplicate Music.app records, including multiple records that point to the same local file and other carefully verified duplicate patterns. A complete Full Library Records index is required so a partial cache cannot produce a misleading zero-result scan.

6.1 Library Duplicates controls and statuses

Scan Library Duplicates **Read-only scan**	Scans the complete indexed Music.app record set for duplicate record groups. It is disabled until the full-record index is complete.
Remove Selected Duplicate Records **Changes Music.app records**	Removes only explicitly selected Music.app duplicate records after review. It does not physically delete the audio file.
Filter **Read-only filter**	Filters the duplicate groups/results.
Show Unused **Read-only filter**	Shows records proven to have no playlist usage. This proof requires complete playlist usage coverage.
All Results **Read-only filter**	Returns to all duplicate groups.
Show More **Read-only view**	Loads more matching groups.
Verify This Group **Read-only targeted verification**	Performs the targeted verification required before higher-risk merge/removal actions are offered.
KEEP **Read-only status**	The record currently favored to remain in the group.

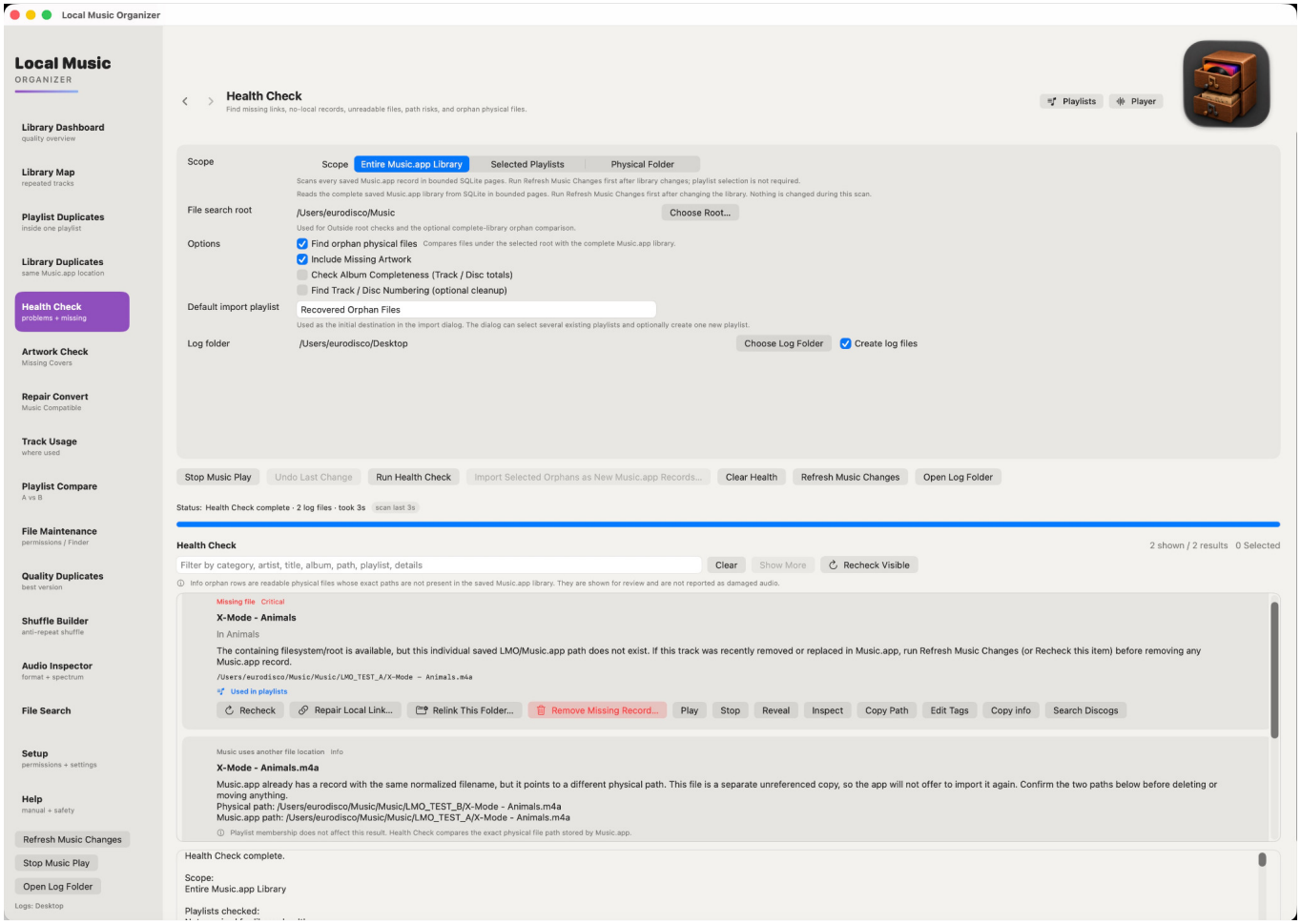
UNUSED DUPLICATE Read-only status	A duplicate record for which complete playlist indexing proves no playlist membership.
NEEDS VERIFIED MERGE Read-only status	A duplicate that cannot safely be removed until its playlist usage/identity is verified and any needed membership is preserved.
PLAYLIST USAGE UNKNOWN Read-only safety status	LMO cannot prove the record is unused because playlist coverage is incomplete. Cleanup is intentionally blocked.
MISSING PHYSICAL FILE Read-only status	The record points to a local path that is no longer present. This is a review condition, not automatic proof that the record should be deleted.
DATA MISMATCH Read-only safety status	Verification found facts that do not agree. LMO blocks automatic cleanup for the group.

6.2 Record-card actions

Play in Music Playback only	Plays the exact Music.app record represented by the card.
Finder Preview Preview only	Previews the local physical file independently of Music.app.
Stop Playback only	Stops playback/preview.
Reveal Navigation only	Reveals the physical file in Finder.
Inspect Read-only analysis	Opens the exact local file in Audio Inspector.
Edit Tags Changes only after save	Opens the exact Music.app record in the metadata editor.
Remove This Record Changes Music.app record	Removes this specific duplicate Music.app library record after the required safety checks/confirmation. The physical file is not automatically deleted.
Verified Merge Then Remove Changes playlists + Music.app record	For a verified merge case, preserves required playlist relationships and then removes the reviewed redundant Music.app record. It is not offered when the verification is incomplete.
Add This Unused To Recovery Playlist Changes playlist entry	Adds the reviewed unused record to the configured recovery/review playlist so it can be inspected before removal.

Quality labels are not mastering judgments Where Library Duplicates shows technical quality information, it is based on measurable file properties. It cannot prove which mastering, mix, pressing, or artistic version sounds best.

7. Health Check



Health Check is a diagnostic workspace for missing links, inaccessible records, path mismatches, orphan physical files, missing artwork, and optional Track/Disc metadata checks. The scan describes what LMO can verify; it does not repair, delete, relink, or import anything by itself.

Health Check principle
A finding is evidence to review, not an automatic repair command. Explicit actions on a result card have a narrower scope than the scan that found the issue.

7.1 Scope and scan controls

Scope - Entire Music.app Library / Selected Playlists / Physical Folder Read-only scope	Chooses whether Health Check examines the complete saved Music.app record set, selected indexed playlists, or audio files under a physical folder. Some options are intentionally available only for Entire Music.app Library.
File search root / Choose Root... LMO setting	Selects the physical folder used for outside-root checks and, when enabled, the complete-library orphan comparison.
Find orphan physical files Read-only comparison	For Entire Music.app Library, compares readable files under the selected root with complete Music.app paths. A file whose exact path is not represented by Music.app is shown for review; this does not mean the audio is damaged.
Include Missing Artwork Read-only scan option	Adds missing-artwork findings for scopes that contain Music.app records.
Check Album Completeness (Track / Disc totals) Read-only live Music.app check	Uses Music.app Track Number / Track Count and Disc Number / Disc Count to identify incomplete or conflicting numbered releases. It does not query the Internet, clear numbering, or add missing tracks.
Find Track / Disc Numbering (optional cleanup) Read-only discovery by default	Finds Music.app records where Track Number, Track Count, Disc Number, or Disc Count is non-zero. The scan itself changes nothing. Cleanup is a separate explicit action on selected rows.

Default import playlist LMO setting	Initial destination for the explicit orphan-import dialog. The dialog can select several existing playlists and optionally create one new playlist.
Run Health Check Read-only scan	Runs the selected scope and enabled diagnostic options. Read-only checks do not delete, relink, import, or rewrite anything.
Import Selected Orphans as New Music.app Records... Changes Music.app library	Imports only explicitly selected orphan physical files after live exact-path verification. If Music.app already references the exact path, LMO reuses that record instead of creating a duplicate.
Clear Health LMO view only	Clears the current Health results from LMO. It does not clear Music.app or delete files.
Refresh Music Changes LMO index	Runs the normal incremental reconciliation after smaller Music.app changes. It updates LMO's saved index rather than rebuilding the whole Library Index.
Recheck Visible / Show More Read-only verification / view	**Recheck Visible** re-verifies only currently visible findings. **Show More** loads additional matching result cards without changing the library.

Album completeness and Track / Disc cleanup are separate operations

Album Completeness is diagnostic only. **Find Track / Disc Numbering** produces selectable numbering findings. When you explicitly choose **Clear Selected Track / Disc Numbering**, LMO targets only Track Number, Track Count, Disc Number, and Disc Count. It live-verifies each record before and after the write. If Music.app temporarily changes editable protected metadata, LMO restores the exact pre-cleanup Music.app values and verifies the final protected fingerprint. Persistent ID, location, and artwork count remain fail-closed safety boundaries. Physical audio-file tags are not rewritten by this cleanup.

7.2 Health result actions

Missing file / Critical Read-only result	The Music.app record still exists, but its saved local path does not currently exist. Treat this as a record-to-file link problem until the current state is verified.
Used in playlists Read-only relationship / navigation	Shows indexed playlist relationships for the record. This is important context before relinking or removing a Music.app record.
Recheck Read-only targeted verification	Rechecks this exact finding against current Music.app and physical-file state.
Repair Local Link... Changes existing Music.app file link	Lets you choose a replacement local file for the same existing Music.app Persistent ID. It is a targeted relink, not a new library import.
Relink This Folder... Changes existing Music.app file links	Uses the missing track's saved parent folder as the old root, asks for the replacement folder, previews relative-path matches, and relinks existing Music.app records. LMO does not move, copy, rename, or delete the physical audio files.
Remove Missing Record... Changes Music.app library record	After confirmation and exact Persistent-ID live verification, removes only that Music.app record. It does not delete a physical audio file.
Play / Stop / Reveal / Inspect / Copy Path / Edit Tags / Copy Info / Search Discogs Playback / navigation / explicit editor	Playback, navigation, and clipboard actions do not change the library. Edit Tags changes supported Music.app metadata only after Save.

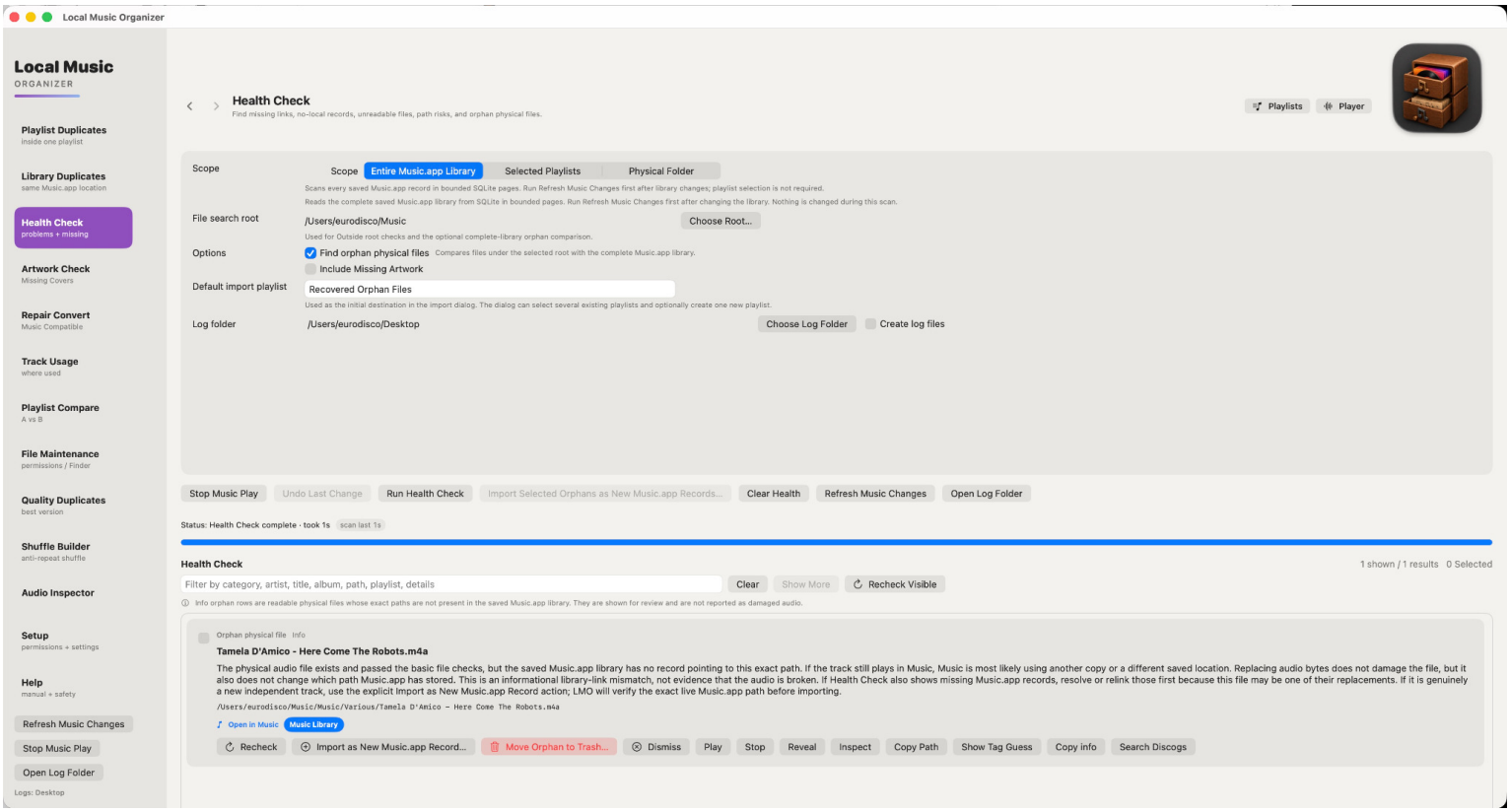
Music uses another file location

This informational result means LMO found a separate physical file with the same normalized filename while Music.app already points to another physical path. LMO does not auto-import that extra copy. Compare the physical path and the Music.app path before any later relink or Trash decision. Section 7.3 continues with the dedicated actions for this result type.

Detailed Health Check safety workflows

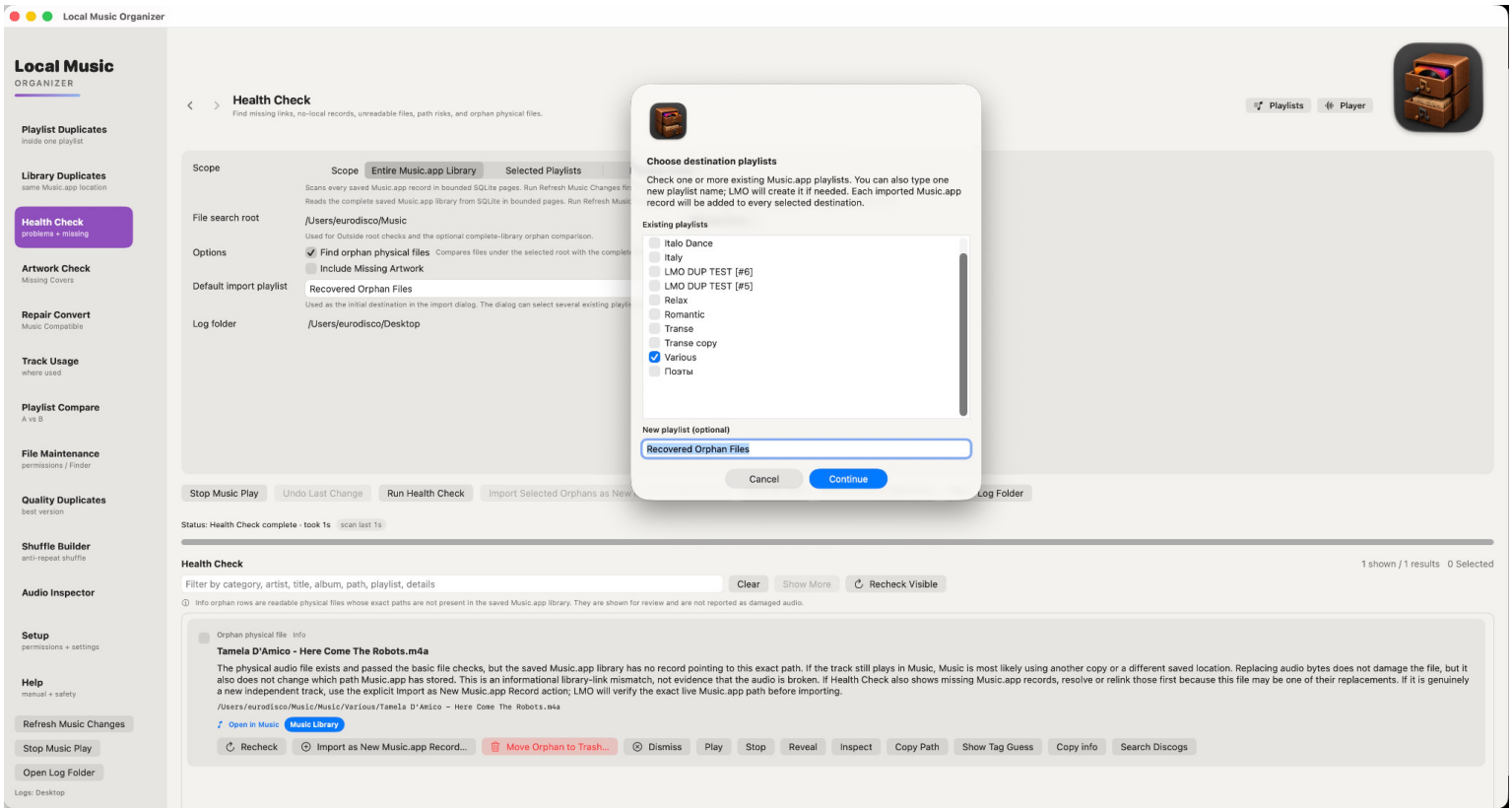
Album Completeness - p.89 Track / Disc Numbering - p.90 Batch Folder Relink - p.91

7.2 Health result actions - orphan physical file card



Orphan physical file **Read-only result label**	A readable physical audio file whose exact path is not present in the saved Music.app library. This is a library-link mismatch, not evidence that the audio file is damaged.
Recheck **Read-only targeted verification**	Rechecks only this visible orphan finding against current file and Music.app state.
Import as New Music.app Record... **Changes Music.app library**	Opens the explicit orphan-import workflow. LMO verifies the exact physical path against live Music.app before importing or reusing an existing exact-path record.
Move Orphan to Trash... **Changes physical file / Trash**	Opens the critical Trash confirmation for this exact orphan file. Before the move, LMO verifies the exact path again against live Music.app and blocks the action if Music.app now references it.
Dismiss **LMO view/state only**	Hides the reviewed orphan finding from the current result context without changing the physical file or Music.app.
Play **Playback only**	Plays the reviewed local file.
Stop **Playback only**	Stops playback started from the reviewed card.
Reveal **Navigation only**	Reveals the reviewed physical file in Finder.
Inspect **Read-only analysis**	Opens the reviewed local file in Audio Inspector.
Copy Path **Clipboard only**	Copies the exact reviewed physical path.
Show Tag Guess **Read-only / clipboard**	Opens the filename-based tag-guess view for a physical-only file that has no matched Music.app record.
Copy Info **Clipboard only**	Copies the visible orphan-result details.
Search Discogs **Navigation only**	Opens a Discogs browser search from the visible metadata. It does not write tags or change Music.app.

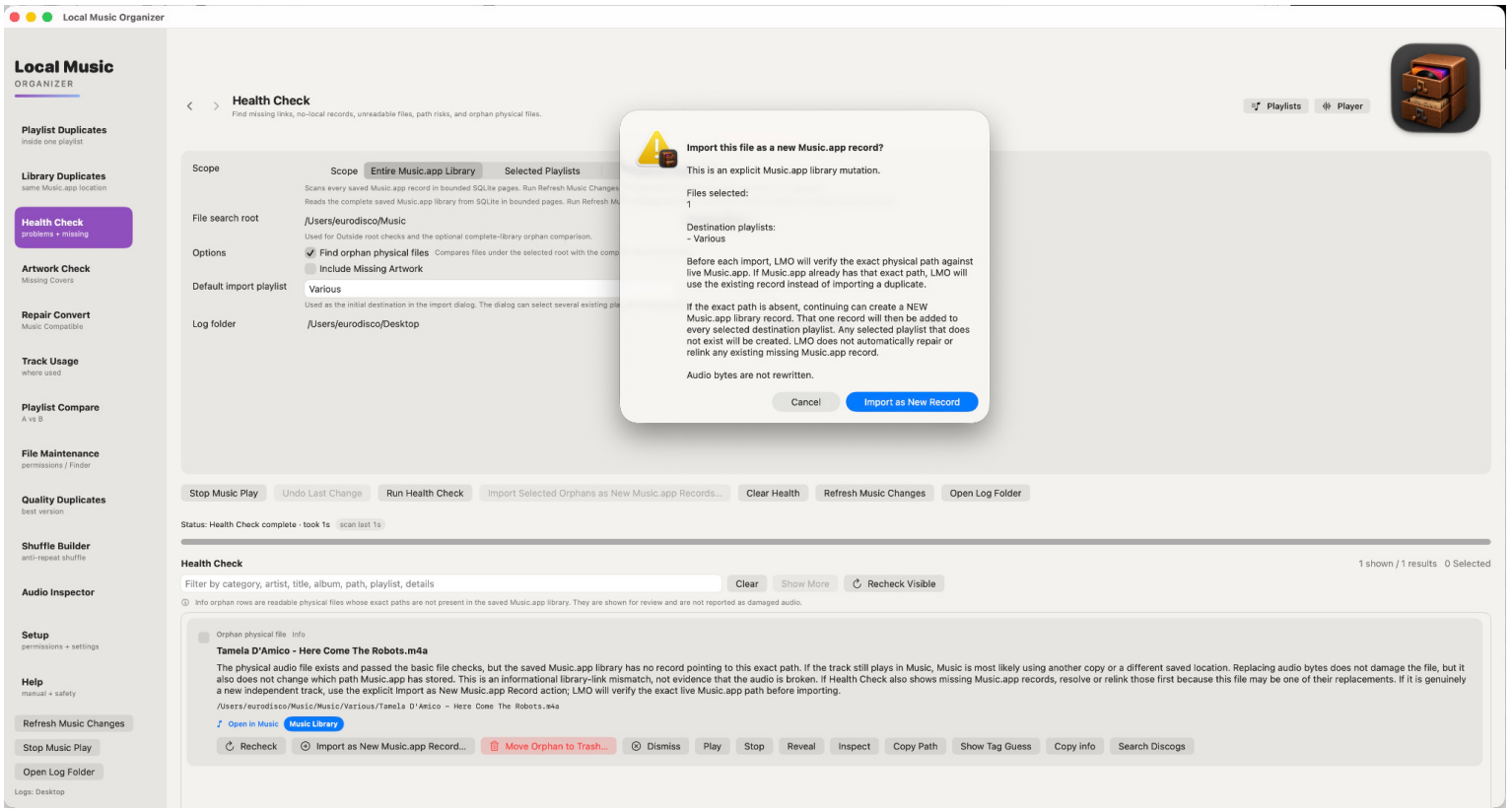
7.2 Health result actions - choose destination playlists



Multiple playlist destinations
The current Default import playlist setting supplies the initial destination for this workflow. The dialog can select one or more existing Music.app playlists and can also name one new playlist. The imported Music.app record is then added to every selected destination.

Existing playlists **Playlist selection**	Checkbox list of existing Music.app playlists. More than one playlist can be checked; checking a box only selects a destination and does not change Music.app yet.
Playlist checkboxes **Read-only selection**	Each checked existing playlist will receive the imported or reused Music.app record after the final confirmation.
New playlist (optional) **Playlist creation field**	Allows one new playlist name to be entered. If that playlist does not exist, LMO creates it before adding the record.
Cancel **No Music.app change**	Closes the destination chooser and returns without importing the file or creating a playlist.
Continue **Proceeds to confirmation**	Keeps the chosen destinations and opens the separate final import confirmation. No import is performed by Continue alone.

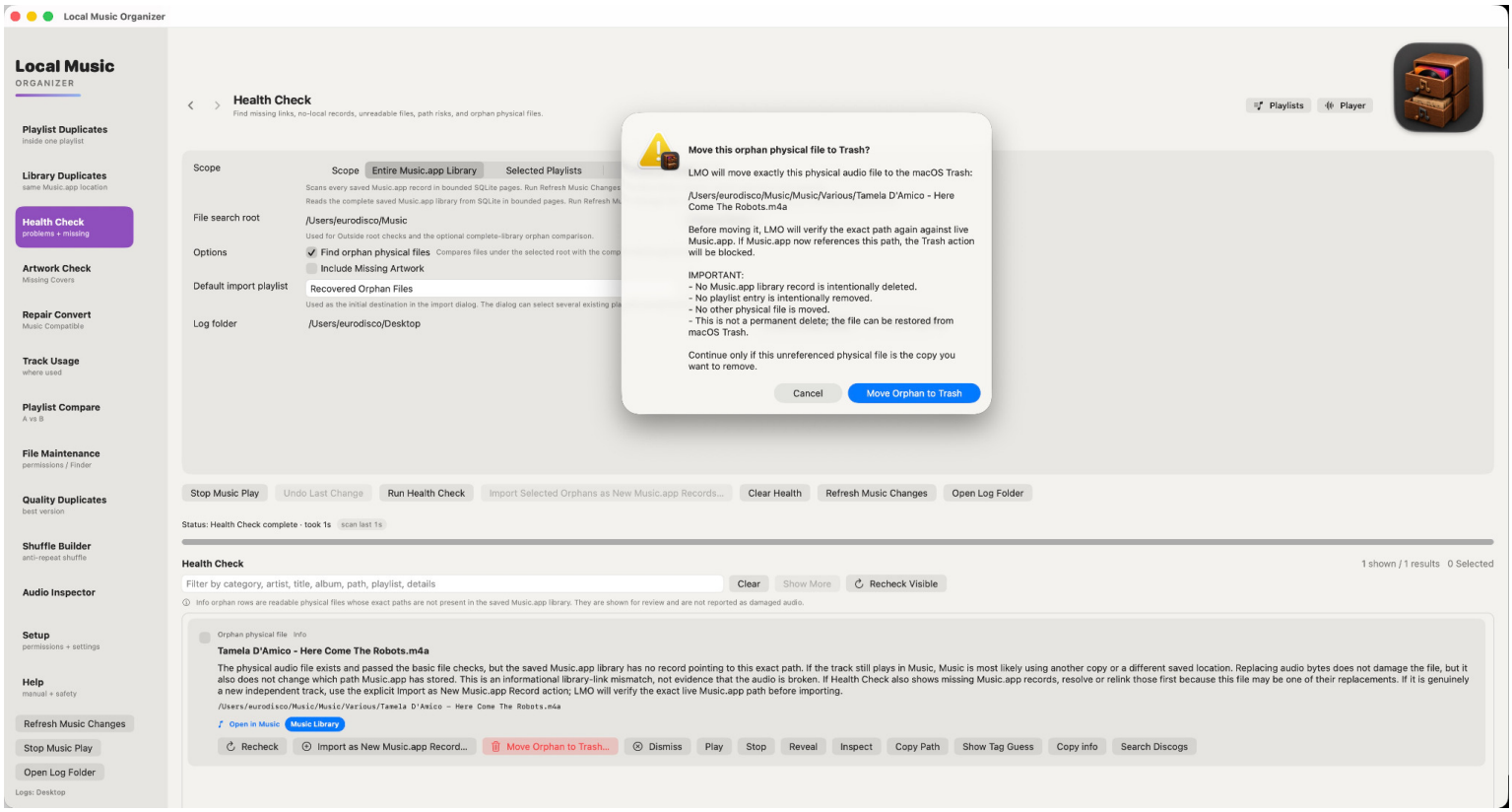
7.2 Health result actions - import confirmation



What this confirmation means
This is an explicit Music.app library mutation. Before each import, LMO verifies the exact physical path against live Music.app. If Music.app already has that exact path, LMO uses the existing record instead of importing a duplicate. If the path is absent, a new Music.app record can be created.

Files selected **Read-only confirmation**	Shows how many explicitly reviewed orphan physical files are included in this import action.
Destination playlists **Read-only confirmation**	Lists every selected destination. The same imported or reused Music.app record is added to each listed playlist.
Exact-path verification **Read-only safety check**	Immediately before import, LMO checks live Music.app for the same exact physical path. An existing exact-path record is reused instead of creating another library record.
Cancel **No Music.app change**	Aborts the import confirmation. No Music.app record or playlist entry is created by Cancel.
Import as New Record **Changes Music.app library**	Confirms the reviewed import. If a new record is required, LMO creates it and then adds that same record to every selected destination playlist. Audio bytes are not rewritten.

7.2 Health result actions - orphan Trash confirmation



Trash safety scope
LMO moves only the reviewed orphan physical file to the macOS Trash. Before moving it, LMO verifies the exact path again against live Music.app. If Music.app now references that path, the Trash action is blocked. No Music.app library record or playlist entry is intentionally removed.

Reviewed file path **Read-only confirmation**	Shows the exact physical audio file that will be moved if the action is confirmed.
Live exact-path check **Read-only safety check**	Immediately before Trash, LMO verifies that live Music.app still does not reference this exact path.
Move Orphan to Trash **Changes physical file / Trash**	Moves only this reviewed orphan file to the macOS Trash. It is not a permanent delete; the file can be restored from Trash.
Cancel **No file change**	Closes the confirmation and leaves the physical file, Music.app records, and playlist entries unchanged.

Common reviewed-card rule
The same interpretation applies wherever LMO presents similar per-item result cards: read-only, playback, navigation, inspection, and clipboard controls do not mutate the library; an explicitly labeled Music.app, playlist-entry, or physical-file action applies only to the reviewed item and uses the confirmation/safety checks provided for that action.

Repair Local Link… **Changes Music.app file link**	Relinks the same existing Music.app Persistent ID to the local file the user chooses, with verification. It is not a whole-library database repair.
Import as New Music.app Record… **Changes Music.app library**	Imports the reviewed orphan file as a new Music.app record.
Dismiss **LMO view/state only**	Hides/dismisses a reviewed orphan finding from the current result context without changing the physical file.
Play **Playback only**	Plays the matching local/Music.app item.
Reveal / Reveal Extra Copy **Navigation only**	Reveals the path represented by the finding. “Extra Copy” distinguishes an unreferenced physical file from the file Music.app currently uses.
Reveal Music.app Copy/Copies **Navigation only**	Opens the alternative current Music.app location(s) when a suspicious extra physical copy is being reviewed.
Inspect **Read-only analysis**	Opens the local file in Audio Inspector.
Fix Invisible Characters **Changes reviewed name/tag only**	Applies the narrowly scoped visible naming/tag correction offered for a verified invisible-character finding.
Copy Path / Copy Info **Clipboard only**	Copies the path or finding details.
Edit Tags **Changes only after save**	Opens Music.app metadata editor for a matched Music.app record.
Show Tag Guess **Read-only / clipboard**	For a physical-only file with no Music.app record, opens a guessed tag view; it cannot save to Music.app because there is no matched record.
Search Discogs **Navigation only**	Opens a browser search from the visible metadata.

7.3 “Music uses another file location” actions

Meaning of "Music uses another file location"
This does not mean that Music.app has lost the song. It means LMO found an additional physical file with the same normalized filename while Music.app already points to another physical path. The extra copy is therefore not auto-imported. The result deliberately separates the path Music.app uses from the extra path so the user can compare them before choosing any destructive action.

TRASH THIS FILE **Read-only safety label**	Label identifying the unreferenced extra physical copy, not the current Music.app file.
KEEP THIS FILE **Read-only safety label**	Label identifying the physical path currently referenced by Music.app.
Trash THIS Extra File… **Changes physical file / Trash**	Moves only the explicitly identified extra physical file to macOS Trash after critical confirmation. It does not remove the current Music.app record automatically.
Keep Both - Dismiss for Now **No file change**	Leaves both physical files untouched and dismisses the finding for the current review.
Prefer Extra Copy… **Review workflow**	Opens the deliberate workflow for preferring the extra copy; it does not silently replace the Music.app link.

7.4 Health Check: record and physical-file decisions

The same results area can offer actions with different targets. Read the target path and Music.app identity before confirming a change.

Recheck Visible	Re-verifies currently displayed Health findings. It is a scoped read-only check and does not repair every hidden or unshown result.
Remove Missing Record...	Removes the explicitly reviewed missing Music.app library record after confirmation and safety checks. It does not move the audio file to Trash. Record deletion is not the same as repairing its saved file link.
Repair Local Link...	Uses a selected readable replacement file to update the existing Music.app record. The operation preserves the verified Persistent ID rather than importing another record.
Relink This Folder...	Batch Folder Relink applies exact relative-path mapping from the old parent to a chosen replacement folder. It checks live record identity and mapped paths; it does not move files.
Import as New Music.app Record...	Available for an eligible orphan physical file. A live exact-path check runs before import; an existing record referring to that path is reused instead of duplicated.
Move Orphan to Trash...	Moves only the specifically reviewed orphan physical file to macOS Trash after critical confirmation and a fresh exact-path Music.app check. It is not a Music.app record deletion.
TRASH THIS FILE / KEEP THIS FILE	The on-screen labels read TRASH THIS FILE - unreferenced extra copy and KEEP THIS FILE - Music.app currently points here. LMO marks two exact paths, not two interchangeable filenames. The Trash action targets only the unreferenced path; the kept Music.app path and playlist entries remain untouched.
Keep Both - Dismiss for Now	Leaves both physical files in place and dismisses that finding for the current review. It does not merge records, change playlists or select a better recording.
Clear Numbering...	A separate Music.app metadata operation for the four Track/Disc number and total fields. Finding incomplete albums is diagnostic; clearing numbering never fills missing tracks.

A file absent from a saved Library Index snapshot is not automatically unreferenced in live Music.app. Import and Trash operations perform additional verification for this reason.

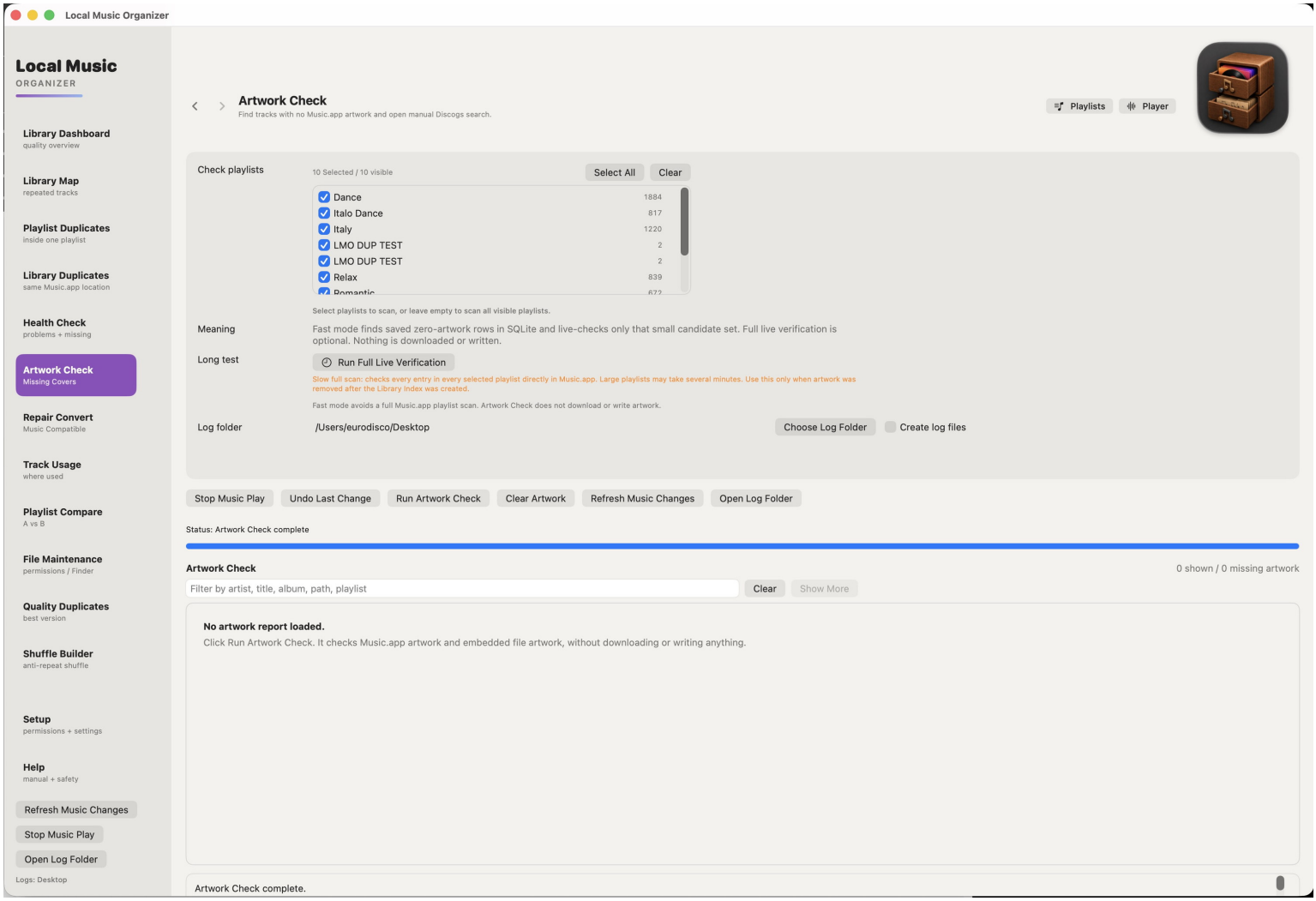
7.5 Health Check: album, numbering and diagnostic controls

Album Completeness and Track / Disc Numbering are separate Health result modes. An album warning is diagnostic until a named metadata operation is selected and confirmed.

Select All Numbered	Selects the currently displayed Track / Disc numbering findings. It does not select unrelated Health issues or change metadata.
Deselect Numbered	Removes the selection from displayed numbering findings without altering tracks.
Clear Selected Track / Disc Numbering...	Requests a confirmed change to Track Number, Track Count, Disc Number and Disc Count for the selected eligible Music.app records. The control is unavailable if this feature is disabled in the current build.
Recheck Album	Runs the Health Check again for the current album review. It is diagnostic; it neither imports a missing track nor clears any numbering.
Open Album	Opens or reveals the reviewed album context in Music.app, when available. It is a navigation operation.
Clear Album Numbering...	Requests numbering cleanup only for the numbered tracks shown for this selected album. This is a separate confirmed metadata change.
Save Health Log...	Exports the current live Health diagnostic timeline when one exists. It can be used while Health Check is still running; it is not a command to stop or finish the scan.
Clear Filter and Show Results	Clears result-list focus restrictions so the broader already-loaded Health findings can be inspected. It does not start an extra scan.
Copy Info / Search Discogs	Copies the displayed album findings or opens a browser search. These actions do not change Music.app numbering.

A disabled cleanup button may reflect an active scan, no eligible selection, or a build-level feature gate. Do not interpret a disabled button as permission to edit protected Music.app fields elsewhere.

8. Artwork Check



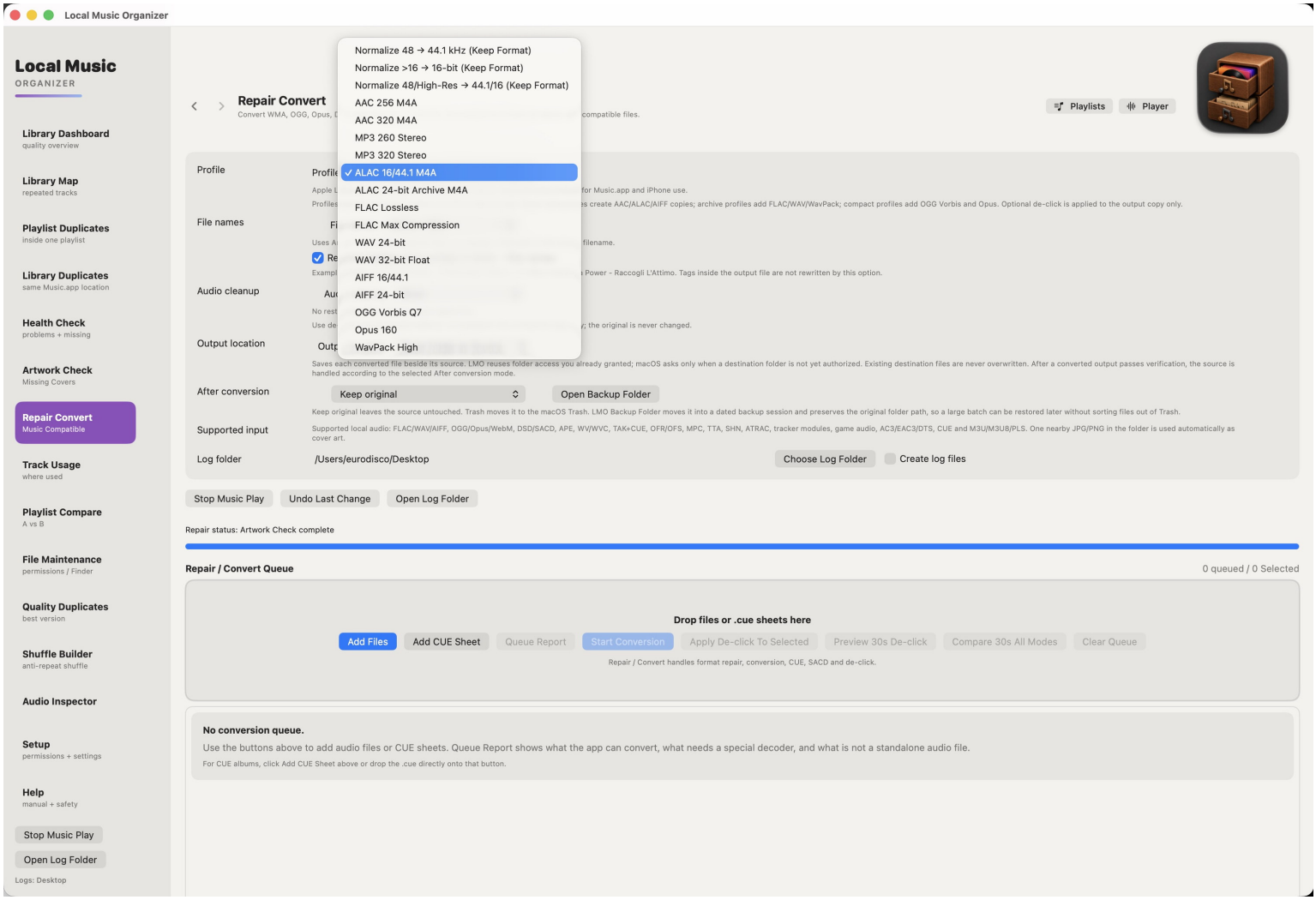
Artwork Check searches for Music.app records that appear to have no cover artwork. The check is diagnostic only: LMO does not download artwork and does not write artwork into Music.app or audio files.

8.1 Artwork Check controls

Playlist selection **Read-only selection**	Chooses the playlist whose saved rows will be checked for artwork state.
Meaning **Read-only display**	Explains the difference between the fast candidate scan and full live verification.
Run Artwork Check **Read-only scan**	Uses indexed rows to identify possible missing-artwork candidates and live-verifies only those candidates as required.
Run Full Live Verification **Read-only Music.app verification**	Slow path that asks Music.app to verify every selected playlist entry live. Use when a complete current verification is more important than speed.
Clear Artwork **LMO view only**	Clears Artwork Check results from LMO.
Filter / Show More / Clear **Read-only view**	Filter and page the current missing-artwork findings.
Result-card Play / Reveal / Inspect / Edit Tags / Copy **Mixed - explicit only**	Use the same exact-record/file actions described elsewhere. Edit Tags changes supported text metadata only after Save; Artwork Check itself never writes cover art.

Important limitation “Missing artwork” means LMO could not confirm artwork for that Music.app record in the chosen check. LMO does not fetch artwork from the internet and does not attempt to rewrite embedded cover images.

9. Repair / Convert



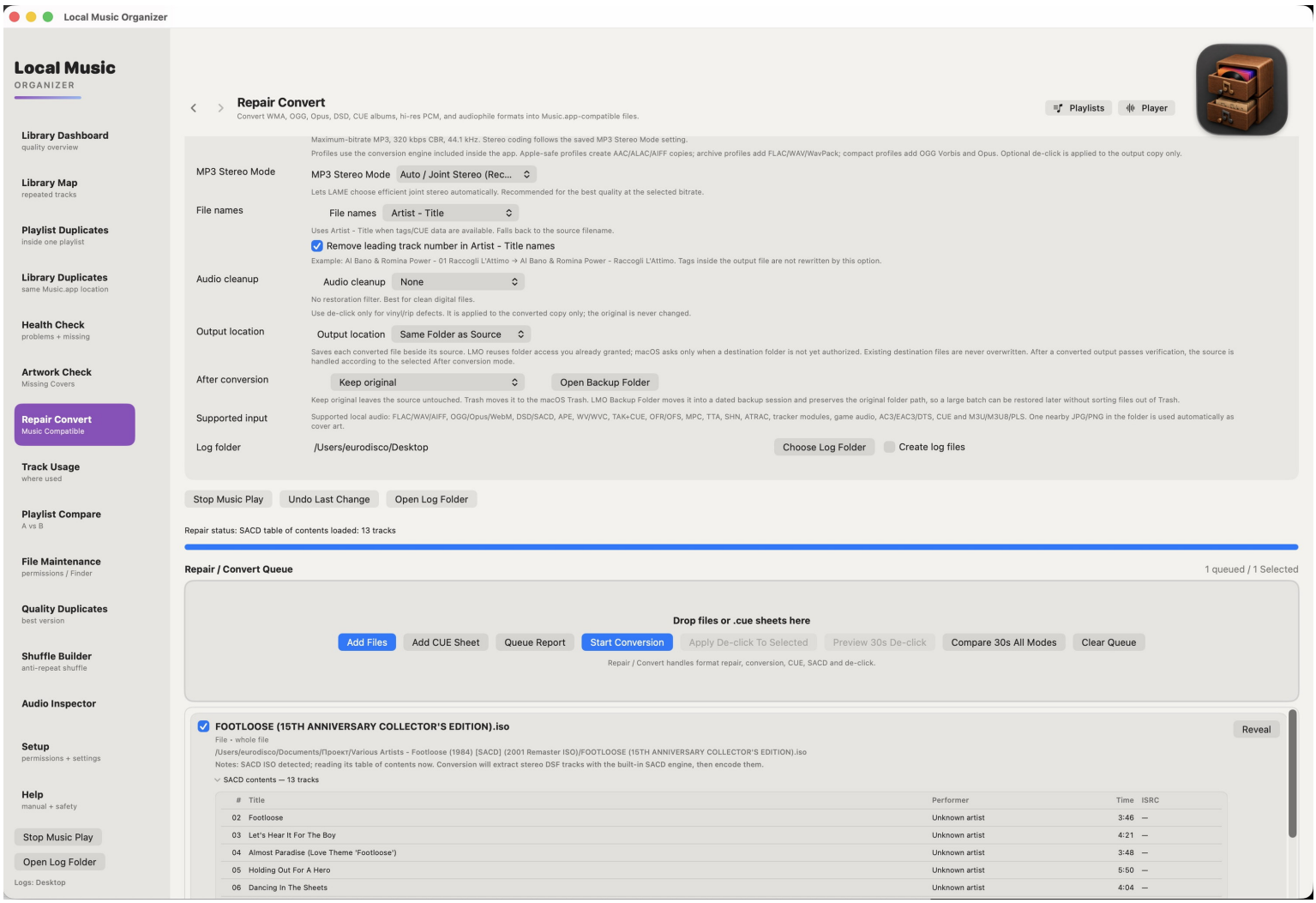
Repair / Convert creates verified output files from selected local audio sources and handles specialist sources such as CUE albums. Conversion, Queue Report, Vinyl De-click, and CUE handling are separate functions even though they share the same queue.

9.1 Conversion profile and naming controls

Profile **LMO conversion setting**	Selects the conversion/normalization target. The profile defines codec/container/sample-rate/bit-depth behavior; it does not start conversion.
Normalize 48 → 44.1 kHz (Keep Format) **Output file only**	Resamples 48 kHz sources to 44.1 kHz while keeping a supported source format family. Lossy MP3/AAC require lossy re-encoding and are warned before conversion.
Normalize >16 → 16-bit (Keep Format) **Output file only**	Reduces supported lossless/PCM sources above 16-bit to 16-bit while keeping their format family.
Normalize 48/High-Res → 44.1/16 (Keep Format) **Output file only**	Combines 44.1 kHz resampling and 16-bit normalization for supported sources while retaining the source format family.
AAC 256 M4A / AAC 320 M4A **Output file only**	Creates stereo AAC M4A listening copies at the stated bitrate and 44.1 kHz.
MP3 260 Stereo / MP3 320 Stereo **Output file only**	Creates CBR MP3 at the stated bitrate and 44.1 kHz. Stereo coding follows MP3 Stereo Mode.
ALAC 16/44.1 M4A **Output file only**	Creates a Music.app-friendly Apple Lossless 16-bit / 44.1 kHz / stereo copy.

ALAC 24-bit Archive M4A Output file only	Creates a 24-bit Apple Lossless archive copy.
FLAC Lossless / FLAC Max Compression Output file only	Creates FLAC output with balanced or stronger lossless compression. FLAC is primarily for workflows outside Music.app.
WAV 24-bit / WAV 32-bit Float Output file only	Creates uncompressed WAV output for archive/editor/intermediate workflows.
AIFF 16/44.1 / AIFF 24-bit Output file only	Creates PCM AIFF output at the stated resolution.
OGG Vorbis Q7 / Opus Output file only	Creates the stated lossy non-Apple listening format.
WavPack High Output file only	Creates high-compression WavPack lossless output when the included engine supports encoding.
MP3 Stereo Mode - Auto / Joint Stereo (Recommended) LMO conversion setting	Lets the MP3 encoder choose efficient joint stereo automatically.
MP3 Stereo Mode - Forced Stereo LMO conversion setting	Encodes left/right independently; intended only for a specific compatibility requirement.
File names - Keep album numbering LMO output naming setting	Keeps the source/CUE track name including leading album track numbers.
File names - Artist - Title LMO output naming setting	Uses Artist - Title when usable metadata/CUE fields exist, otherwise falls back to the source name.
Remove leading track number in Artist - Title names LMO output naming setting	When Artist - Title naming is selected, removes leading numeric track prefixes from the generated output name.

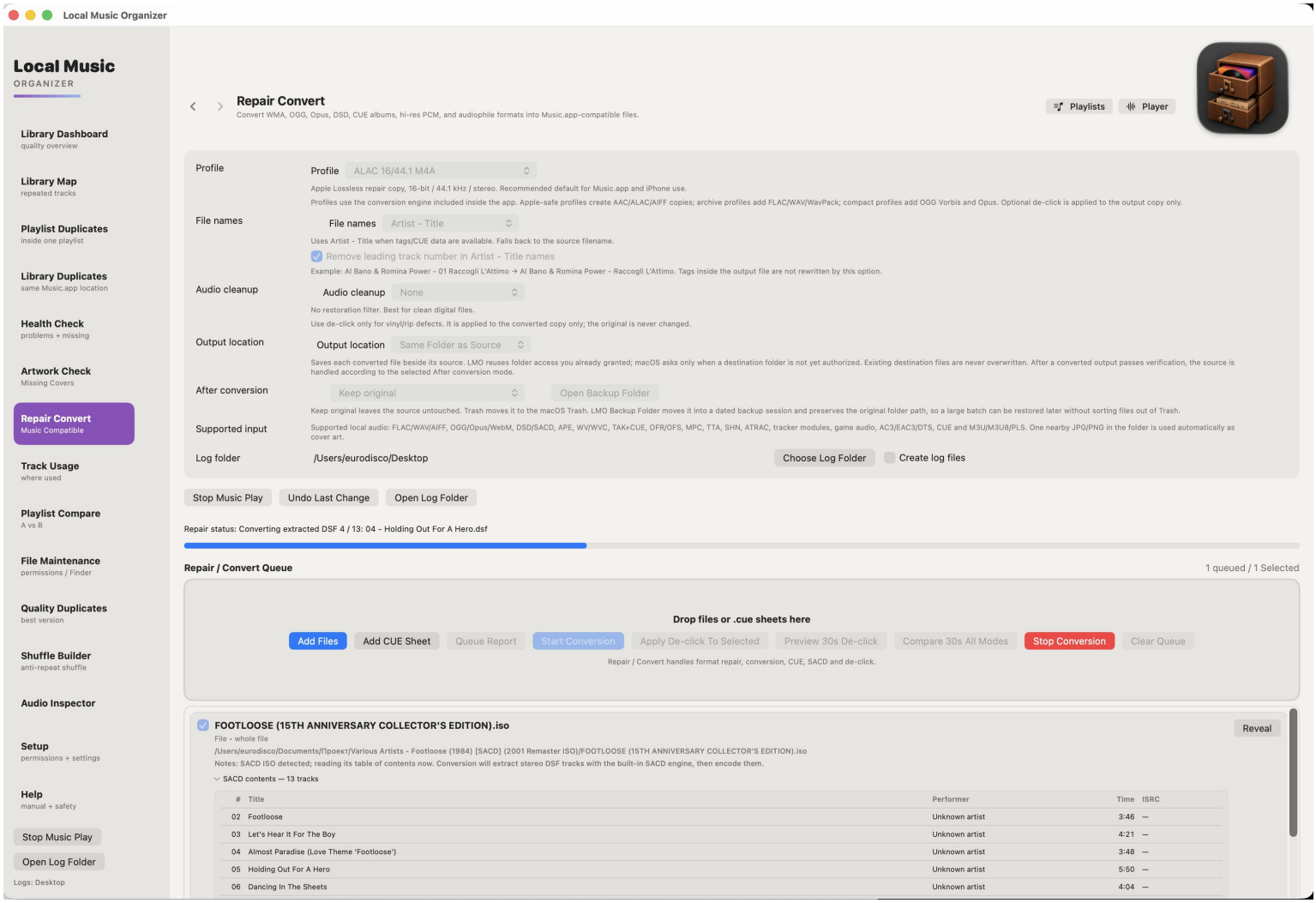
9.2 Output, cleanup, and source handling



Audio cleanup - None **Output setting**	No restoration filter is applied.
Vinyl De-click Light **Output file only**	Light de-click intended for mild ticks while preserving transients.
Vinyl De-click Medium **Output file only**	Careful stronger de-click intended as the normal vinyl-cleanup starting point.
Vinyl De-click Strong **Output file only**	Stronger multi-pass cleanup for noisier rips; sharp attacks can be softened, so review is important.
Vinyl De-click Rescue **Output file only**	Most aggressive rescue mode. It is the slowest/most destructive de-click option and should be auditioned before use.
Output location - Music Converted Folder **Output setting**	Saves converted files in the selected conversion destination.
Output location - Same Folder as Source **Output setting**	Saves each verified output beside its source. Existing destination files are never silently overwritten.
Choose... / Open **LMO setting / navigation**	Chooses or opens the current Music Converted Folder destination.
After conversion - Keep original **Source unchanged**	Leaves the source file in place after output verification.
After conversion - Move original to Trash **Changes physical source / Trash**	After the converted output passes verification, moves the source to macOS Trash.

After conversion - Move original to LMO Backup Folder Changes physical source location	After successful output verification, moves the source into LMO's backup folder rather than Trash.
Open Backup Folder Navigation only	Opens the LMO backup folder in Finder.

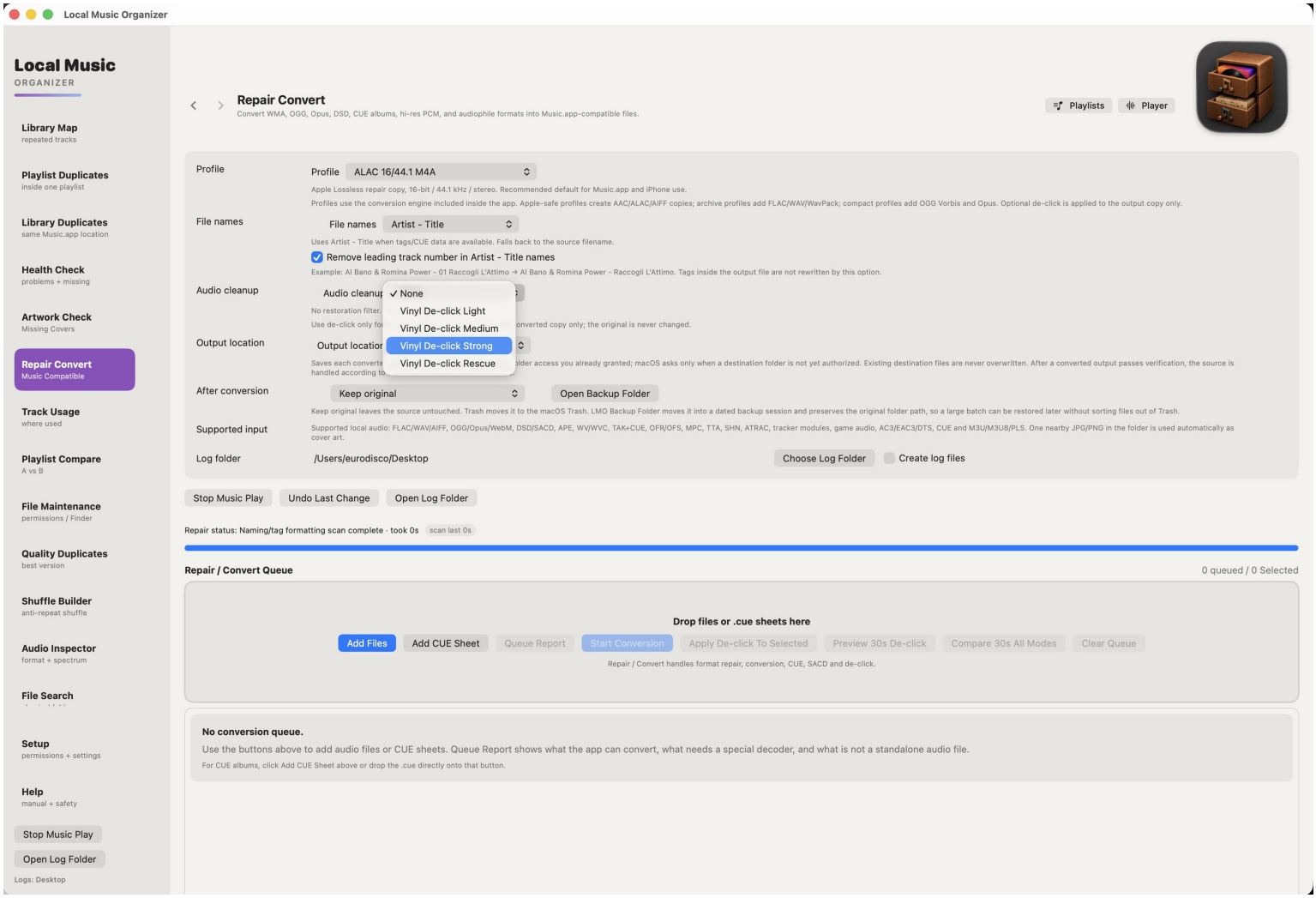
9.3 Queue controls



Drop area Queue only	Accepts supported local audio sources/CUE sheets for the Repair / Convert queue. Adding a source does not convert it.
Add Files Queue only	Opens a file picker and adds supported local audio files to the queue.
Add CUE Sheet Queue only	Reads CUE FILE references and automatically locates sibling album audio after any needed folder authorization. You normally do not select each file; source files are unchanged.
Queue Report Read-only report	Creates an informational report for the current queue. It reports source kind, support/decoder status, and conversion decision. It does not convert files.
READY Read-only status	Queue Report status indicating the source is ready for the selected conversion path.
NEEDS TOOL Read-only status	Queue Report status indicating a required bundled/available decoder/tool path is not ready.
NOT STANDALONE Read-only status	Queue Report status for an item that is not a standalone audio source by itself.

LIMITED Read-only status	Queue Report status for a source with limited handling.
SPECIAL DECODER Read-only status	Queue Report status indicating a specialist decoder path is needed.
BLOCKED Read-only status	Queue Report status indicating conversion is intentionally blocked for the current item/condition.
Start Conversion Creates output / may handle source	Starts conversion only for the selected queue items using the current profile/output/cleanup/original-handling settings.
Stop Conversion Stops conversion	Stops the active encoder process. Already verified completed outputs retain the selected After conversion handling; an unverified current source remains protected.
Clear Queue Queue only	Clears pending queue items and related preview state from LMO. It does not delete source files.
Reveal Navigation only	Reveals the queued source file in Finder.

9.4 Vinyl De-click, CUE, and specialist-source controls



Apply De-click To Selected Creates output file	Runs the chosen de-click mode for selected queue files and creates cleaned copies at the selected destination. The selected source files are not overwritten by this action.
Preview 30s De-click Temporary preview output	Creates/auditions a 30-second sample for the currently selected de-click mode so the restoration strength can be judged without changing the source.
Compare 30s All Modes Temporary preview outputs	Creates 30-second comparison samples for Light, Medium, Strong, and Rescue. The original remains unchanged.

CUE track rows Queue only until conversion	Represent track ranges defined by the CUE sheet, including title/artist/album information when present. Conversion extracts the selected ranges from the referenced source audio.
SACD contents disclosure Read-only source info	For recognized SACD ISO sources, expands the readable disc/table-of-contents information when the specialist decoder can obtain it. Reading the table does not extract audio.

Queue Report is not conversion The correct button name is Queue Report. Its purpose is to explain the queue and decoder path. Conversion begins only with Start Conversion.

CUE files and album-folder permission

When a CUE is selected or dropped, LMO reads its FILE references and tries to resolve the matching source audio in the album folder. macOS may ask you to authorize that folder. Successful permission normally avoids selecting individual source tracks. If sandbox access is denied, LMO reports the problem; it does not bypass macOS permissions or modify the original audio.

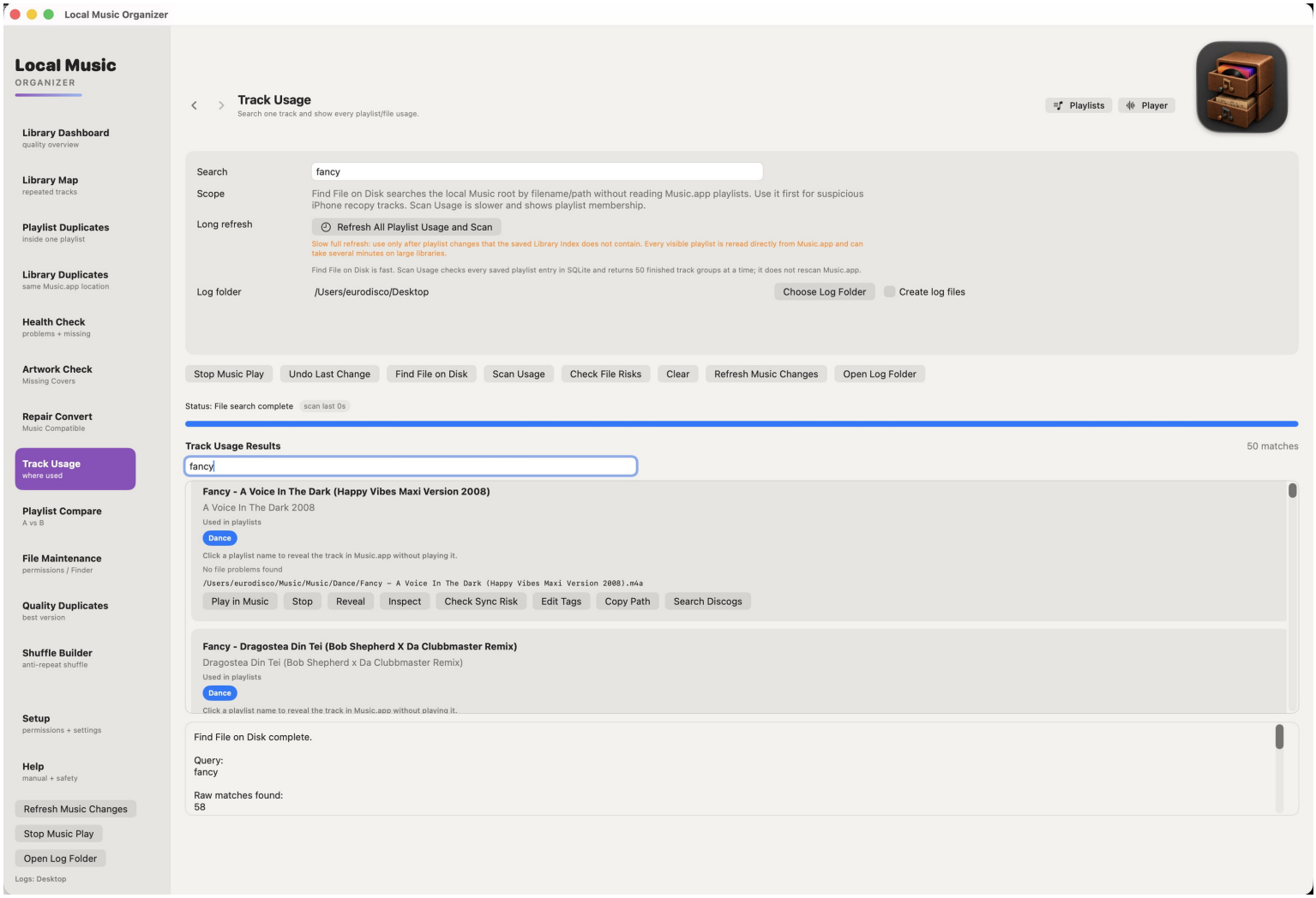
9.5 De-click preview samples and their files

De-click preview actions let you listen before deciding whether to create a full cleaned copy. Preview and comparison actions do not rewrite the source audio.

Preview 30s De-click	Creates a saved 30-second preview with the currently selected de-click strength. Requires an eligible selected queue item and an active de-click mode.
Compare 30s All Modes	Creates comparison excerpts for Light, Medium, Strong and Rescue. Select the weakest setting that removes objectionable clicks without softening attacks.
Preview sample card	Displays the saved sample filename, de-click mode, original source and output folder. This is a new output file, not a replacement of the source.
Play Sample	Plays the saved sample through LMO local preview when it exists. This is useful for comparing the results of individual modes.
Stop	Stops current playback from the sample card. It does not cancel an unrelated conversion job.
Reveal Sample / Reveal Folder	Reveal Sample locates one excerpt in Finder; Reveal Folder opens the directory holding the created excerpts.
Copy Path	Copies the precise physical path to the selected sample; neither the sample nor the source file is changed.
Delete Sample	Deletes that saved preview file and removes its card. This action uses direct file removal, not Move to Trash, so retain any sample you still need.
Clear Samples	Clears the preview cards and current preview selection in LMO. It does NOT delete preview files from their folder. To remove a sample file, use Delete Sample.
Apply De-click To Selected	Creates cleaned output for the selected eligible queue files using the chosen mode. Original handling remains governed separately by the selected conversion/output policy.

A queue reset also clears the on-screen preview state. It is not a substitute for reviewing the preview folder and deleting files that are no longer wanted.

10. Track Usage



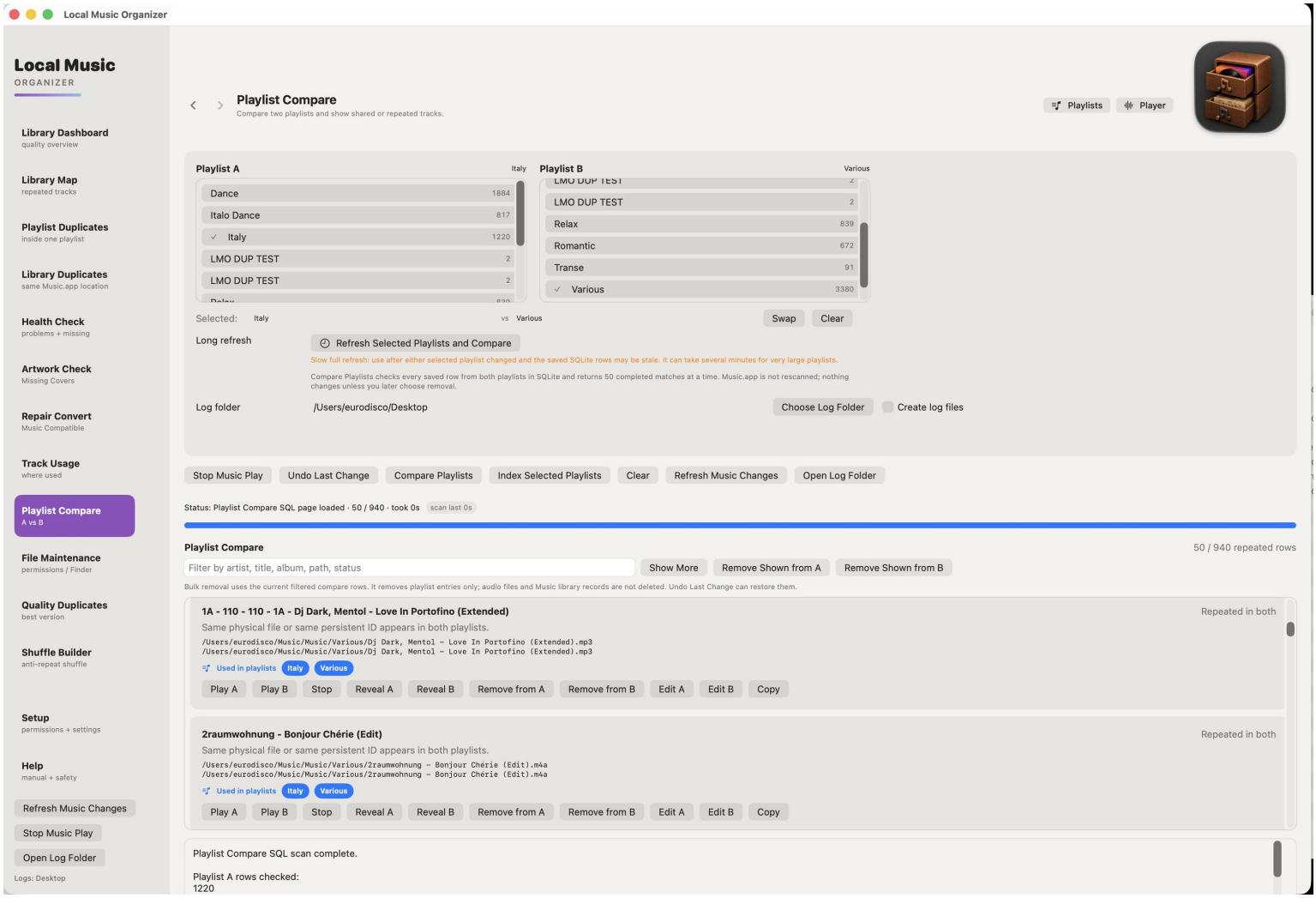
Track Usage answers “where is this track used?” by searching indexed Music.app records and saved playlist membership. It can also start from a physical-file search and then test only the matching rows for playlist use or file risks.

10.1 Track Usage controls and results

Search field **Read-only query**	Accepts artist, title, album, filename, path, or Persistent ID.
Scan Usage **Read-only scan**	Fast path: searches the saved SQLite playlist-usage index. It does not rescan every playlist from Music.app.
Refresh All Playlist Usage and Scan **LMO index then read-only scan**	Slow path: refreshes all visible playlist membership from Music.app, saves it, then runs the usage search.
Find File on Disk **Read-only filesystem search**	Searches the configured physical-file area for the requested item when the physical file rather than playlist usage is the starting point.
Check File Risks **Read-only targeted verification**	Runs read-only diagnostics on the matching rows for duplicate paths, repeated-copy risk, unusual file metadata, and related sync/file concerns.
Clear **LMO view only**	Clears Track Usage search/results in LMO.
Filter / Show More **Read-only view**	Filters and pages the current usage results.
Playlist chips **Navigation only**	Show the indexed playlists that contain the track. Selecting a chip reveals the playlist destination rather than changing membership.

Scan Usage now Read-only scan	For a physical result whose playlist usage has not yet been checked, performs the targeted usage lookup.
Play in Music Playback only	Plays the matched Music.app record.
Preview Preview only	For a physical-only result, previews the local file without creating a Music.app record.
Stop Playback only	Stops playback.
Reveal Navigation only	Reveals the file path in Finder.
Inspect Read-only analysis	Opens Audio Inspector for the local file.
Edit Tags Changes only after save	Opens the Music.app metadata editor when an exact Music.app record is available.
Copy Path Clipboard only	Copies the physical path.
Search Discogs Navigation only	Opens an external Discogs search from the visible metadata.

11. Playlist Compare



Playlist Compare compares every saved row from two selected playlists and shows what is shared or present on only one side. It can use the fast saved index or explicitly refresh the two selected playlists first.

11.1 Playlist Compare controls

Playlist A / Playlist B **Read-only selection**	Choose the two playlists to compare.
Swap **Read-only selection**	Exchanges Playlist A and Playlist B so one-sided results can be interpreted from the opposite direction.
Clear **LMO view only**	Clears the selected comparison/result state.
Compare Playlists **Read-only scan**	Fast path: compares all saved rows from both selected playlists directly in SQLite and loads results in pages.
Index Selected Playlists **LMO index**	Builds missing saved membership for the two selected playlists without indexing every visible playlist.
Refresh Selected Playlists and Compare **LMO index then read-only scan**	Slow path: rereads both selected playlists from Music.app, saves complete snapshots, then compares them.
Filter **Read-only filter**	Filters comparison results by artist/title/album/path/status.
Show More **Read-only view**	Loads additional comparison rows.

Remove Shown from A / Remove Shown from B **Changes playlist entries**	Bulk-removes only the currently shown/filtered playlist entries from the indicated playlist after review. Physical files are not deleted.
Play A / Play B **Playback only**	Plays the exact record from the corresponding side.
Reveal A / Reveal B **Navigation only**	Reveals the local path for the corresponding side.
Remove from A / Remove from B **Changes playlist entry**	Removes that exact occurrence from the indicated playlist only.
Edit A / Edit B **Changes only after save**	Opens the exact Music.app record on that side in the metadata editor.
Copy **Clipboard only**	Copies the comparison row information.

11.2 Playlist comparison: saved and live data

Track Usage and Playlist Compare have fast SQLite paths and deliberate slower Music.app refresh actions. Use the slower choice only when the saved membership snapshots may be stale.

Track Usage - Find File on Disk	Finds local file candidates for a track identity. This path is separate from checking saved playlist membership.
Track Usage - Scan Usage	Checks the already saved SQLite playlist entries and returns finished track groups in batches; it does not reread Music.app.
Refresh All Playlist Usage and Scan	Slow Track Usage action: reads every entry in every visible playlist from Music.app, saves the refreshed membership rows and then reruns Track Usage. Requires search text and may take several minutes.
Playlist A / Playlist B	Choose the two source playlists for Playlist Compare. The ordinary fast comparison checks their saved SQLite membership rows.
Swap / Clear	Exchanges the two selected playlists or clears the comparison selections. Neither operation changes Music.app playlist contents.
Refresh Selected Playlists and Compare	Slow Playlist Compare action: rereads the two selected playlists directly from Music.app, saves both snapshots to SQLite and then compares them.
Compare Playlists	Fast path over the saved membership snapshots. It returns completed matches in pages and never removes a playlist entry by itself.
After manual playlist editing	If the source playlist was changed after indexing, refresh only the affected selected playlists rather than rebuilding unrelated full library records.

A fast empty comparison from an incomplete or stale snapshot is not proof that no relationship exists in live Music.app. Check playlist-usage coverage first.

12. File Maintenance

File Maintenance keeps permission checks, Finder attributes, naming fixes, and embedded audio-tag cleanup in one place. Scanning never changes files. Review a result and choose a modifying action only when you want to apply it.

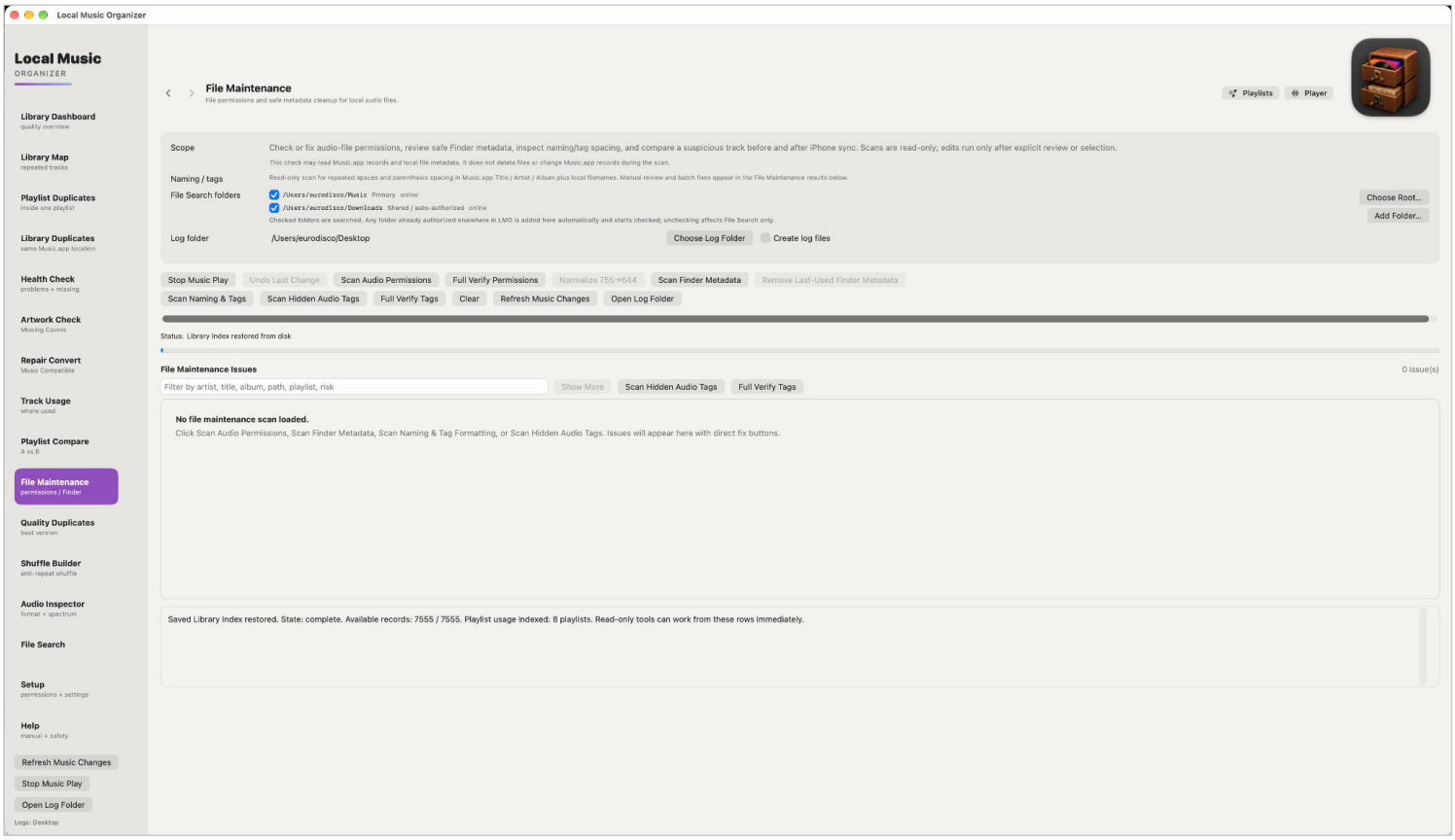


Figure 1. The starting screen before a maintenance scan. Scope and scan buttons are above the result area; the result area starts empty until a scan runs.

12.1 Scope card and maintenance toolbar

Scope / Naming / tags	Explanations of what LMO inspects. Naming checks compare Music.app text fields and local filenames; the explanation is not a second scan menu.
File Search folders	Authorized roots and checked File Search inclusion. Unchecking a root here does not delete its files or revoke previously granted folder access.
Log folder / Create log files	Choose where optional operation reports are written. The checkbox enables log creation; it does not run a scan.
Scan Audio Permissions / Full Verify Permissions	Read file permissions; Full Verify performs a new thorough check. Both are read-only.
Scan Finder Metadata / Scan Naming & Tags	Start separate read-only Finder-xattr and spacing/naming scans. Their corrections do not run as part of Hidden Audio Tags.
Scan Hidden Audio Tags / Full Verify Tags	Find eligible embedded fields; Full Verify forces direct file reinspection instead of relying on unchanged-file cache results.
Stop Music Play / Undo / Clear / Refresh / Logs	Playback stop, supported undo, clear current results, update LMO's saved Music.app index, or open the log folder. Hidden-tag removal has no Undo after success.

12.2 Hidden audio tag result example

Run Scan Hidden Audio Tags to populate the maintenance results. The scan reports physical-file fields, not every value Music.app may still hold in its own library.

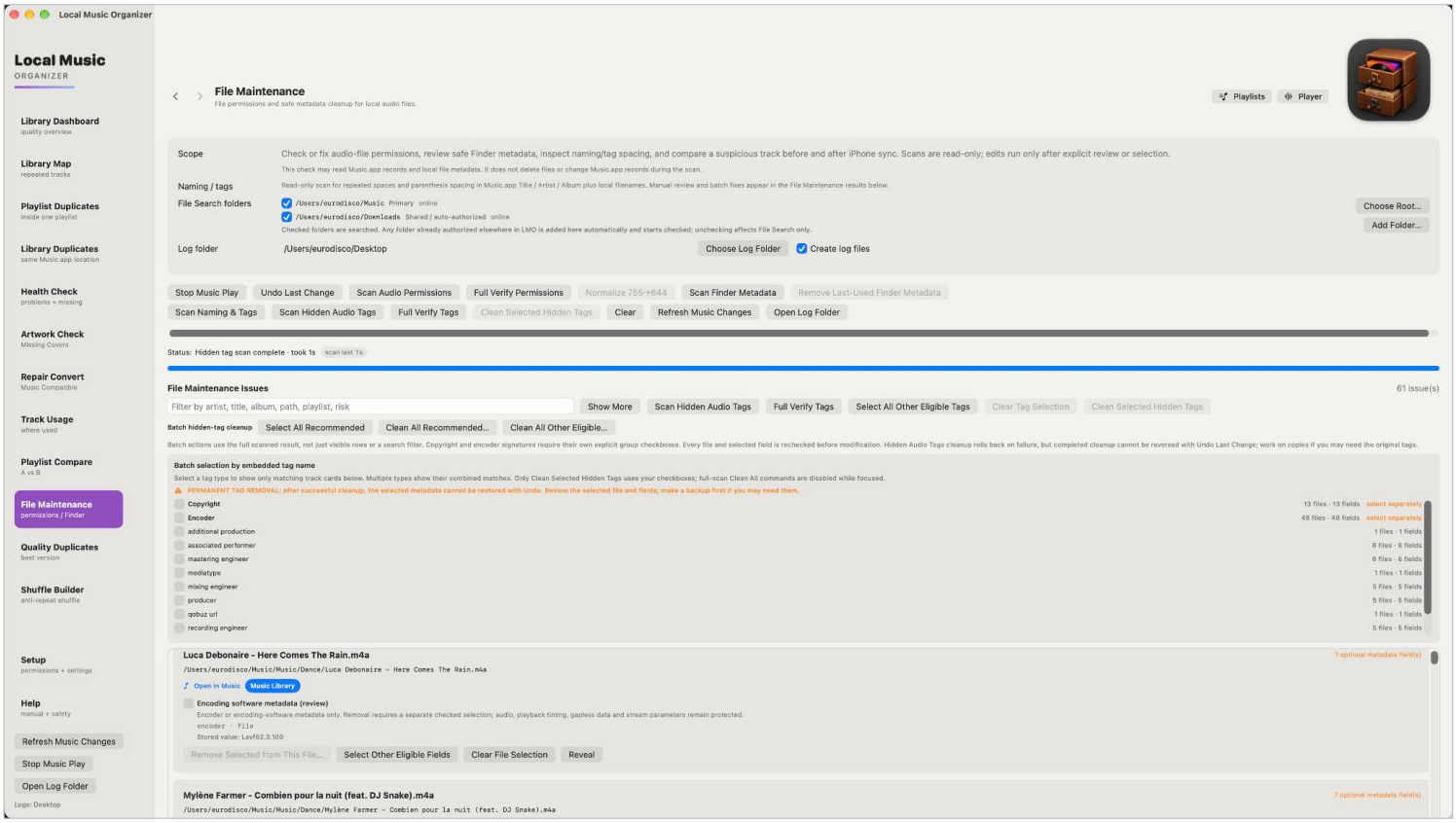


Figure 2. The general Hidden Audio Tags results: batch buttons, a tag-name group list, and individual file cards. The batch-cleanup controls live inside the results area; there is no separate duplicate upper batch menu.

Tag-name group list	Each row shows a tag type and how many files/fields match it. Click a name to focus matching cards; use its checkbox to select that group for cleanup.
Copyright and Encoder	They may appear in the same general list but require deliberate review. Copyright removal changes the physical file only; an old Music.app record can still display its stored value.
Per-file result cards	Each card identifies the physical file, stored field and value. Check only the individual fields you want removed; Reveal locates the file in Finder.
Select All Recommended / Other Eligible	Bulk selection is limited to the indicated eligible groups. Special or protected data must not be assumed selected. Review the actual checked fields before cleanup.
Show More / search / focus	These change which cards you see, not what file is modified. Check the selections and the confirmation count; a focused group may disable full-scan batch commands.
Clean Selected Hidden Tags	Runs only after the user explicitly selects supported fields and confirms. A successful operation permanently removes those embedded fields; Undo Last Change cannot restore them.

Important: scans and selections are not edits

Reading a tag or checking its box never rewrites audio. Before a cleanup, review file count, tag values and the confirmation. Keep a backup if the removed metadata may be needed later.

Second example: M4A Encoded with (©too)

An optional, separate M4A-only result panel appears when a physical ©too atom is found. It is not a duplicate of the general Hidden Audio Tags list.

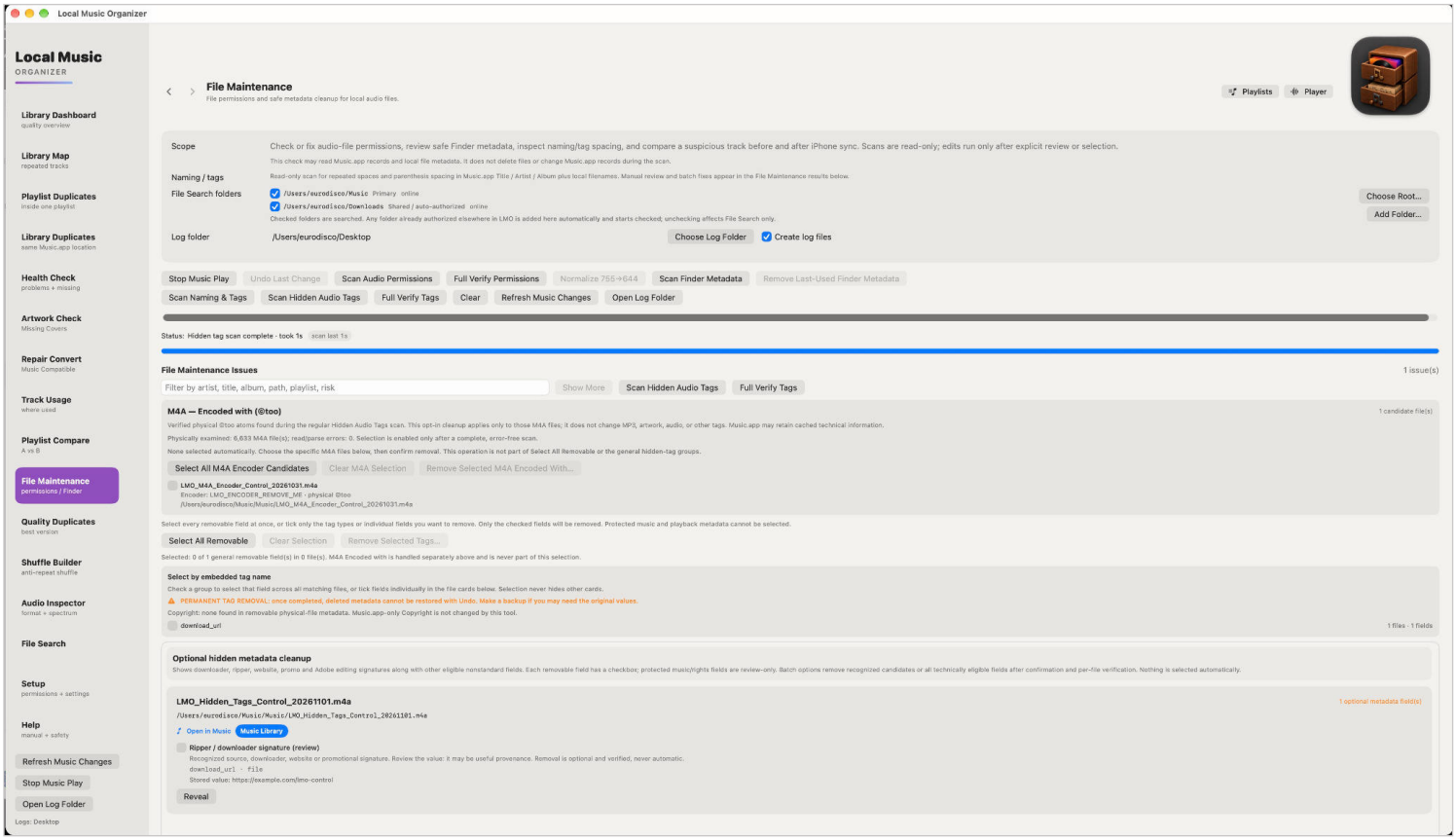


Figure 3. A scan showing an exact M4A encoder candidate alongside the normal Hidden Audio Tags results. The dedicated M4A panel is shown only for a verified physical match.

Physical M4A ©too scan	The normal scan also inspects real M4A atoms. It displays the physical-file count and any read/parse errors; a zero means no matching removable atom was found, not that Music.app cannot display an encoder.
M4A candidate checkbox	Explicitly select one or more candidates in the dedicated panel. The M4A encoder action is separate from Select All Removable and normal hidden-tag cleanup.
Remove Selected M4A Encoded With	Removes only verified ©too metadata from the chosen M4A files. It does not change MP3 MPEG headers, codec, audio samples, gapless timing, artwork or unrelated tags.
Normal hidden-tag list below	The general list can still show downloader URLs, Copyright, encoding signatures and other eligible fields. Its checkboxes and cleanup controls are independent of the M4A-only panel.
Why controls are disabled	A cleanup button remains unavailable if nothing is checked or a required verification stage has not succeeded. Run Full Verify Tags if you need a fresh physical check.

Copyright: physical file versus Music.app record

When LMO removes physically present Copyright metadata, Music.app can still display an old Copyright value saved in its library. In the verified test, refresh, tag edits and relinking did not erase it; removing and re-importing the cleaned file did, but created a new record with a new Date Added and Persistent ID. LMO does not re-import automatically.

12.3 File Maintenance: Review Finder Metadata

The Finder metadata review window is more selective than general metadata cleaners. It shows all discovered extended attributes, but editing is restricted to explicitly allowed entries.

Scan Finder Metadata	Scans the selected physical-file scope for Finder extended attributes. The result is read-only until a separate modifying action is requested.
Review Metadata...	Opens one-file Review Finder Metadata with the file name/path, attribute names, decoded date where recognized, raw bytes and byte length.
Allow-listed / Read only	Only the recognized allow-listed Finder last-used attribute may be edited or removed. Protected, unknown, audio, artwork and embedded music metadata are shown read-only.
Last-used date	For a recognized 16-byte timestamp value, the date/time picker provides a human-readable edit of that one attribute. It is not the filesystem creation or modification date.
Raw 16-byte value	Manual input is accepted only when the field contains exactly 32 hexadecimal characters. Invalid lengths or characters prevent Save Raw Hex.
Save Date	Writes the chosen last-used date to the allow-listed extended attribute for this file, subject to the operation verification.
Save Raw Hex	Writes exactly the reviewed 16-byte hexadecimal value to that same allow-listed attribute. It is not a general-purpose metadata editor.
Remove Attribute	Removes the allow-listed attribute from the reviewed file; this is not a command to erase all xattrs or embedded audio tags.
Cancel	Closes Review Finder Metadata without applying a draft value.

If an allow-listed value has an unknown layout, the window disables date and raw editing; safe removal may still be offered. The wider list of attributes remains visible for inspection.

12.4 File Maintenance: selection, batch and naming

Physical permissions, Finder metadata, Music.app spacing and physical filenames have different scopes. The scan for one category never authorizes writing another category.

Select visible / Clear Selection	Select visible checks only currently displayed Finder metadata findings. Clear Selection deselects without changing any file.
Select all matching / Clear Selection	In Naming & Tag Formatting, Select all matching selects every issue matching the current filter, even beyond the first 50 rendered cards. Clear Selection deselects them. This differs from Select visible in the Finder Metadata results, which selects only the currently displayed Finder metadata findings.
Edit Selected Date...	Opens Edit Last-Used Date in Batch for the selected rows. Only com.apple.lastuseddate#PS is targeted, and original values are captured for supported Undo before writing.
Remove Selected	Removes the safely allow-listed Finder attribute from the selected files, not arbitrary protected attributes.
Remove Safe Metadata From This	Applies the allow-listed Finder metadata cleanup to the one displayed physical file.
Full Verify Permissions	Forces the more thorough permission verification pass instead of relying on the reusable fast check. No permission is changed by verification alone.
Fix All Permissions / Fix This To 644	Applies a reviewed file-permission correction to 644 for eligible items. It does not rename, retag, relink or convert the file.
Review / Edit...	Opens a naming/tag formatting draft. Check the proposed Music.app text fields and physical filename independently before applying either.
Fix Tag Spacing	Applies only the supported Music.app text spacing correction for this row. It does not rename the physical audio file.
Rename Filename Safely	Applies only the supported verified same-folder physical rename for this row. It does not by itself request other tag-spacing changes.
Batch Fix Selected...	Opens the reviewed multi-row naming/tag editor. Music.app tag edits and safe filename renames remain distinct selectable actions.

Verify the path and Music.app Persistent ID before any file rename. A read-only scan may report issues even when the right action is to keep the existing value.

12.5 Embedded audio tags: review and verified cleanup

Use this workflow for physical metadata inside local audio files. Music.app records, playlist membership and LMO's saved SQLite index are different layers. A metadata scan does not prove that a Music.app-only field is present in the audio file.

A simple, safe workflow

1. Scan Hidden Audio Tags (or Full Verify Tags). 2. Review tag-name groups and individual values. 3. Select supported fields or a deliberate group. 4. Check the confirmation and run Clean Selected Hidden Tags. 5. Read the result and optional log.

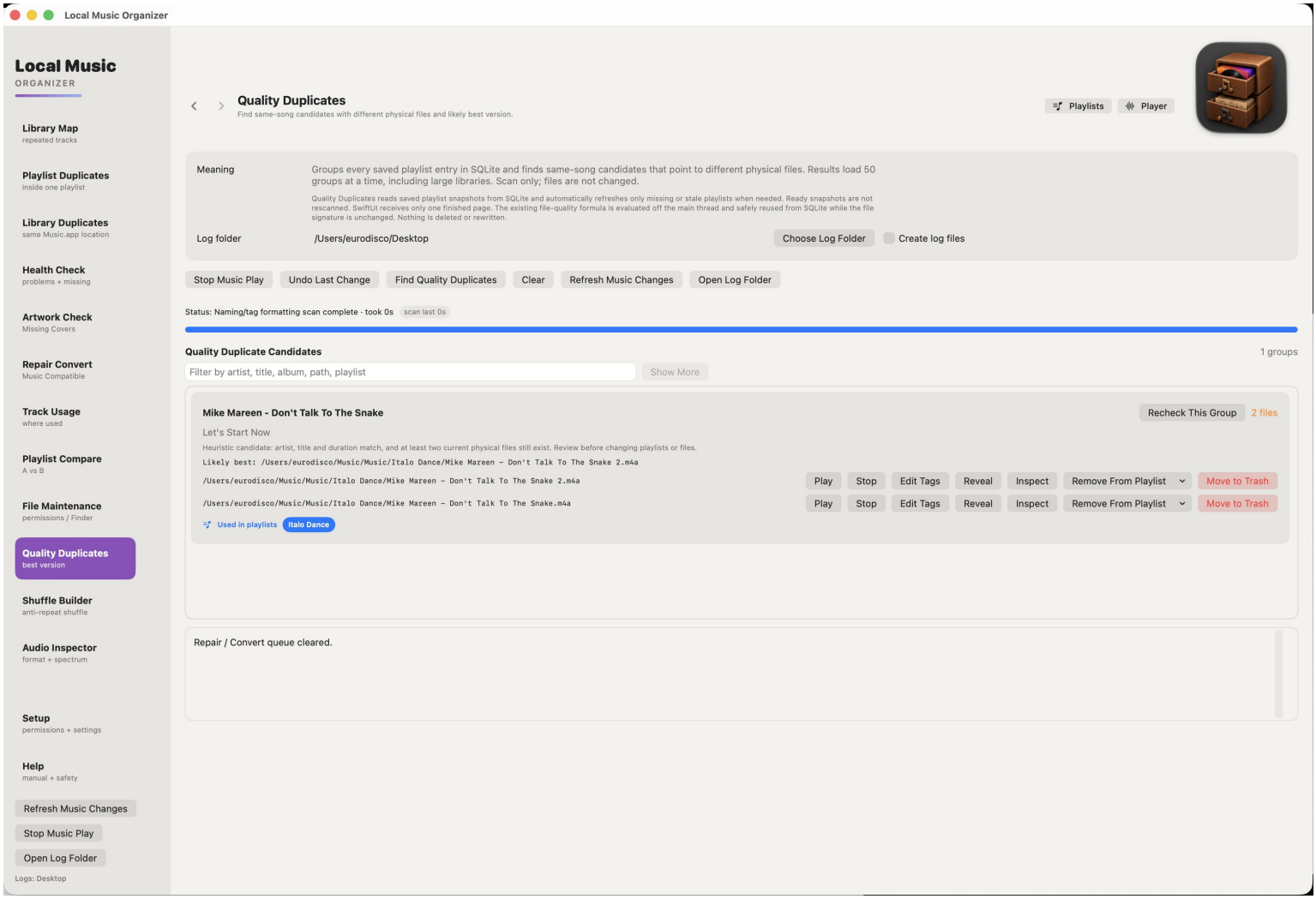
Scan versus Full Verify	The normal scan can reuse typed SQLite metadata for unchanged files and checks physical M4A ©too atoms. Full Verify Tags rereads the source files when you need independent evidence. Neither scan modifies audio.
General groups and cards	The tag-name list groups matching file fields. A group focus controls what cards are visible; checked boxes, not visibility or Show More, control Clean Selected Hidden Tags.
Suggested versus optional	Recognized downloader/source/ripper/promo signatures may be recommended for review. Nonstandard fields are not automatically bad. Legal, provenance and core music data may deserve preservation.
Batch shortcuts	Select All Recommended and other-eligible controls act on their defined classes. Clean All Recommended / Clean All Other Eligible are full-scan commands; while a group display focus is active, use selected-field cleanup instead.
Copyright metadata	Copyright refers to real MP3 TCOP or supported M4A/MP4 rights metadata. LMO can physically remove a selected field, but cannot guarantee removal of its previously stored display value from an existing Music.app record.
Encoder versus M4A ©too	A general Encoder metadata field is not the same as MPEG/LAME gapless or codec data. Verified M4A ©too candidates have their own separate section and action; MP3 stream encoder/gapless information is not cleared by that action.
Protected data and exceptions	Title, artist, album, actual artwork, media type, lyrics and gapless/playback timing are protected. Only explicitly supported narrow exceptions can be selected. Unsupported ID3 frame/MP4 atom layouts are refused.
Before and after cleanup	LMO checks the exact selected fields and original file, creates a temporary rollback copy, removes eligible metadata without audio re-encoding, and verifies retained metadata and artwork. Failed operations reject or roll back the candidate.
Cancellation and Undo	Stop requests cancellation at safe boundaries: already completed files remain cleaned. A completed embedded-tag removal is permanent and cannot be restored using Undo Last Change; the temporary rollback copy is removed after success.
Scope of other tools	Scan Finder Metadata targets supported Finder extended attributes, not embedded tags. Naming & Tags addresses Music.app text and filenames. Refresh Music Changes updates the LMO index, not stale Copyright in Music.app.

When Music.app still displays Copyright

If the file has no Copyright tag but Music.app still shows it, repeating physical cleanup will not help. Re-import can remove the stale display but may change Date Added, Persistent ID, play statistics and playlist membership. Do not bulk re-import as part of this maintenance workflow.

Review before removal. Copyright and provenance can be meaningful. Having an eligible field does not mean its removal is recommended. No arbitrary raw tag editor is provided.

13. Quality Duplicates



Quality Duplicates compares different physical versions of what appears to be the same recording across selected playlist data. It helps choose which physical copy deserves closer attention; it does not know which mastering or mix is artistically preferable.

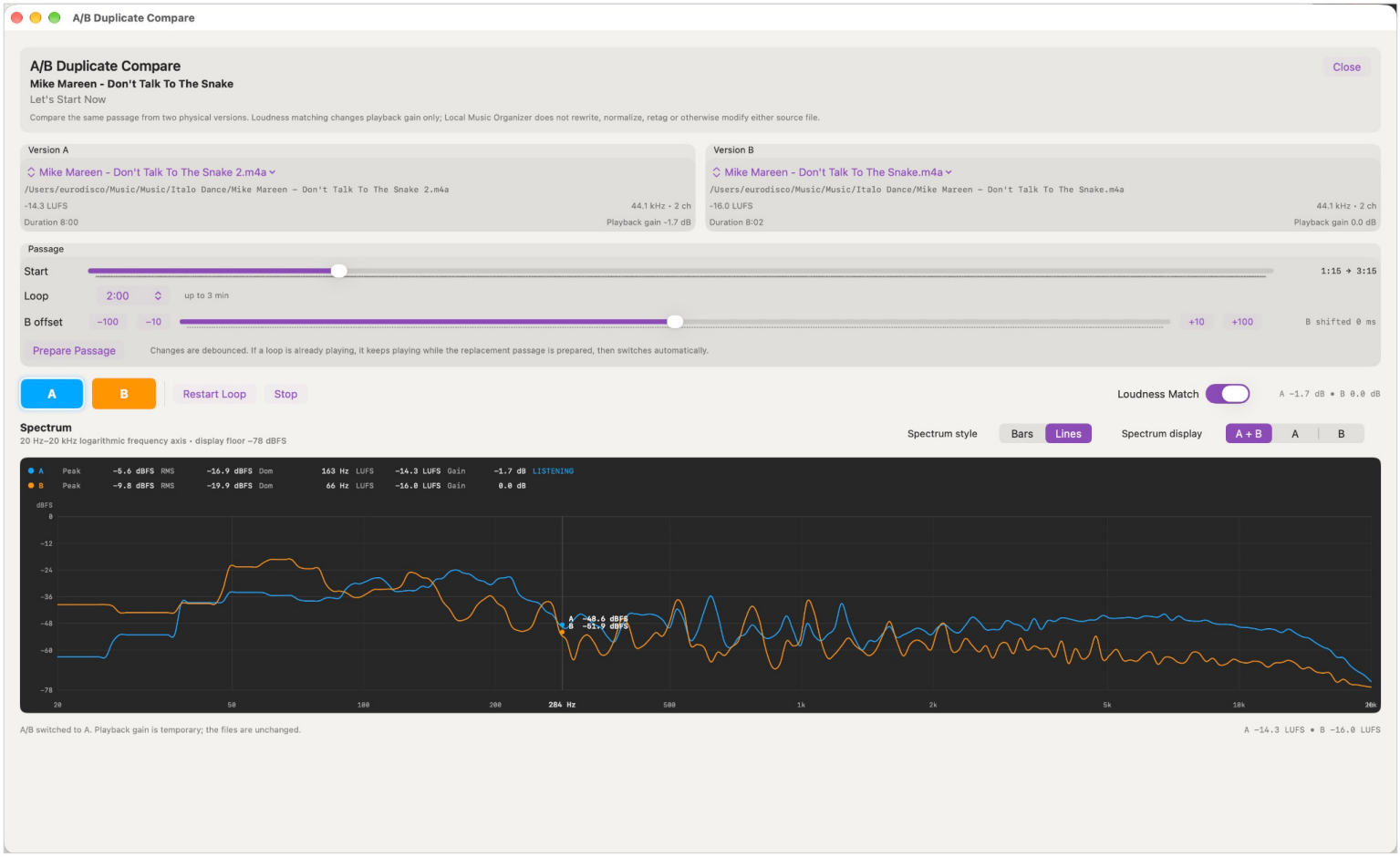
13.1 Quality Duplicates controls and actions

Find Quality Duplicates **Read-only scan**	Groups candidate same-song/different-file rows using saved playlist data and verifies current file existence/technical facts as needed. The scan is read-only.
Clear **LMO view only**	Clears Quality Duplicates results.
Filter / Show More **Read-only view**	Filters and pages the current duplicate-version groups.
Recheck This Group **Read-only targeted verification**	Re-verifies the current group against present indexed/file state.
Likely Best **Read-only ranking**	Technical ranking based on measurable file properties. It is a candidate for review, not proof of better mastering, mix, pressing, or sound quality.
Playlist chips **Navigation only**	Show which selected playlists reference each physical version.
Play / Stop **Playback only**	Play or stop the selected physical version.
Edit Tags **Changes only after save**	Opens Music.app metadata editor for a matched record.

Reveal **Navigation only**	Reveals the physical version in Finder.
Inspect **Read-only analysis**	Opens the physical file in Audio Inspector.
Remove From Playlist **Changes playlist entry**	Removes the chosen occurrence from one selected playlist. It does not delete the physical file.
Move to Trash **Changes physical file / Trash**	Moves the explicitly chosen physical file to macOS Trash. It does not automatically remove Music.app records or playlist entries, so use only after relationship review.

13.2 A/B Duplicate Compare

A/B Duplicate Compare opens from a reviewed Quality Duplicates group and places two physical versions in a separate resizable listening window. It keeps the ear comparison and the measured spectrum visible together. Preparing a passage, offsetting B, loudness matching, and switching A/B are playback-only operations: neither source file is rewritten, normalized, retagged, moved, or otherwise changed.

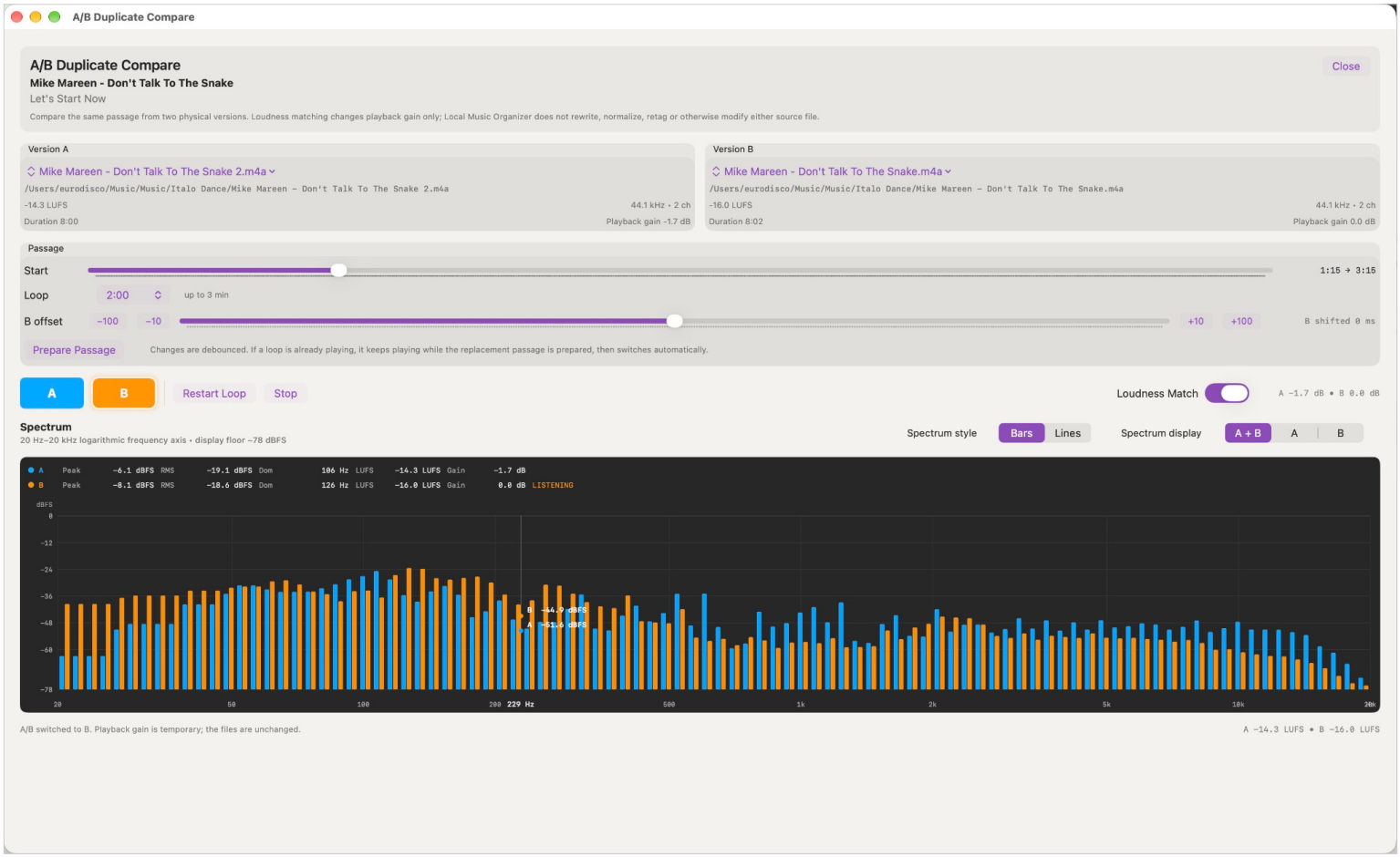


Example - Lines view. A remains blue and B remains orange; LISTENING identifies the heard side without changing color identity.

Version A / Version B **Read-only selection**	Choose the two physical files being compared. Each side shows the file path, measured loudness, sample rate/channels, duration, and any temporary playback gain. A and B must be different paths.
Start **Playback preparation only**	Moves the beginning of the shared comparison passage. The available range is constrained so the chosen passage still exists in both files after any B offset is applied.
Loop **Playback preparation only**	Chooses the passage length from 5 seconds up to 3 minutes. A length that cannot fit inside the common remaining duration is not available.
B offset / -100 / -10 / +10 / +100 **Playback alignment only**	Shifts only Version B relative to A, from -2000 to +2000 ms in 10 ms steps. Use it to compensate for a different lead-in, edit point, or small timing offset. Positive values start B later; negative values start B earlier.
Prepare Passage **Playback preparation only**	Builds the current A/B passage. Control changes are debounced. If a loop is already playing, the current loop continues while the replacement is prepared and then switches automatically.

13.2 A/B Duplicate Compare - spectrum and interpretation

The two views below use the same live measurement data. Bars make side-by-side band differences easy to scan; Lines make the overall spectral shape easier to compare. The spectrum is evidence for review, not a mastering verdict: use it together with matched listening, alignment, and the source-file facts shown above.



Example - Bars view with A + B displayed together. The fixed blue/orange pair is the same in buttons, bars, lines, legend markers, and hover values.

Interpretation

Peak/RMS and the live spectrum describe the current audio; LUFS and Gain support loudness-matched comparison. A visible difference can be meaningful, but it does not by itself prove that one mastering, mix, or pressing is artistically preferable.

A / B **Playback only**	Selects the side you hear. A uses fixed blue (#00A8FF) and B fixed orange (#FF9500); the active side is marked separately as LISTENING. The A and B keys provide the same switch.
Play Loop / Restart Loop / Stop **Playback only**	Starts the prepared passage, restarts it from the beginning while already playing, or stops the dedicated comparison playback.
Loudness Match **Playback gain only**	Temporarily adjusts playback gain to reduce simple loudness bias between versions. The A/B gain readout shows the applied dB values. Source audio and tags are unchanged.
Spectrum style - Bars / Lines **Read-only display**	Changes only the presentation of the live spectrum. Bars show paired A/B bands; Lines overlay smooth A/B curves. Both are derived from the same live spectrum measurement.
Spectrum display - A + B / A / B **Read-only display**	Chooses which spectrum is drawn. A + B keeps both visible for direct comparison; A or B isolates one side. The separate LISTENING indicator shows which side is actually being heard.
Peak / RMS / Dom / LUFS / Gain / LISTENING **Read-only measurements**	Peak and RMS are live dBFS level measures; Dom shows the dominant frequency region; LUFS is the measured loudness used for comparison; Gain is the temporary matching adjustment; LISTENING marks the heard side.
Graph / hover readout **Read-only measurement**	Uses a logarithmic 20 Hz-20 kHz frequency axis with a -78 dBFS display floor. Hovering shows the selected frequency and the A/B dBFS values at that position.
Status row / Close **Status / navigation**	The status line reports preparation and A/B switching state and repeats the A/B loudness values. Close ends the comparison window and its playback engine; it does not change either source file.

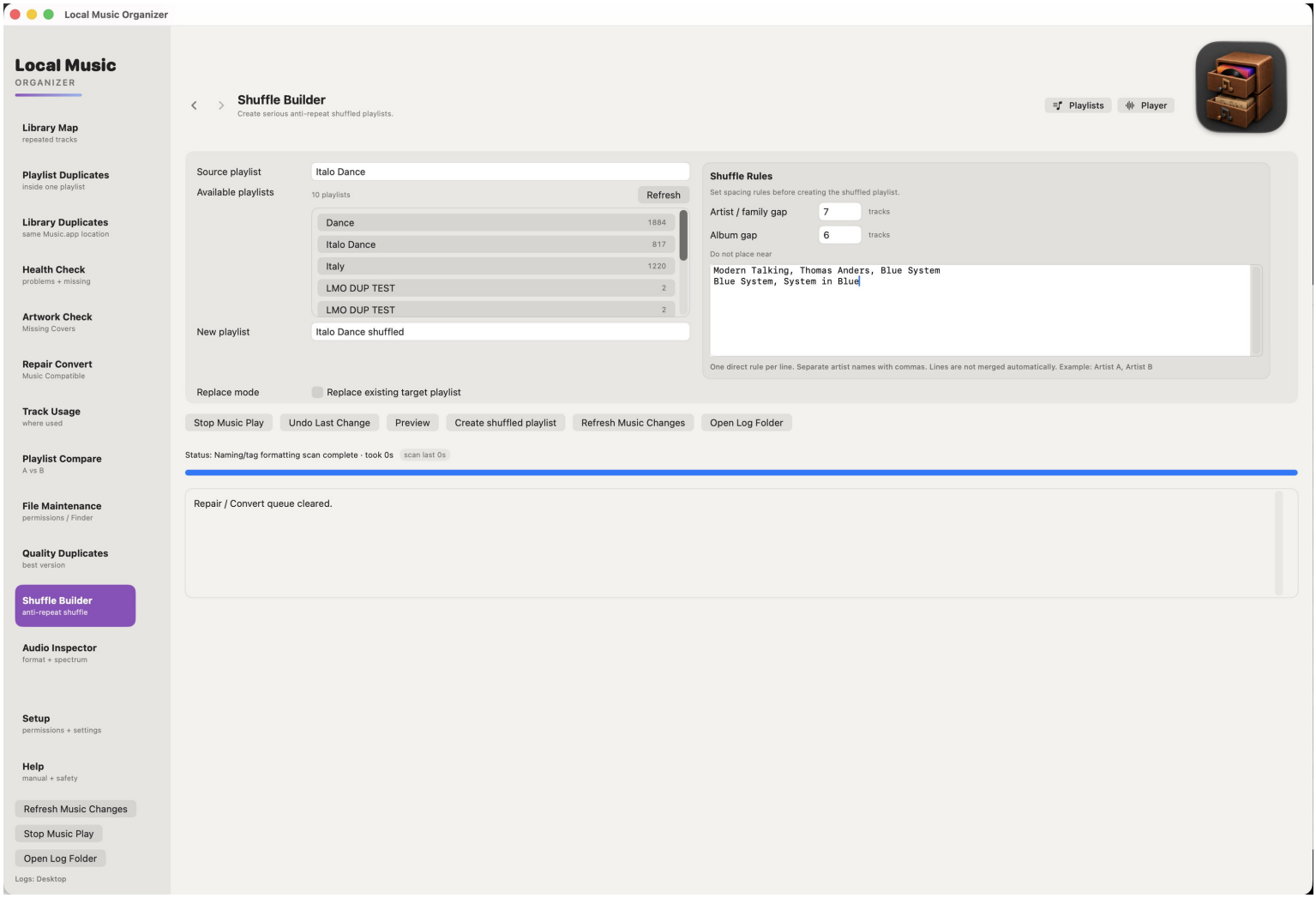
13.3 Quality Duplicates: comparison entry and scope

A candidate quality ranking is a reason to inspect files, not an instruction to remove one. The Quality Duplicates group has distinct controls for verification, listening comparison, playlist entry and physical-file actions.

Recheck This Group	Updates the current group before acting on any version. The result may change when the saved index or the physical files have changed.
A/B Compare...	This is the exact button in a reviewed Quality Duplicates group. It opens the listening/spectrum comparison when at least two physical file paths are present; it is disabled while an operation runs or when the group contains fewer than two locations.
A/B comparison only	A/B passage preparation, spectrum Bars/Lines and loudness matching do not write either audio file. Listening to a preferred version is not proof that another version can be safely deleted.
Remove From Playlist	Removes only the specifically selected playlist occurrence. It is different from deleting a Music.app library record or moving a physical file to Trash.
Move to Trash	Targets the explicitly chosen physical file, after confirmation and applicable safety checks. Examine playlist usage and the exact path before choosing it; it is not part of the A/B comparison itself.
Likely Best	Technical file-property ranking is only a review aid. Different mixes, masters and intentional low-fidelity sources may sound preferable despite a lower technical score.

Use A/B Compare... before making a decision about similar-looking files. Keep playlist-entry, library-record and physical-file operations separate.

14. Shuffle Builder



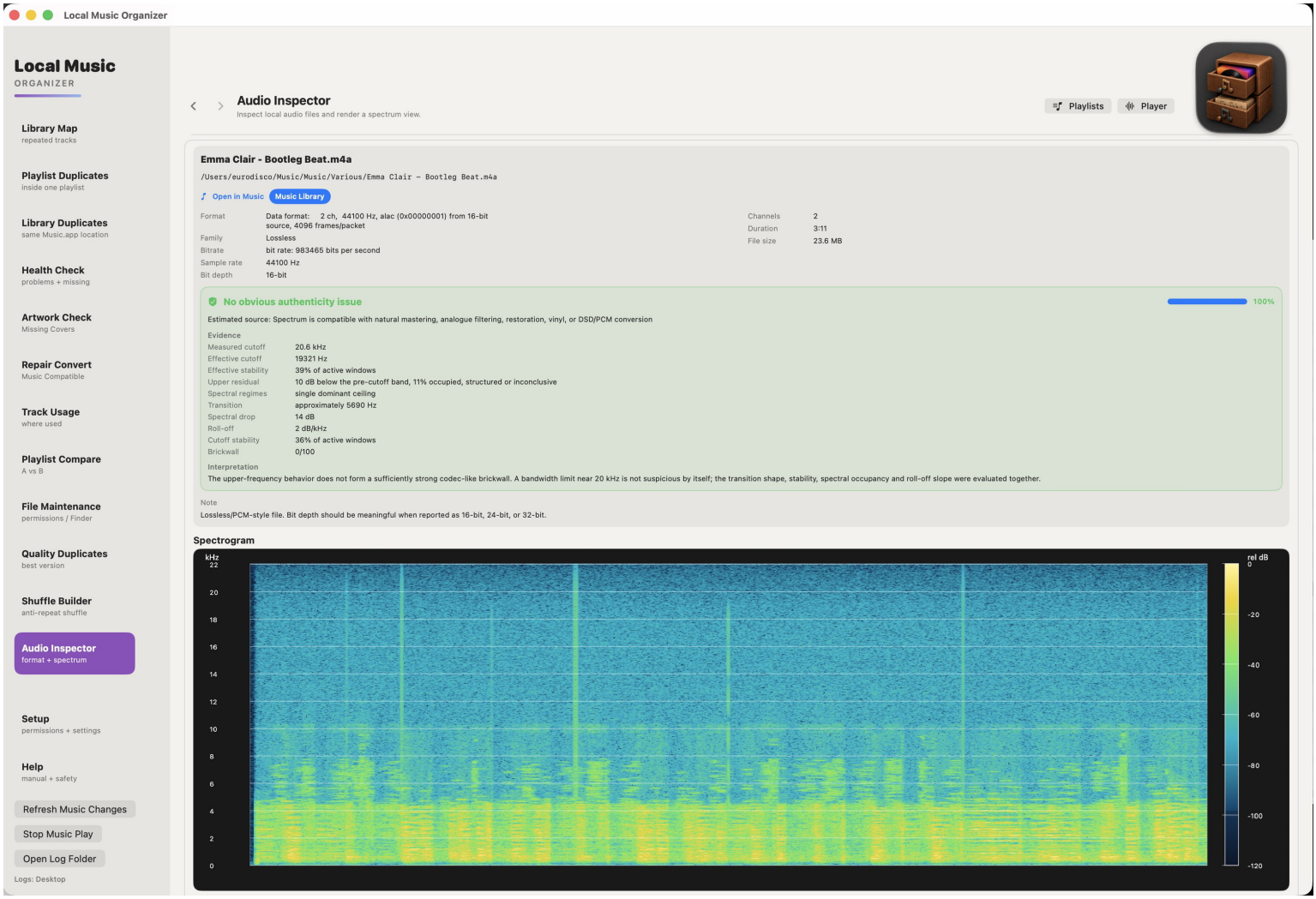
Shuffle Builder creates a new ordering from a source playlist while trying to maintain spacing between the same artist/family, album, and user-defined “do not place near” groups. Preview is non-destructive; Create is the action that writes the target playlist.

14.1 Shuffle rules and actions

Source playlist **Read-only selection**	Chooses the Music.app playlist whose tracks will be rearranged.
Available playlists **Read-only selection**	List of visible playlists available as the source.
New playlist **LMO input**	Name of the target playlist that will receive the shuffled order.
Artist / family gap **Shuffle rule**	Minimum preferred separation between the same artist/family key. Default is 7. The builder scores remaining violations if a perfect arrangement is impossible.
Album gap **Shuffle rule**	Minimum preferred separation between tracks from the same album/cover key. Default is 6.
Do not place near **Shuffle rule**	One direct group per line. Names can be separated by comma, semicolon, pipe, or equals. A line A,B,C makes those names a direct group. Separate lines A,B and B,C do not automatically imply A,C.
Replace existing target playlist **Playlist-write setting**	Allows the named target playlist to be replaced/rewritten by the create action. Leave it off when you do not intend to replace an existing target.
Preview **Read-only preview**	Builds and scores candidate shuffled orders without creating or changing the target playlist.

Create shuffled playlist **Changes playlist**	Creates/writes the target Music.app playlist using the best arrangement found under the current rules.
Stop **Stops playlist creation**	When creation is in progress, stops further additions. Entries already added to the target remain.
Shuffle quality / remaining issues **Read-only report**	The result/log reports artist, title, album/cover spacing violations and a quality score so unavoidable compromises are visible.

15. Audio Inspector



What Audio Inspector is for

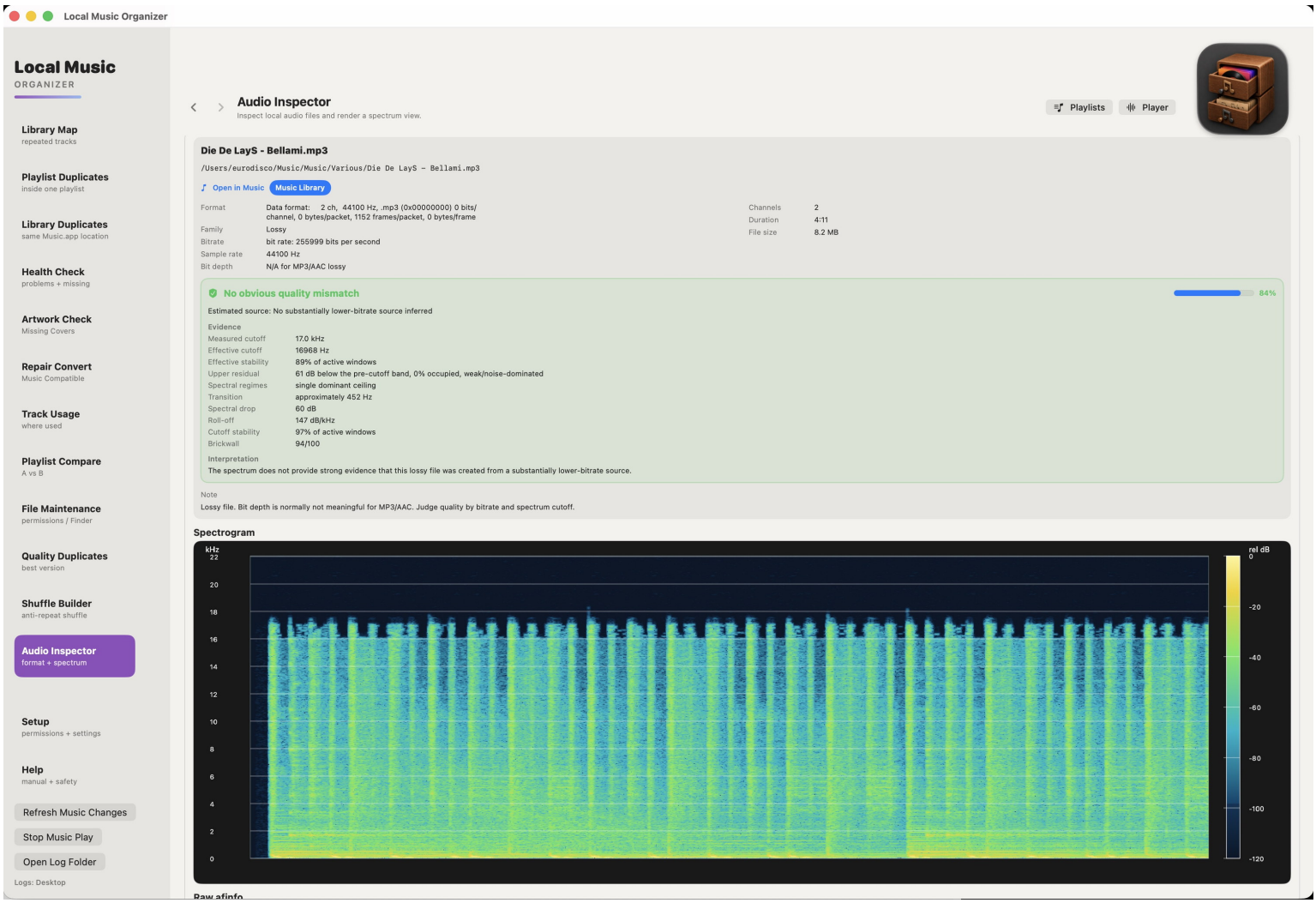
Audio Inspector examines one readable local audio file in detail. It shows format and technical properties and can render a spectrogram so you can inspect frequency content visually. It is useful when a file looks suspicious in Dashboard, Quality Duplicates, Health Check, File Search, or another result list.

Typical workflow

Open or Inspect a local file, review its technical facts, then use the spectrogram and the authenticity summary as evidence. If the file is matched to Music.app, the inspector can also provide the relevant library/navigation context. The full Library Index is not required just to inspect a directly chosen local file.

Safety and interpretation

Inspection is read-only. A spectrogram or authenticity estimate can flag something worth reviewing, but it cannot prove a particular mastering, source, or encoding history by itself. The detailed Single File controls continue on the following pages.



Audio Inspector examines one local audio file without requiring a Library Index. It reports container/codec information, duration, channels, technical characteristics, authenticity evidence, and a spectrogram. It does not “repair” the file merely by inspecting it.

15.1 Single File mode

Mode - Single File **Read-only selection**	Chooses the one-file inspection workspace.
Choose Audio File **Read-only file selection**	Opens the macOS file picker and loads one local audio file for inspection.
Drop area **Read-only file selection**	Accepts a local audio file by drag and drop.
Clear Inspector **LMO view only**	Clears the loaded inspection result.
Format / family / bitrate / sample rate / bit depth / channels / duration / file size **Read-only analysis**	Technical facts reported from the selected local file and available system/bundled analyzers.
Authenticity assessment **Read-only analysis**	Summarizes evidence that may suggest retranscoding/upscaling/bandwidth limitations. It is evidence, not proof of source history.
Spectrogram **Read-only analysis**	Visual frequency-over-time representation of the selected audio. The picture is diagnostic evidence; mastering filters and old recordings can resemble codec cutoffs.
Raw technical output **Read-only display**	Shows the underlying technical information returned by the local analyzer for advanced review.

Music.app match/reveal controls **Navigation only**	When the local file is matched to a Music.app record, lets you reveal/open the relevant library context without creating another record.

What to look at first

Start with format/family, bitrate, sample rate, bit depth, channels, duration, and file size. Then compare the **Authenticity assessment** with the spectrogram. These are complementary pieces of evidence, not independent repair commands.

How to interpret authenticity evidence

A bandwidth limit, roll-off, or unusual spectrum can be worth reviewing, but old recordings, mastering filters, restoration, analogue sources, and legitimate production choices can produce similar shapes. Treat the result as a technical lead, not proof of a file's history.

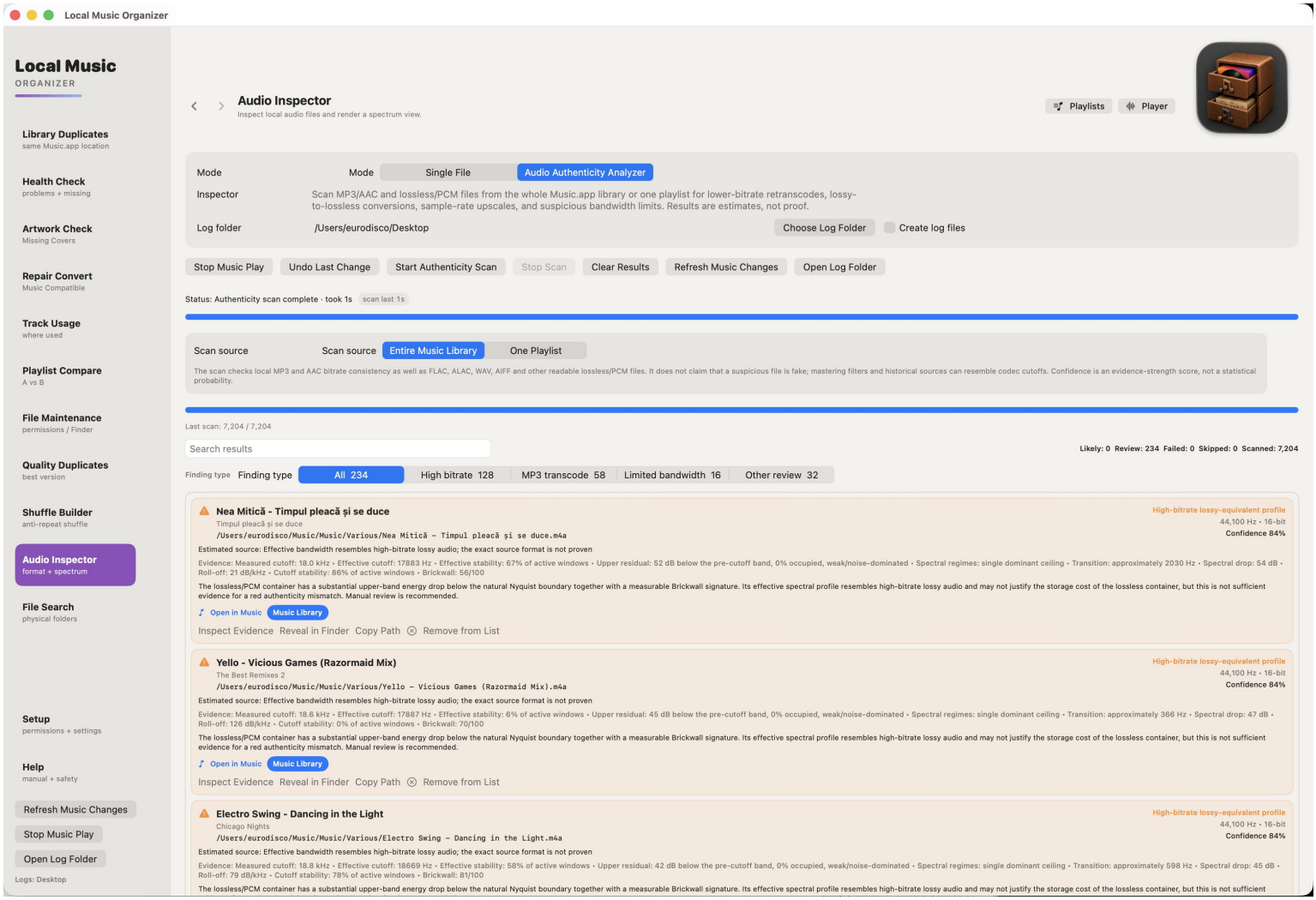
Music.app match controls

When the inspected local file matches a Music.app record, the match/reveal controls let you open the relevant library context. They do not create another Music.app record. If there is no match, Audio Inspector still works as a local-file analysis tool.

Spectrogram excerpt position (version 2.9)

For files longer than 1:02 (62 seconds), the Single File spectrogram reads its analysis excerpt starting at 1:00 rather than at the introduction. For files of 1:02 or less, it starts at 0:00. The separate authenticity assessment still samples across the full recording; neither operation modifies the file.

16. Audio Authenticity Analyzer



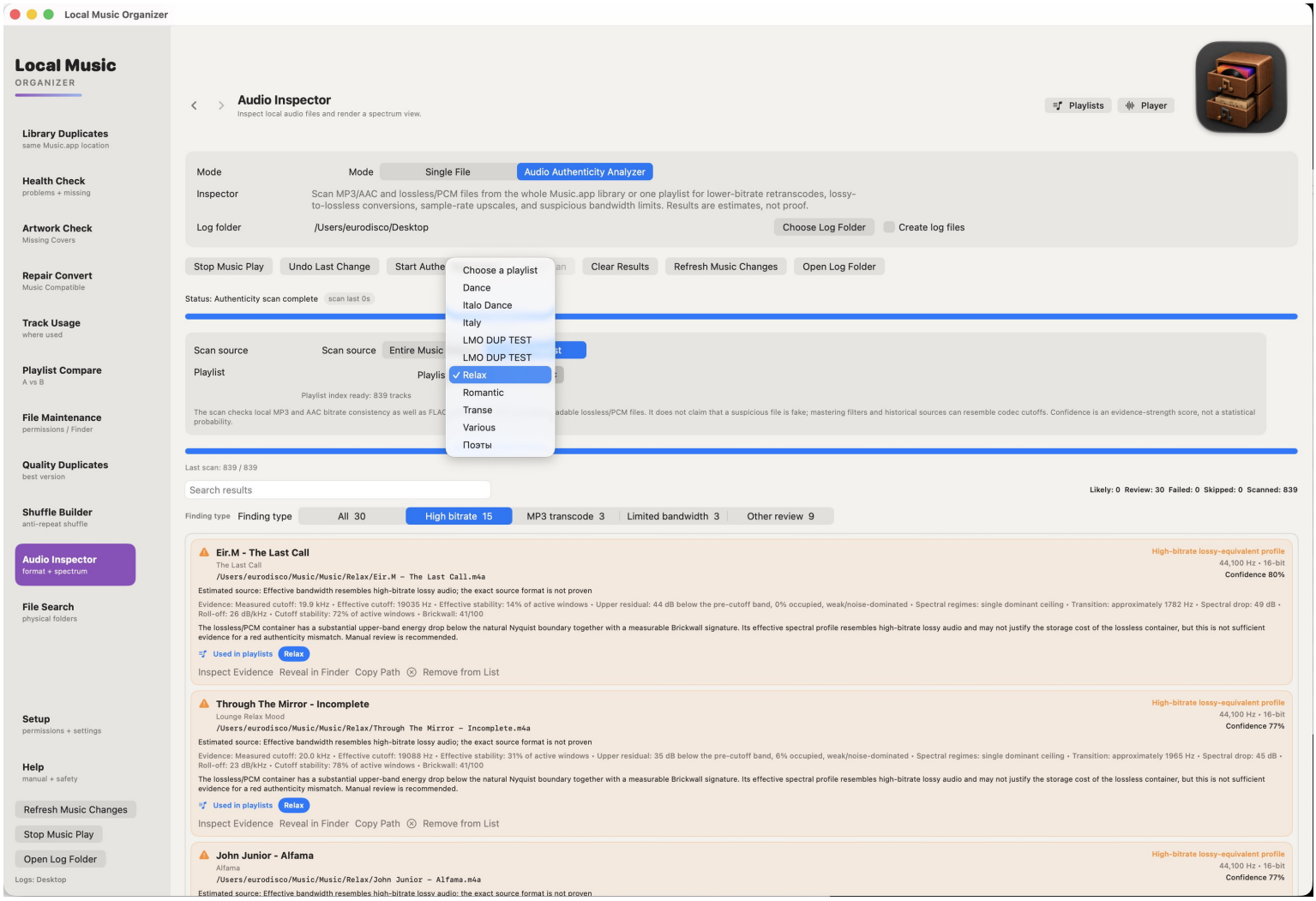
Audio Authenticity Analyzer is the batch mode inside Audio Inspector. It searches an indexed library or one indexed playlist for evidence of lower-bitrate retranscodes, lossy-to-lossless conversions, sample-rate upscales, and suspicious bandwidth limits. Results are estimates, not proof.

16.1 Batch authenticity controls

Mode - Audio Authenticity Analyzer **Read-only selection**	Switches Audio Inspector from one-file inspection to the indexed batch analyzer.
Scan source - Entire Music Library **Read-only scan source**	Uses available saved Library Index rows as the batch source.
Scan source - One Playlist **Read-only scan source**	Uses one selected indexed playlist as the source. The playlist must have a usable saved snapshot.
Playlist **Read-only selection**	Chooses the playlist when One Playlist is selected.
Start Authenticity Scan **Read-only analysis**	Runs the local/indexed authenticity analysis for the selected source.
Stop Scan **Stops scan only**	Requests cancellation of the batch analysis.
Clear Results **LMO view only**	Clears authenticity findings from LMO.

Search results **Read-only filter**	Filters the loaded findings by visible metadata/path.
Likely / Review / Failed / Skipped / Scanned counters **Read-only display**	Summarize current analysis outcomes. These are classifications of evidence/processing state, not delete/repair commands.
Finding type - All / High bitrate / MP3 transcode / Limited bandwidth / Other review **Read-only filter**	Filters findings by the evidence category.

16.2 Authenticity result cards



Category / severity / confidence **Read-only display**	Describe the type and strength of the evidence. Confidence is evidence strength, not a mathematical probability that the source history is known.
Estimated source **Read-only estimate**	LMO's best technical interpretation of the observed spectral/codec evidence.
Evidence **Read-only evidence**	Lists measurable observations that triggered the classification.
Interpretation **Read-only guidance**	Explains how the evidence should be understood and its limitations.
Inspect Evidence **Read-only analysis**	Opens the exact file in the detailed one-file Audio Inspector so the spectrogram/metrics can be reviewed.
Reveal in Finder **Navigation only**	Reveals the local file.

Copy Path **Clipboard only**	Copies the file path.
Remove from List **LMO view only**	Removes the finding from the current LMO result list only. It does not delete or modify the file.
Skipped row - Copy Path / Open Parent Folder **Navigation only**	Lets you locate an item that could not be analyzed and inspect why it was skipped.

Authenticity is evidence, not provenance proof Codec cutoffs, vintage sources, mastering filters, noise reduction, resampling, and production choices can produce similar spectral signatures. Use these findings to decide what to inspect, not as an automatic reason to replace or delete a file.

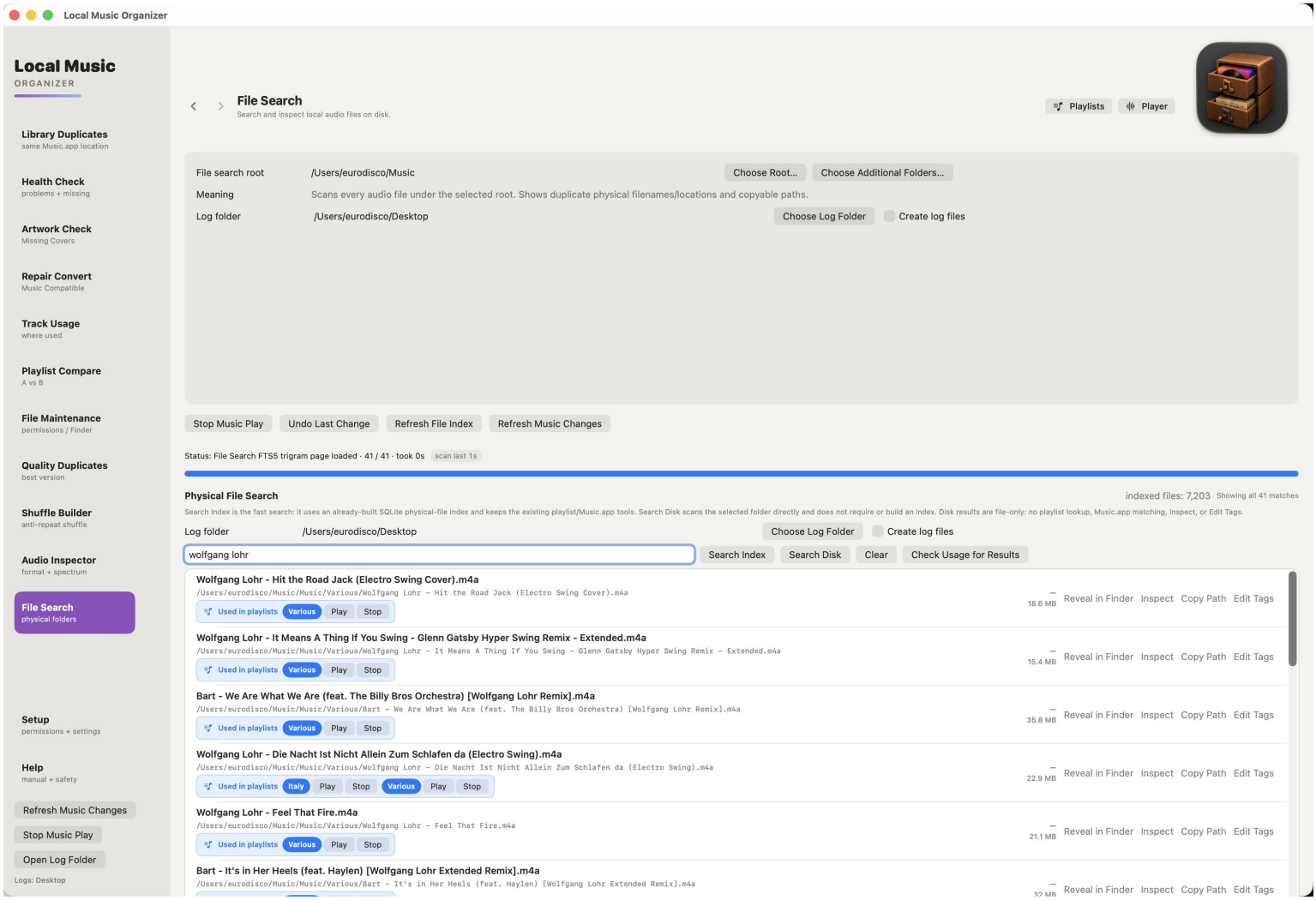
16.3 Audio Authenticity: skipped paths and record removal

Skipped files and authenticity findings are different. The scanner may show a stale Music.app record when its saved physical path is missing, but an unavailable external volume must not be treated as a deleted file.

Skipped before audio analysis	Shows how many records could not be analyzed. A skipped record is not an authenticity verdict and is not proof of a bad or fake source.
External storage offline	Shows a grouped offline-volume warning. Records on currently offline external storage are excluded from the stale-record deletion action.
Remove stale Music.app record...	A modifying action in the skipped-source area only when the storage is online and the saved local path is still missing. It requires the full edition and explicit critical confirmation.
What exactly is removed	The action targets one verified Music.app library record by Persistent ID. It does not delete the physical audio file, move it to Trash or directly remove another record.
Live safety checks	Immediately before deletion, LMO re-reads the exact Persistent ID. It blocks the action if Music.app now reports a different accessible path, the file reappeared, or the storage is offline.
Saved Library Index	After a verified removal, LMO retires the corresponding saved index entry and requests Refresh Music Changes. Failure of that follow-up refresh is reported separately.
Skipped row - Copy Path / Open Parent Folder	Navigation actions for investigating a skipped source; they do not modify the Music.app record.
Remove from List	Dismisses an analyzed result card from the current LMO results only. Do not confuse this harmless view action with Remove stale Music.app record... in the skipped-source area.
More than twelve stale candidates	Only a limited number of stale-path removal buttons are shown here. Health Check is the appropriate place for a complete missing-record review.

An offline disk, an incorrectly mapped path and an intentionally deleted file require different decisions. Reconnect or repair the storage path before considering record deletion.

17. File Search



File Search has two deliberately different search paths. Search Index is fast and uses an explicitly built physical-file SQLite index. Search Disk walks the selected folder directly. LMO does not silently build a physical index merely because you typed a search.

17.1 File Search controls

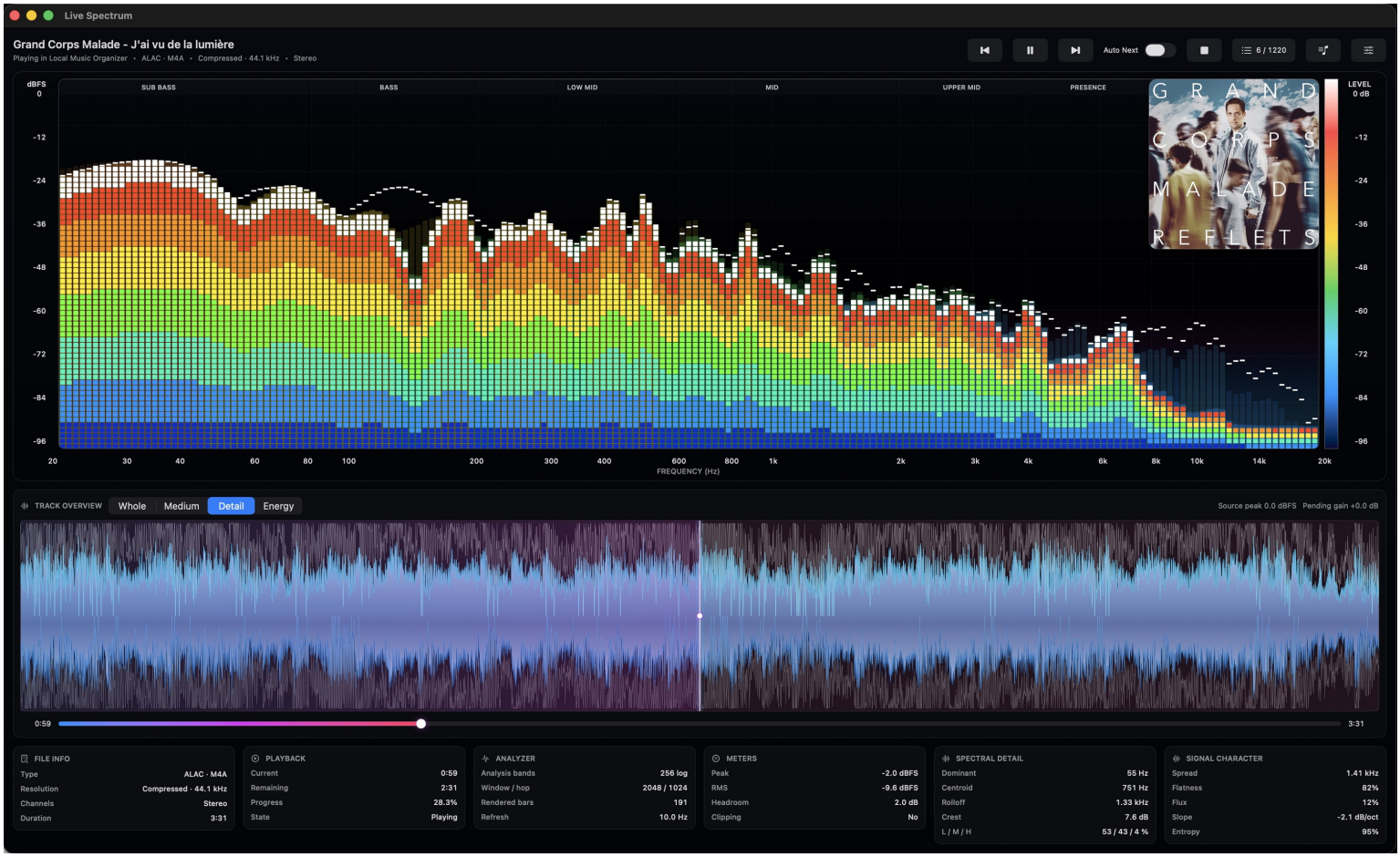
File search root **Read-only display**	Shows the current primary physical search folder.
Choose Root... **LMO setting**	Chooses the primary physical search root.
Choose Additional Folders... **LMO setting**	Chooses the additional File Search folder root. Network/NAS behavior is not validated by this guide.
Refresh File Index **LMO physical index**	Explicitly builds or refreshes the physical-file index from the selected folders. It does not search playlists and does not change audio files. On very large libraries this is never started silently.
Refresh Music Changes **LMO index**	Incrementally reconciles LMO's Music.app index; it is separate from the physical-file index.
Search field **Read-only query**	Search text for filename/path and indexed metadata where available.
Search Index **Read-only indexed search**	Fast search in the already-built physical SQLite index. If no physical index exists, LMO reports that fact instead of building one automatically.
Search Disk **Read-only filesystem scan**	Directly walks the selected folder for matches. It does not require or build the physical index. Direct disk results are intentionally file-only.

Clear **LMO view only**	Clears the current File Search results/query.
Check Usage for Results **Read-only targeted usage scan**	For indexed results, checks playlist membership for the current matches instead of rescanning unrelated files.
Show More **Read-only view**	Loads more indexed/direct result rows.

17.2 Indexed and direct-disk result actions

Indexed playlist chips **Navigation only**	Show playlists associated with an indexed physical result. A chip reveals/opens that playlist context.
Play from playlist / Play **Playback only**	Plays the exact Music.app record represented by the playlist match.
Stop Music Play **Playback only**	Stops Music.app playback.
Reveal in Finder **Navigation only**	Reveals the physical file.
Inspect **Read-only analysis**	Opens the physical file in Audio Inspector. Available on indexed/matched results, not intentionally on the minimal direct-disk path.
Edit Tags **Changes only after save**	Opens the exact Music.app record in metadata editor when a match is known.
Copy Path **Clipboard only**	Copies the physical path.
Direct Search Disk row **Read-only / navigation**	Intentionally offers only file-oriented actions such as Reveal in Finder and Copy Path. It does not imply a Music.app match, playlist usage, or tag editor.

18. Live Spectrum Player



Fast shortcut: Option + Play opens LMO playback
Hold the Option key while clicking a Play control for a track that has a readable local file. LMO resolves the local path and sends that item to its own Live Spectrum Player instead of using the normal Play behavior. When possible, the player also builds its Previous/Next queue from the current playlist or the current result set. A normal click keeps the button's standard behavior. If no readable local file can be resolved - for example a cloud-only or missing item - the Live Spectrum Player cannot play it.

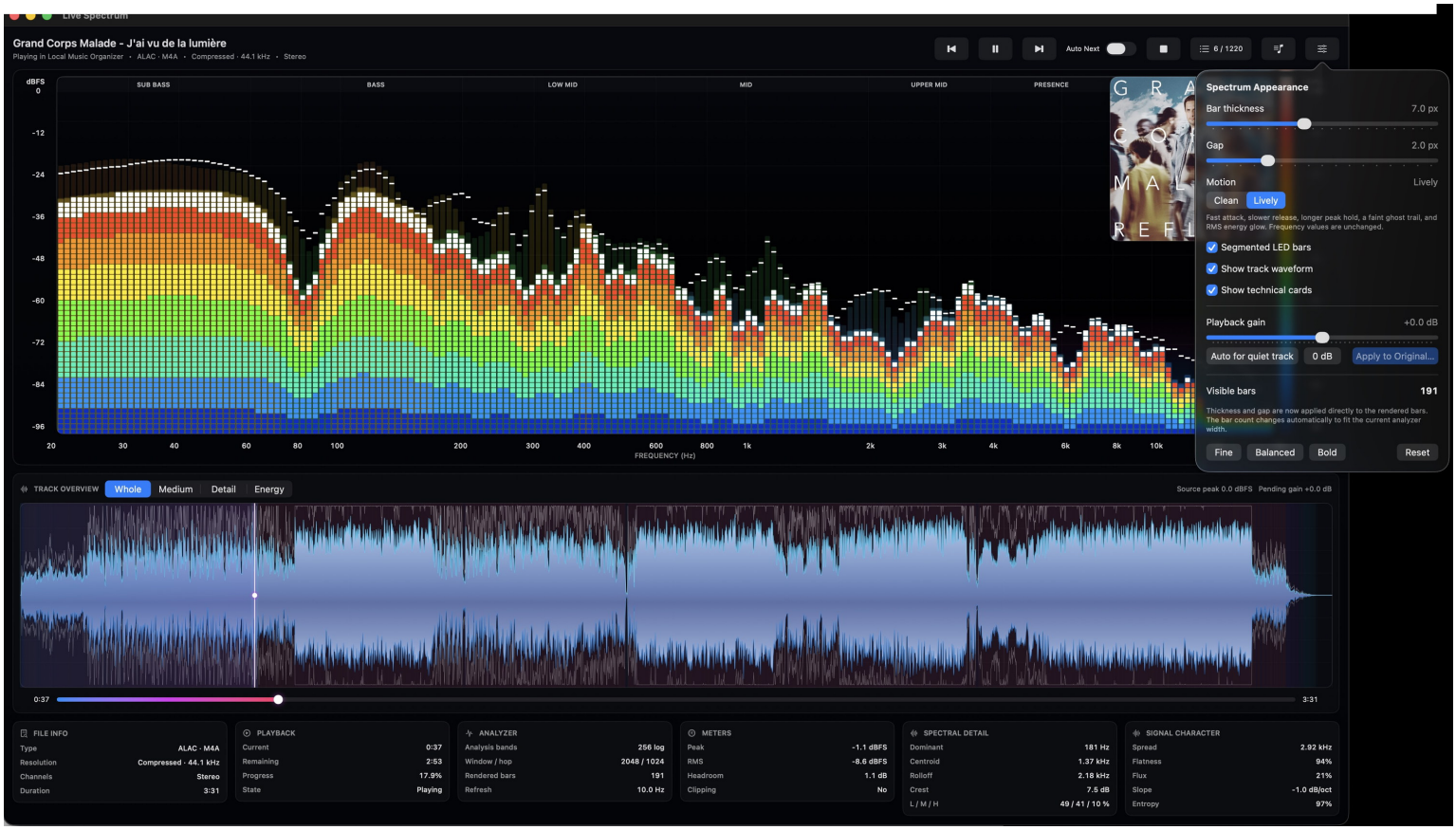
The Player button opens/restores LMO's local Live Spectrum window. It plays local audio through LMO, shows real-time spectrum/waveform information, and can maintain a playlist-derived queue. Playback controls and visual settings do not modify the source file unless the separate Apply to Original File gain action is explicitly confirmed.

18.1 Original: transport and queue controls

Previous track **Playback only**	Moves to the previous queue item when a queue exists.
Play / Pause / Resume **Playback only**	Starts, pauses, or resumes the current local track.
Next track **Playback only**	Moves to the next queue item.
Auto Next **Player setting**	Controls whether the player automatically advances when the current track finishes.
Stop **Playback only**	Stops local player playback.
Current Track List **Read-only navigation**	Opens the queue popover, showing current position and the tracks in the active queue.

Switch Playlist Read-only queue selection	Inside the queue popover, selects another Music.app playlist as the queue source. Loading a queue does not rewrite that playlist.
Play next automatically Player setting	Queue-popover equivalent of Auto Next.
Queue row Playback/navigation	Selecting a row plays that local queue item. Hover/context actions can reveal the file or copy its path.
Music.app Playlists Read-only navigation	Opens the playlist hierarchy popover used to load a playlist-derived local queue.
Refresh Playlist Hierarchy Read-only Music.app query	Rereads the visible Music.app playlist hierarchy for the navigator.

18.2 Original: Spectrum Appearance controls



Bar thickness Player visual setting	Changes the rendered width of spectrum bars.
Gap Player visual setting	Changes spacing between rendered spectrum bars.
Motion - Clean Player visual setting	Uses strict analyzer motion without the decorative trail/energy glow. Frequency values are unchanged.
Motion - Lively Player visual setting	Uses faster attack/slower release, peak hold, ghost trail, and RMS glow while leaving frequency values unchanged.
Segmented LED bars Player visual setting	Turns the segmented-bar visual treatment on/off.
Show track waveform Player visual setting	Shows/hides the track overview waveform below the spectrum.
Show technical cards Player visual setting	Shows/hides the technical analysis cards.

Playback gain Playback only	Temporary player gain from -12 dB to +12 dB. Moving this control changes playback level in LMO; by itself it does not rewrite the file.
Auto for quiet track Playback only	Sets a suitable Playback gain automatically from the currently measured source peak. It still does not write the file.
0 dB Playback only	Returns Playback gain to unity.
Apply to Original File Changes original physical file	Explicitly applies the current gain to the original local audio file after confirmation and verification. This is a real physical-file operation; lossy formats may require re-encoding.
Visible bars Read-only display	Shows the current estimated number of spectrum bars that fit at the chosen thickness/gap.
Fine / Balanced / Bold Player visual setting	Convenience presets for bar thickness and gap.
Reset Player visual setting	Restores the default spectrum appearance preset values.

18.3 Original: waveform and technical cards

TRACK OVERVIEW Read-only display / seek	Waveform overview for seeking and understanding the track's amplitude envelope.
Whole / Medium / Detail / Energy Player visual setting	Changes the waveform/detail presentation mode. It does not alter audio.
Click/drag waveform Playback only	Seeks playback to the chosen position in the local file.
Technical cards Read-only analysis	Report source/engine/analyzer, level meters, spectral details, and signal character when enabled.

Original header shortcuts

Volume slider Playback only	The speaker icon and horizontal slider set LMO playback volume. This is separate from Playback gain in Spectrum Appearance.
Music.app Playlists Read-only navigation	The music-note-list button opens the Music.app playlist/folder hierarchy without stopping playback.
Elektronika Mini Player Navigation / playback companion	The overlapping-rectangles button shows or hides the separate Elektronika Mini Player.
Spectrum appearance Player visual setting	The sliders button opens Spectrum Appearance, including Player Skin, Bar thickness, Gap, Motion, Segmented LED bars, waveform, technical cards, Playback gain, visible-bar presets, and Reset.

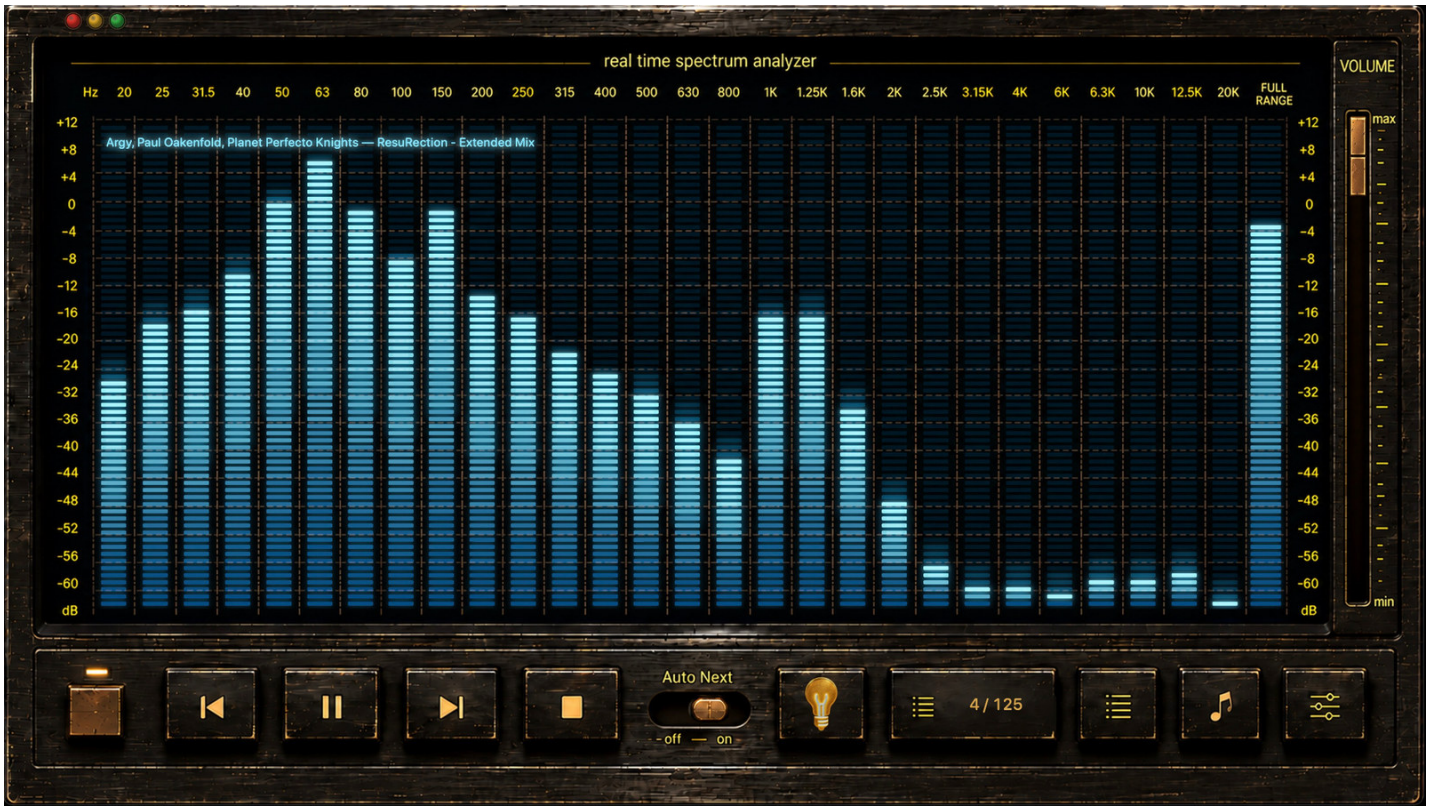
Positive Gain: full-track clipping preflight (version 2.9)

Apply to Original File first scans the complete source when Gain is positive. The confirmation shows the measured sample peak and projected peak after Gain. If the result is at or near 0 dBFS, choose Cancel - Reduce Gain or explicitly Apply Anyway. If the scan fails, no replacement is offered. Negative Gain keeps the usual confirmation; a sample-peak check is not a true-peak guarantee.

Selecting track text and opening Audio Inspector (version 2.9)

Track titles can be selected and copied in ORIGINAL, CLASSIC, GLASS and compact player layouts. Right-click supported titles or queue entries for Copy actions and Open in Audio Inspector. Inspector access requires an available local file. These shortcuts do not alter audio or playback.

18.4 Classic: controls and skin switching

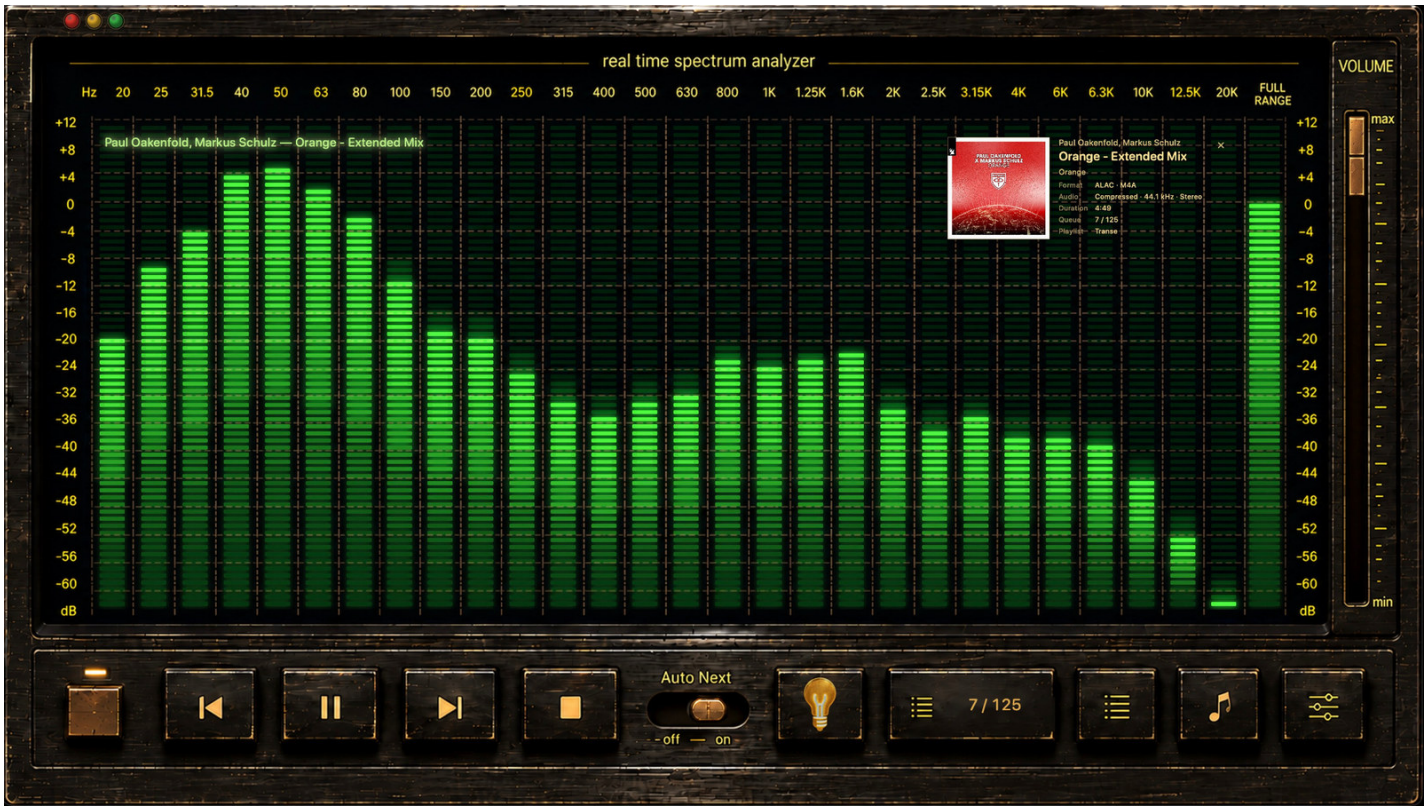


Switch between Original, Classic and Glass
Original -> Classic or Glass: sliders button (Spectrum appearance) -> Player Skin -> desired skin.
Classic -> another skin: rightmost sliders hardware button -> Player Skin -> desired skin.
Glass -> another skin: PLAYER · GLASS at the top right -> desired skin.

GLASS: pages 76-78. Original Mini: 79; GLASS Mini: 80; CLASSIC small-window: 81.

Red / Yellow / Green LEDs **Window controls**	Close the Live Spectrum window, minimize it, or enter/exit native macOS fullscreen.
Power **Playback / window**	Starts the Classic shutdown cue. After about one second LMO stops local playback, hides the player, and returns focus to the Organizer.
Previous **Playback only**	Moves to the previous item in the active playback queue when one exists.
Play / Pause **Playback only**	Pauses the current track or resumes the loaded track.
Next **Playback only**	Moves to the next item in the active playback queue when one exists.
Stop **Playback only**	Stops local playback while leaving the Classic player window open.
Auto Next **Player setting**	Controls the same automatic queue advance used by Original: when enabled, the next queue item is started after the current track ends.
Volume **Playback only**	The vertical hardware fader on the right controls LMO playback volume.

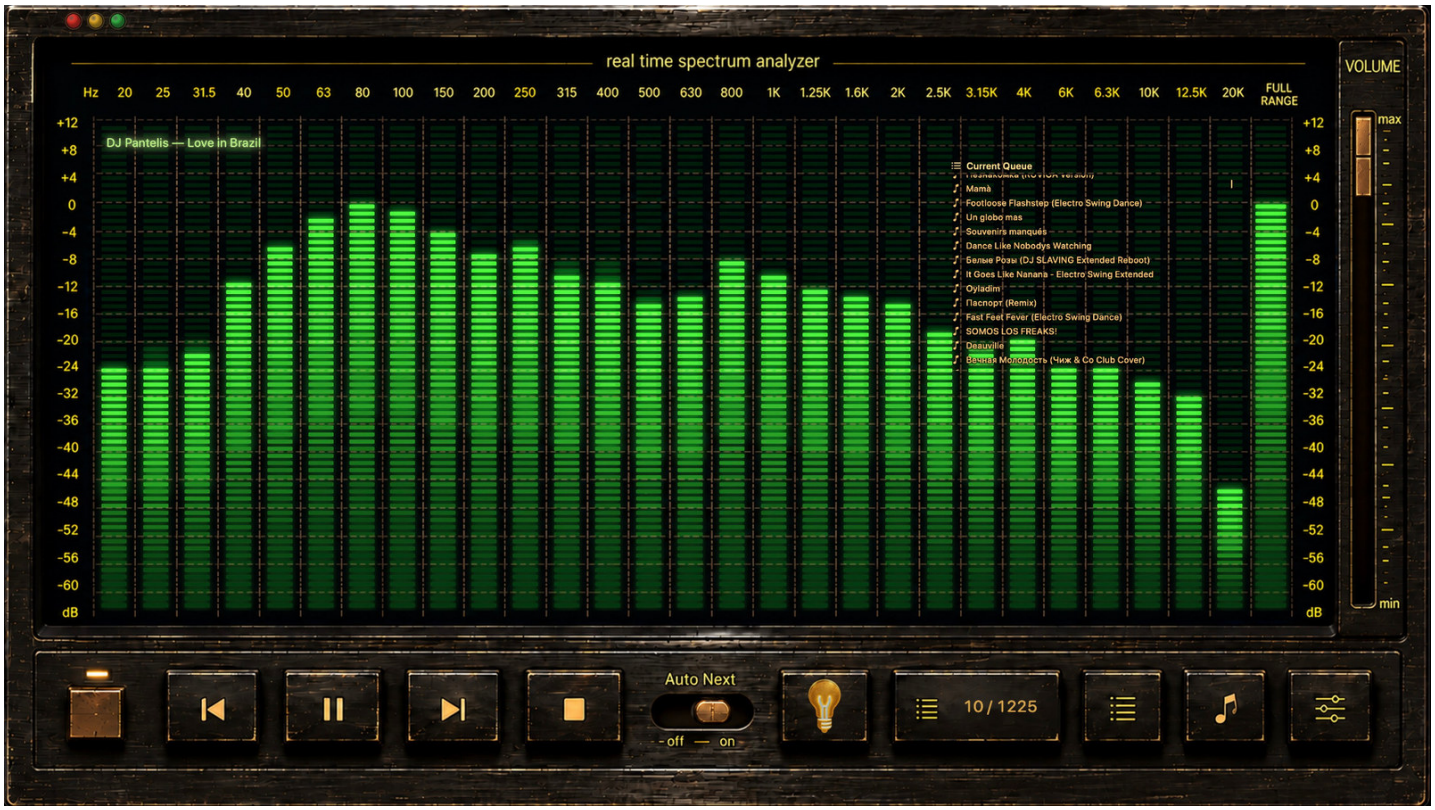
18.5 Classic: displays and information



Lamp / phosphor button **Player visual setting**	Toggles the Classic spectrum between its cyan/blue and green phosphor presentation. The current track title follows the selected phosphor mode; audio is unchanged.
Queue button **Navigation / playback**	Opens Current Queue. The large button displays the current queue position such as 7 / 125 when available, otherwise QUEUE.
Playlists button **Read-only navigation**	Opens the Music.app playlist/folder hierarchy used by the Classic player.
Track Card (music note) **Read-only display**	Opens the current-track card with artwork, artist, title, album, format, audio resolution/channels, duration, queue position, and playlist when those values are available.
Track Card X **Read-only display**	Closes only the track-information card.
Settings (sliders) **Player setting**	Opens the Player Skin selector used to switch between Original, Classic, and Glass.
Small-window overlays **Window behavior**	When Queue or Playlists is open and the Classic window becomes small, that panel moves to a separate companion panel below the player. If the window grows again and the panel is still open, it returns inline.

Classic keeps one utility overlay active at a time: opening Queue, Playlists, Track Card, or Settings closes the other utility overlays.

18.6 Classic: Current Queue

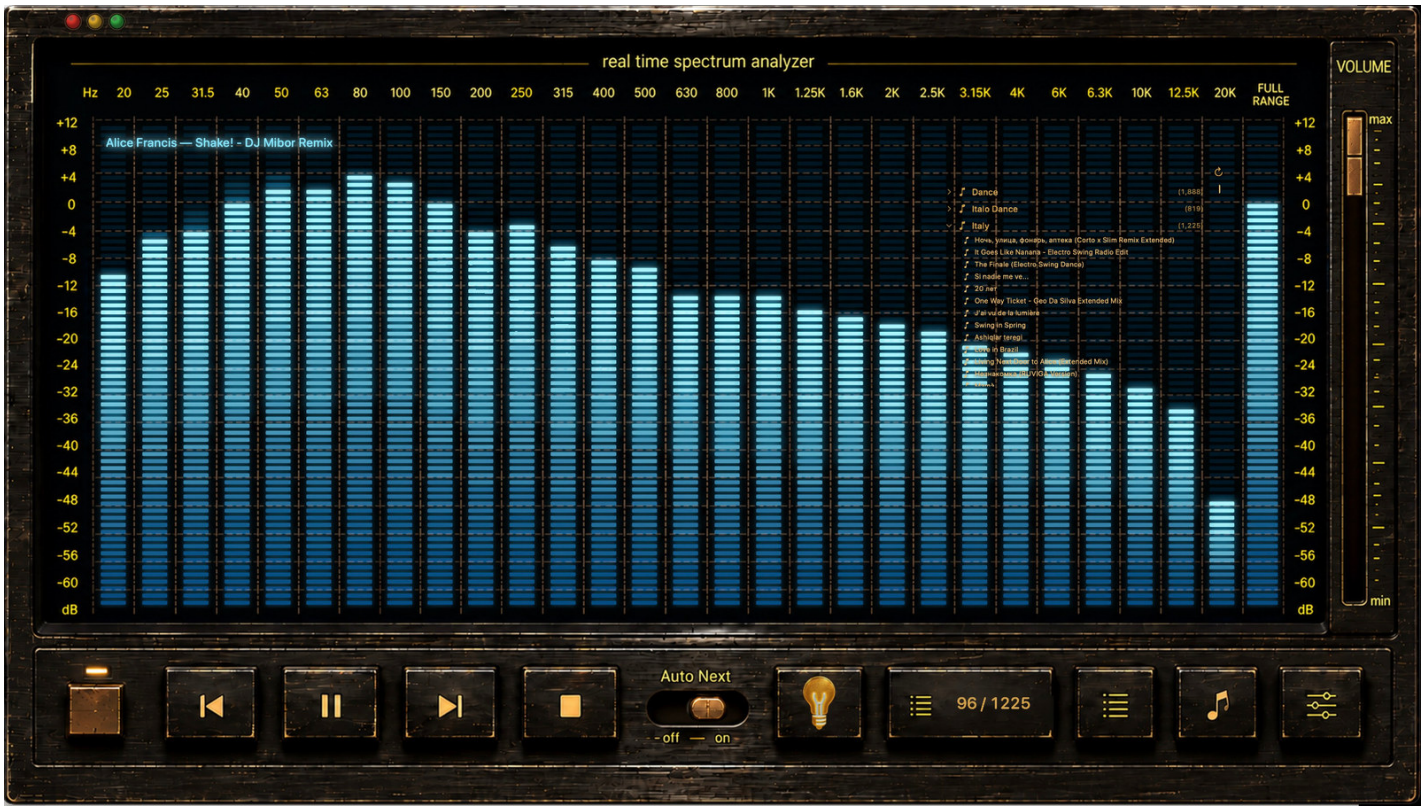


Open Queue **Read-only navigation**	Press the large Queue hardware button. The panel appears over the analyzer at desktop/fullscreen size.
Current Queue **Read-only display**	The header identifies the active playback queue used by Previous, Next, and Auto Next.
Queue row **Playback / navigation**	Selecting a row plays that exact local queue item and makes it the current queue position.
Speaker icon **Read-only status**	Marks the queue item that is currently selected for playback.
Queue position **Read-only display**	The large hardware button shows the current position and total item count when a queue exists.
Scrolling **Read-only navigation**	Use the mouse wheel/trackpad or drag the gold hardware-style scrollbar. Scrolling does not change playback until a row is selected.
Small-window mode **Window behavior**	If the player is reduced while Queue is open, the queue moves to the detached companion panel; enlarging the player restores it inline when it is still open.

Current Queue context menu (version 2.9)

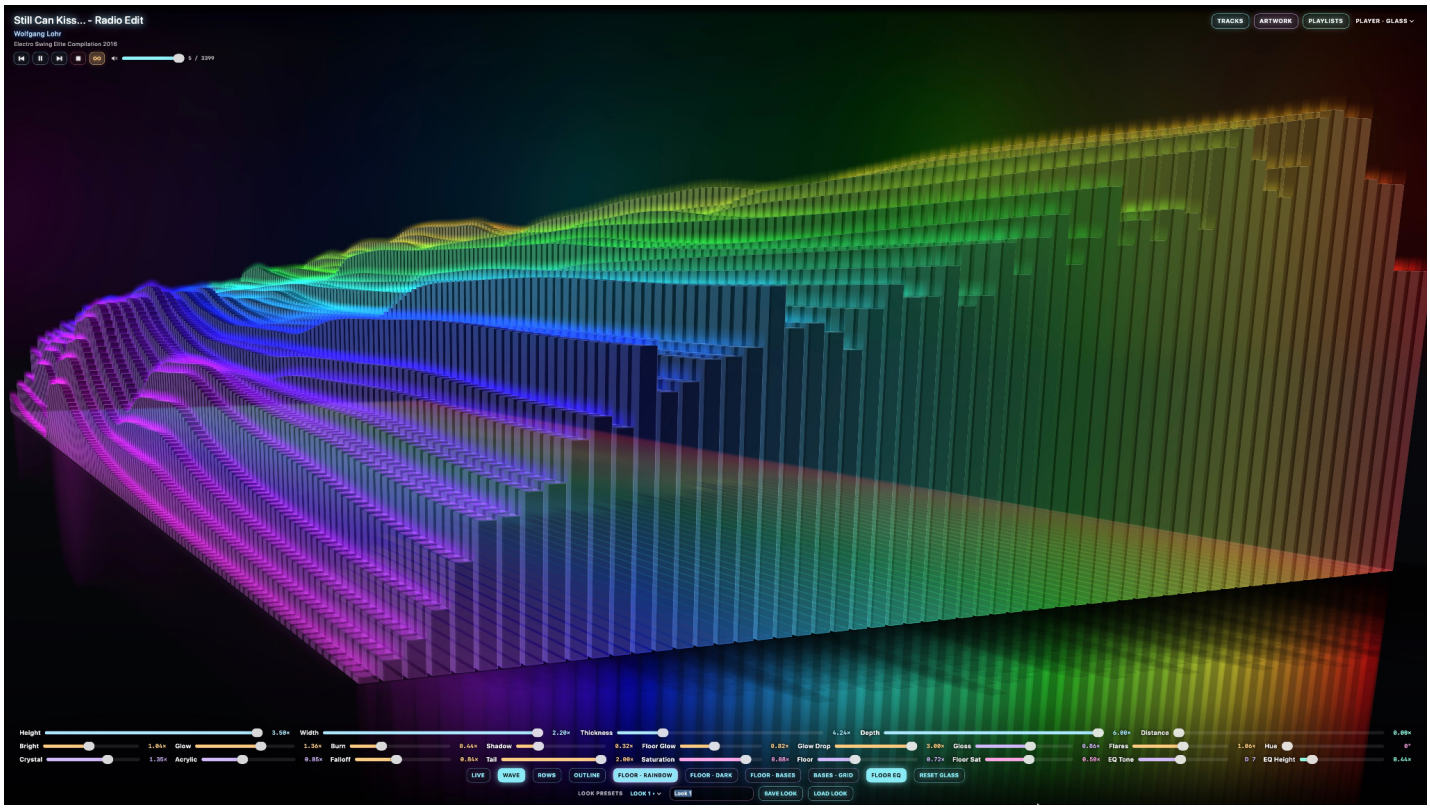
Right-click a supported queue entry to access track-copy actions or Open in Audio Inspector for its local file. A normal click still selects and plays the queue item. The shortcut does not edit the file or Music.app playlist.

18.7 Classic: Playlist navigator



Open Playlists **Read-only navigation**	Press the playlist-list hardware button. Classic preloads the lightweight playlist hierarchy shortly after the player opens when no hierarchy is already available.
Refresh arrow **Read-only Music.app query**	Forces a refresh of the visible Music.app playlist/folder hierarchy.
Folder disclosure **Read-only navigation**	Expands or collapses a Music.app playlist folder. Empty folders are labeled Empty Folder.
Playlist chevron **Read-only navigation**	Expands or collapses one playlist. Expanding requests that playlist's browsable track list.
Track count **Read-only display**	The fixed right-hand count column shows the playlist track count in parentheses when the count is available.
Expanded track row **Playback / queue selection**	Selecting a track is the point at which the browsed playlist becomes the active playback queue; LMO then plays the selected track.
Before track selection **Read-only browsing**	Merely expanding a playlist does not replace the current playback queue.
Scrolling **Read-only navigation**	Use the mouse wheel/trackpad or the gold scrollbar to move through folders, playlists, and expanded tracks.

18.8 Glass: WAVE, LIVE and navigation



Glass example in WAVE mode: the current FFT history forms a deep moving field while the player controls remain available along the bottom edge.

Glass is the three-dimensional Metal spectrum display. The depth modes are LIVE, WAVE and ROWS. The same view also supports filled prisms, OUTLINE, floor lighting, floor EQ and saved LOOK presets. Transport, queue, Auto Next and volume remain shared with Original, Classic and Mini. Display settings do not change the source audio.

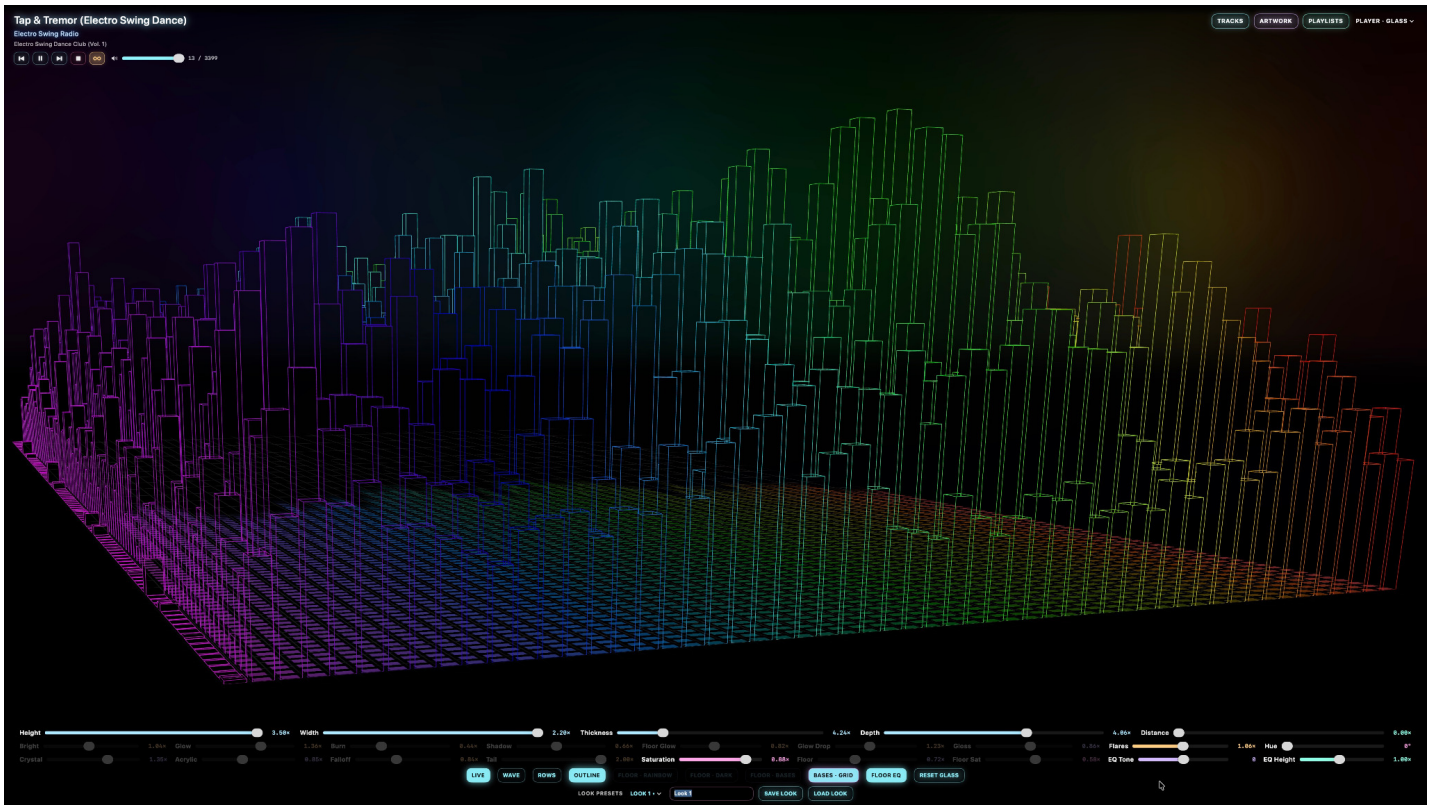
Track title / artist / album Read-only display	Shows the currently loaded local track information. The queue position is also shown when an active queue provides it.
Previous / Play-Pause / Next / Stop Playback / queue	Previous and Next move through the active queue. Play-Pause toggles the loaded track. Stop stops local playback while leaving the Glass window available.
Auto Next / Volume Player setting / playback	The infinity button toggles the same shared automatic queue advance used by the other player skins. The speaker slider changes LMO playback volume and does not rewrite the source file.
TRACKS Playback / navigation	Opens the current queue overlay. Selecting a queue row uses the same active Player queue; opening this utility closes the other Glass utility overlays.
ARTWORK Read-only display	Shows the artwork panel for the loaded track when artwork is available.
PLAYLISTS Read-only navigation	Opens the Music.app playlist/folder navigator used by the Player. A selected playlist can become the playlist-derived local queue without rewriting that Music.app playlist.
PLAYER · GLASS Player skin selector	Opens the skin menu. Switches among Original, Classic, and Glass while keeping the Player session; changing skin is presentation only.
Height Player visual setting	Changes the vertical scale available to the spectrum prisms.
Width Player visual setting	Changes the spectrum field footprint across the viewer-facing horizontal axis. When the view is rotated, the control remains viewer-relative rather than changing the audio analysis.
Thickness Player visual setting	Changes the physical thickness of the rendered prism/bar geometry.
Depth Player visual setting	Extends or compresses the 3D field away from the current viewer-facing side. It changes presentation depth, not the FFT values.
Distance Camera setting	Changes camera distance/zoom. It is a camera control and is not used as a brightness or material attenuation control.
Mouse drag / scroll Camera navigation	Horizontal drag orbits the Glass field around its vertical axis. Scrolling changes the same camera-distance control used by Distance.
LIVE Depth presentation	Each depth row uses current spectrum information with gentle per-row spatial variation.
WAVE Depth presentation	Uses recent FFT history as depth: the newest response is near the viewer and older response travels away through the field.
ROWS Depth presentation	Every depth row uses the same current FFT frame, producing a repeated current-time spectrum through depth.

18.9 Glass: materials, light and floor

Glass separates prism material, floor illumination, and downward cascades. Floor Glow and Glow Drop are intentionally independent. Floor Glow exists on the physical floor inside the spectrum footprint and can be cut by the moving prism shadows; Glow Drop is the separate colored cascade that falls downward from the spectrum. This separation is what allows a restrained reflective floor, a strong cascade, or any balance between them.

Bright Player visual setting	Controls the overall light/exposure of the filled Glass prisms. It does not move the camera or change the analyzer values.
Glow Player visual setting	Strengthens the guided/internal prism glow and luminous energy around the live spectrum.
Burn Player visual setting	Raises the hottest edge/core highlights and energetic accents while preserving the underlying spectrum motion.
Shadow Player visual setting	Controls the dark projected prism shadow on the physical floor. Shadow is independent from Floor Glow and can cut through it as the spectrum moves.
Floor Glow Player visual setting	Live colored reflected light on the floor plane only. It follows prism activity and LIVE/WAVE/ROWS, stays inside the real spectrum footprint, and moving shadows break it into the low drifting "ground-mist" / pozemka effect. It never increases Glow Drop.
Glow Drop Player visual setting	Independent colored cascades falling downward from the spectrum perimeter. Higher values increase optical reach and strength; it is not the physical floor reflection and is not Floor Glow.
Gloss Player visual setting	Controls the physical floor clear-coat/reflection quality. It does not act as a hidden Floor Glow or Glow Drop brightness control.
Flares Player visual setting	Controls the additional light-flare accents associated with the prism field.
Hue Player visual setting	Rotates the spectrum palette around the color wheel. The displayed value is shown in degrees.
Saturation Player visual setting	Changes the chroma of the main spectrum palette without changing the FFT geometry.
Crystal Material setting	Moves the prism material toward a clearer, more faceted crystalline appearance with stronger edge/facet character.
Acrylic Material setting	Moves the filled prism body toward a denser, more opaque colored-acrylic presentation.
Falloff Material/light setting	Controls how strongly the internal prism light fades from the base toward the top.
Tail Player visual setting	Controls the luminous afterglow extending above the prism tops. It is a top-light effect, separate from the downward Glow Drop layer.
Floor Floor setting	Controls surface/base/grid exposure for the selected normal floor presentation. OUTLINE intentionally keeps the floor structural instead.
Floor Sat Floor setting	Controls the chroma of the selected floor surface/base/grid while keeping the intended exposure relationship.
FLOOR · RAINBOW Floor presentation	Colored lacquer floor. Shadow, Floor Glow, and Gloss remain separate controls.
FLOOR · DARK Floor presentation	Dark spectral/graphite lacquer floor. Floor controls the exposure while Floor Sat controls the remaining spectral tint.
FLOOR · BASES Floor presentation	Uses illuminated square/rectangular foundations under individual prisms. The main prism material itself is not replaced by this floor mode.
BASES · GRID Floor presentation	Black/neutral floor with a thin spectral perspective grid and no broad rainbow floor illumination. This is also the structural floor used by OUTLINE.

18.10 Glass: OUTLINE, FLOOR EQ and LOOK presets

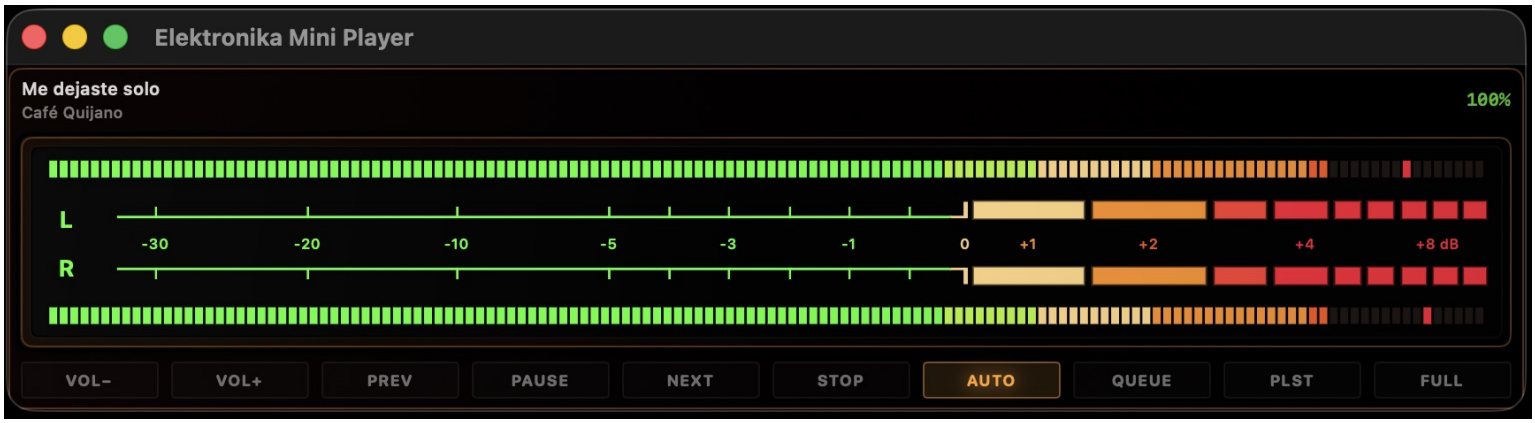


OUTLINE example: pure spectral contour prisms on the structural grid floor. Filled-glass body controls are intentionally removed from the look.

OUTLINE is a complete alternate presentation, not a filter over the filled Glass body. It automatically uses BASES · GRID so the contour columns and their floor structure remain readable. FLOOR EQ remains available and changes to short spectral vector segments. Saved LOOK slots can store both filled-Glass and OUTLINE presentations.

OUTLINE **Player visual setting**	Toggles pure colored contour prisms in the vector/neon presentation. Enabling it automatically selects BASES · GRID; filled glass, broad floor material, Glow Drop, and Tail-style filled effects are intentionally not used.
FLOOR EQ **Player visual setting**	Toggles a separate one-row current-FFT analyzer laid flat on the floor. It is not a shadow of any 3D depth row. In OUTLINE it is drawn as short spectral vector blocks on the real grid.
EQ Tone **FLOOR EQ setting**	Available while FLOOR EQ is on. Center is the authored balance; left makes the analyzer deeper/darker and right makes it brighter. Tone changes optical contrast, not FFT geometry.
EQ Height **FLOOR EQ setting**	Controls the analyzer reach across floor depth. Higher values extend the one-row response farther, with a hard ceiling that keeps it inside the real 3D field.
RESET GLASS **Player visual setting**	Restores the authored Glass showcase controls/camera. It resets the live Glass presentation and does not delete the five saved LOOK slots.
LOOK 1 ... LOOK 5 **Saved visual preset**	Selects one of five independent Glass LOOK slots. A slot can hold a complete preferred visual setup rather than changing playback or the source file.
Look name **Saved visual preset**	Editable name for the currently selected LOOK slot. The name is stored with the user preset.
SAVE LOOK **Saved visual preset**	Saves the current Glass visual state into the selected LOOK slot, including the relevant geometry, depth presentation, material, floor, EQ, and OUTLINE choices.
LOAD LOOK **Saved visual preset**	Loads the selected saved Glass state. Loading a LOOK changes the presentation only; it does not replace the track, queue, or audio file.

18.11 Original: Elektronika Mini Player



Open the Elektronika Mini Player

From Original, click the overlapping-rectangles button to show or hide the dedicated Elektronika Mini. It shares the current local track, queue, volume and Auto Next. The FULL button returns to the full Player without restarting playback. GLASS Mini and small-window CLASSIC behave differently; see the following pages.

Title / artist / volume Read-only display	The upper strip shows the current track title, artist when available, and the current LMO playback volume as a percentage.
Stereo L / R meter Read-only live meter	The two long outer LED rows are the live left/right level indicators. The center markings provide the fixed reference scale from the green range through the amber/red overload end.
VOL- / VOL+ Playback only	Changes LMO playback volume in 5 percentage-point steps. It does not rewrite the source file and is separate from Spectrum Appearance playback gain.
PREV / NEXT Playback / queue	Moves through the active Player queue. A direction is disabled when no corresponding queue item exists.
PLAY / PAUSE Playback only	The button label follows the current state. It pauses/resumes the loaded track and becomes unavailable when no track is loaded.
STOP Playback only	Stops local playback using the same Player stop action while keeping the Mini Player available.
AUTO Player setting	Toggles the shared Auto Next state. The button remains illuminated while automatic queue advance is enabled.
QUEUE Playback / navigation	Opens Current Queue in a compact popover. Selecting a row immediately plays that exact queue item and closes the popover.
PLST Music.app navigation	Opens the Music.app playlist/folder hierarchy. The popover can refresh the hierarchy; selecting a playlist loads it as the Player queue.
FULL Window navigation	Shows or restores the full Live Spectrum Player while playback and queue state continue unchanged.
Window behavior Window behavior	Original Elektronika Mini is a fixed-size floating window (860 x 222 pt). It cannot be freely resized. It may appear across macOS Spaces. Closing the final restorable Player window during audible playback keeps a restore target available.

ORIGINAL compact layout (not Elektronika Mini)

Resizing the main ORIGINAL Live Spectrum window below 800 pt wide OR below 445 pt high switches to its compact spectrum layout in the SAME resizable window. Playback, volume and queue continue uninterrupted; enlarging the window restores the full ORIGINAL layout. Elektronika Mini above is a separate fixed-size floating window opened by its own button.

18.12 GLASS Mini Player

GLASS has no Mini button in the full window. Finish shrinking the GLASS window to 720 pt wide or less, OR to 420 pt high or less. After live resizing ends, LMO opens the dedicated GLASS Mini automatically.

Automatic entry	The resize threshold is evaluated after the macOS live-resize operation, outside fullscreen. Moving the window or changing camera distance is not the Mini trigger.
Separate fixed window	GLASS Mini is a borderless, floating transparent window with a fixed content size of 560 x 220 pt. It cannot be freely resized.
Spectrum presentation	Shows exactly one live 3D spectrum row. There is no physical floor, floor shadow, Glow Drop or Tail. Full-size GLASS depth modes and LOOK controls are not exposed in Mini.
Previous / Play-Pause / Next	Operate on the current shared LMO playback queue. Prev/Next are unavailable when there is no corresponding queue item; Play is unavailable with no loaded track.
Speaker and volume slider	Change the shared LMO playback volume from 0 to 100%. This is not the Original Spectrum Appearance playback-gain control and does not alter source audio.
H / height multiplier	Changes only GLASS Mini prism height, from 0.45x to 5.25x. Width, thickness and depth are fixed in this Mini layout. The height setting is saved separately.
Playlists button	Opens the Mini playlist/folder panel. Browse and choose a local playlist-derived queue without modifying that Music.app playlist.
Expand icon (Full Player)	The diagonal-arrows button, whose tooltip reads Full Player, leaves Mini and restores the large Live Spectrum window. The loaded track and queue continue.
Not present in GLASS Mini	There is no separate Stop, Auto Next, full Queue, FLOOR EQ, WAVE/ROWS selector or LOOK editor in the compact control strip. Use full Player controls for those operations.

The full GLASS window and GLASS Mini share playback state, but the compact 3D geometry and the H slider are intentionally different from full GLASS settings.

18.13 CLASSIC in a small window

CLASSIC does not replace its instrument panel with the Original Elektronika Mini or GLASS Mini when you resize it. The same Classic hardware face is scaled as a whole.

Resize behavior	The original complete Classic instrument face remains visible and is fitted to the available window. Resizing changes its display scale, not spectrum type or button geometry.
Queue and Playlists threshold	Below 1180 pt width OR 665 pt height, an open Queue or Playlists overlay can move to a detached companion panel below the player.
Return to larger size	When the Classic window becomes large enough, a still-open detached Queue or Playlists panel returns inside the instrument face.
Panel closed while small	Closing a detached panel while in compact size prevents it from reopening automatically when the window grows again.
Playback and queue	Classic retains the same player, queue, volume and Auto Next state through resizing; no replacement track or new queue is created.
CLASSIC hardware buttons	Power, previous, play/pause, next, stop, Auto Next, phosphor, Queue, Playlists, Track Card and Settings retain the functions described on the preceding Classic pages.
Switch player skin	Use the Classic Settings hardware control to choose Original or GLASS. Switching skin is not the same operation as reducing the Classic window.

The word "Mini" means different things across skins: ORIGINAL has a compact layout inside its resizable window AND the separate Elektronika Mini; GLASS has a separate transparent Mini; CLASSIC scales its complete hardware face.

19. Help and Support

The built-in Help page is a short reminder. This manual expands it into a control-by-control reference. Help controls copy support text, show open-source notices, and provide the same incremental Music.app refresh/log-folder shortcuts used elsewhere.

19.1 Help controls

Copy Help Text **Clipboard only**	Copies the built-in Quick Help text to the clipboard.
Copy Safety Notes **Clipboard only**	Copies the built-in safety summary to the clipboard.
Open Source Licenses **Navigation only**	Opens the bundled open-source license notices for included components such as FFmpeg/FFprobe, LAME, WavPack, DSD-related tools, and OpenMPT.
Refresh Music Changes **LMO index**	Incrementally reconciles LMO's saved Music.app index after small changes.
Open Log Folder **Navigation only**	Opens the selected log folder.

19.2 Quick Help - corrected interpretation

• Start with Setup, then build the Library Index. Leave Include playlist usage enabled on the first full build when you want usage/relationship tools ready. • Result counts are review counts. They do not mean “delete/fix everything.” • Health Check and Audio Authenticity are diagnostic evidence. They do not change files merely by scanning. • Playlist cleanup changes playlist membership only. Physical files remain unless a separate physical-file action explicitly says otherwise. • Edit Tags changes supported Music.app fields only after Save to Music. • Physical file changes are explicit. Current LMO does include clearly labeled operations that can move a file to Trash, rename a file, rewrite selected embedded metadata, apply gain to the original, or handle a Repair/Convert source after verified output. Those are not hidden side effects of a scan. • For support logs: enable the requested log option before reproducing the problem, then export diagnostics and include the relevant operation log.

19.3 License documents and diagnostic boundaries

Support controls should be interpreted separately from commands that change the music collection.

Open Source Licenses	Opens bundled notices for the open-source components included with LMO. This is a license/documentation operation only.
License Information...	Opens the LMO licensing information window. It does not verify or change Music.app records and is not a scan.
Open Full License Files	Opens bundled license texts from Help > Open Source Licenses, not License Information.
Copy Help Text / Copy Safety Notes	Copies static help or safety guidance to the clipboard. No library file, playlist membership or tag is modified.
Export Diagnostics	Creates a support report of current application state and technical settings; it is different from the optional file, scan or index operation log.
Known local-source boundary	A missing or cloud-only Music.app item is not an offline playable source in the LMO Player. The program does not obtain or transcode the online stream as a substitute.

For a reproducible problem, export diagnostics and the log belonging to that operation. Avoid assuming that a warning about a file path means the physical file has been deleted.

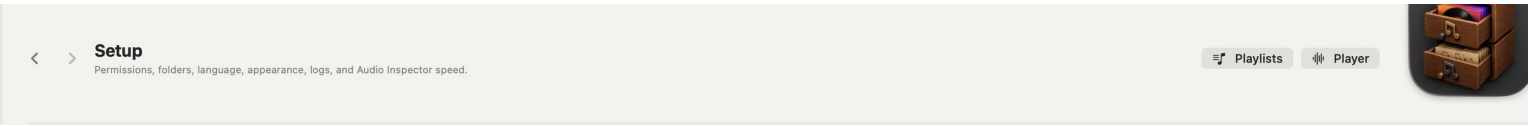
Trial, full unlock and restoring purchases

Start Free 3-Day Trial begins when the free App Store transaction is confirmed. It enables all tools for three days, without automatic charge. Full Version is a one-time unlock, not a subscription. Restore Purchases asks the App Store to restore existing entitlements. These purchase controls do not modify Music.app, local audio files or the Library Index.

20. Global Controls and Common Editors

These controls appear across multiple sections. They are documented once here so their meaning is consistent wherever they appear.

20.1 Global navigation and top-right buttons



Persistent top-right shortcuts
The cropped current-interface view above shows the controls that stay at the top right of every main LMO section. Playlists opens the Music.app hierarchy navigator. Player opens/restores the independent Live Spectrum Player without launching a scan. The Back/Forward arrows on the left navigate LMO screen history; they are not Undo.

Option + Play **Local-player shortcut**	Hold Option while clicking a Play control that can resolve a readable local audio file. LMO opens/uses the Live Spectrum Player and plays the physical file locally with the spectrum analyzer. When the current context provides a playlist or result ordering, that context becomes the player queue for Previous/Next. Normal Play keeps its ordinary behavior, usually Music.app playback. If the item has no accessible local file, spectrum playback is unavailable and LMO does not fabricate or download one.
Option + supported Preview **Local-player shortcut**	For local Preview controls routed through LMO's file preview path, holding Option uses the Live Spectrum Player instead of the normal Finder-style Quick Look preview. This requires a readable physical file.

Back / Forward arrows **Navigation only**	Return to previously visited LMO tool screens and restore their remembered scroll position where available. They do not undo library changes.
Playlists **Read-only navigation**	Permanent top-right button beside the LMO app icon on every main section. It opens LMO's navigator for the actual Music.app playlist/folder hierarchy without changing the current tool. The hierarchy can be refreshed; a selected playlist can be opened/revealed in Music.app or used as player context where the visible UI offers that action.
Player **Navigation / local playback**	Permanent top-right button beside Playlists and the LMO app icon on every main section. It opens or restores the separate Live Spectrum Player window immediately; pressing it does not start a library scan. The player is LMO's own local-file playback/analyzer window.
Stop Music Play **Playback only**	Stops Music.app/local playback; it does not stop an unrelated scan unless that scan has its own Stop control.
Undo Last Change **Changes depend on last action**	Restores the latest supported reversible action. Undo is a convenience recovery layer, not a replacement for a Music library/media backup.
Open Log Folder **Navigation only**	Opens the user-selected local log folder.

20.2 Music.app metadata editor

Title / Artist / Album / Album Artist / Composer / Genre / Year / Track # / Disc # / Comments **Read-only until save**	Editable supported Music.app metadata fields for the exact matched record. Opening the editor does not change anything.
Save to Music **Changes Music.app metadata**	Writes the reviewed supported fields to the exact Music.app record.
Cancel **No change**	Closes the editor without applying the draft.
Copy Artist + Title **Clipboard only**	On a physical-only guessed-tag draft with no Music.app Persistent ID, copies the guessed Artist/Title/Album instead of writing Music.app.

20.3 Naming & Tag Formatting review window

Title / Artist / Album / Album Artist / Composer / Genre / Filename **Read-only until apply**	Shows the proposed spacing cleanup fields for review.
Update Music.app tags **Selection only until apply**	Chooses whether the reviewed Music.app text-field spacing corrections will be written.
Rename physical filename **Selection only until apply**	Chooses whether the reviewed same-folder/same-extension physical rename will be performed.
Apply Reviewed **Changes / Changes selected metadata/file name**	Applies only the options currently selected in the review window. Filename rename uses Persistent ID/artwork/filesystem verification and rollback guards.
Cancel **No change**	Closes without applying the draft.

20.4 Batch Naming & Tag Formatting window

Fix Music.app Title / Artist / Album spacing **Selection only until apply**	Controls whether selected batch rows update the supported Music.app spacing fields.
Rename physical filenames safely **Selection only until apply**	Controls whether selected rows with filename suggestions perform verified safe renames.
Apply to Selected **Changes selected metadata/file names**	Runs the chosen batch actions only for the selected reviewed rows.
Cancel **No change**	Closes without applying the batch.

Backup remains essential Undo and rollback protect specific supported operations, but they are not a substitute for a current backup of an important Music.app library and its local media files.

20.5 Editing Music.app tags: Normalize Spaces

Normalize Spaces is a draft-preparation control in the Music.app metadata editor. It is not the same as File Maintenance batch formatting or a physical filename rename.

Normalize Spaces	Trims leading/trailing whitespace and reduces runs of ordinary spaces or tabs in supported editable Music.app text fields. It changes the editor draft only.
Which fields are normalized	Title, Artist, Album, Album Artist, Composer, Genre, Year, Track Number and Disc Number are included in this editor operation.
Comments are excluded	The free-form Comments editor is not automatically normalized because repeated or leading spaces may be meaningful.
Save to Music	Writes reviewed supported draft fields to the exact matched Music.app record. Without this action, Normalize Spaces has not changed Music.app.
Cancel	Discards the current unsaved draft. It does not rename the physical file or move the track between playlists.
Physical-only guessed metadata	When there is no verified Music.app Persistent ID, the draft can be copied as Artist/Title/Album but cannot be saved as an existing Music.app record.
Separate filename action	Use File Maintenance > Rename Filename Safely or the reviewed naming/batch window for an actual physical rename; metadata-editor spacing is not a filename operation.

A single Music.app record can have different text in its Music.app metadata and its embedded audio tags. Saving in this editor does not imply an embedded-tag rewrite.

20.6 Playlist navigator and contextual buttons

The permanent Playlists shortcut opens a navigator over Music.app folders and playlists. Browsing the hierarchy does not by itself edit a playlist or start a new playback queue.

Load Playlist Hierarchy	Shown when the local Player playlist browser does not yet have a hierarchy. Requests the current Music.app folder and playlist catalog.
Load Playlists	This is the exact empty-state button in the Classic playlist browser when no hierarchy is loaded. It requests the Music.app catalog. Elsewhere the Player navigator may use the longer label Load Playlist Hierarchy for the corresponding request.
Loading Music.app hierarchy...	A progress state while the catalog request is active; not a button to start playback. Wait for the hierarchy or an explicit error.
Folder and playlist disclosure	Expands or collapses browse rows. Opening a folder is navigation; selecting a track or a queue-loading action is the later playback choice.
Play in LMO	On a reviewed track inside the global playlist navigator, starts that local track through the LMO Live Spectrum Player.
Requires Music.app: View > as Songs	The warning alongside certain playlist-reveal chips describes a Music.app navigation prerequisite. In Music.app, choose View > as Songs when required for revealing a particular track in its playlist; this message is not a command to edit metadata.
Reveal in Music	Locates the exact matched Music.app track in the selected playlist. This requires a Music.app Persistent ID; the control is disabled without one.
Reveal in Finder	Locates the matched physical file. The action is unavailable when the track has no readable local path.
Copy Path	Copies the displayed local path without editing it. The control is unavailable when the path is empty.
Apply Reviewed Changes	In the Naming & Tag Formatting review dialog, commits only the selected Music.app tag correction and/or safe physical filename rename. The action stays disabled when both options are off.

Play in LMO and Reveal in Music are deliberately different: the first uses LMO local-file playback, the second navigates Music.app. Playlist browsing, selected queue playback and physical-file editing are three separate actions.

Safety, Privacy, and Support Reference

Offline/local media boundary
LMO processes the documented library information, reports, and local-file work on this Mac. It does not collect analytics or tracking data for the workflows described here. Subscription streams, online-only audio, and streaming services are not supported source workflows in this release.

Keep a current backup before maintenance of an important library. A result count is evidence to review, not an instruction to delete or clean everything. Scan-specific log controls create support files only when explicitly enabled. Export Diagnostics is a separate support summary.

Open-source components. FFmpeg/FFprobe, LAME, WavPack, DSD Nexus, OpenMPT and their bundled license texts are listed by the visible Open Source Licenses control in Help.

21. Album Completeness

Album Completeness is the read-only Health Check sub-scan for numbered releases. It is intentionally separate from Track / Disc cleanup: one reports evidence, the other is an explicit metadata action.

Read-only by design

Album Completeness does not add missing tracks, clear numbering, rewrite tags, relink Music.app records, move audio files, or query the Internet. It only evaluates Track / Disc numbering already exposed by Music.app.

21.1 What the check uses

Track Number / Track Count Read-only Music.app metadata	Used to identify numbered releases whose visible track totals are incomplete or internally inconsistent.
Disc Number / Disc Count Read-only Music.app metadata	Used to identify multi-disc numbering that is incomplete or conflicting.
Affected Artist - Album release Focused evaluation	When the optional newly-added-album watcher is enabled in Setup, LMO checks only releases affected by the detected import event. The Health Check option remains a separate manual diagnostic.

21.2 Interpreting a result

A finding means the currently visible Music.app numbering suggests an incomplete or conflicting numbered release. It is not proof that an audio file is damaged, and it does not tell LMO to invent missing titles or import anything automatically.

Separate operation

To review records with non-zero Track / Disc numbering, or to clear those four fields,
use Health Check > Find Track / Disc Numbering. This operation is described on page 90.

22. Track / Disc Numbering

Find Track / Disc Numbering is a separate optional Health Check scan. Discovery is read-only. The mutating action appears only after you explicitly select numbering findings and choose Clear Selected Track / Disc Numbering.

Find Track / Disc Numbering Read-only discovery	Finds Music.app records where Track Number, Track Count / Total Tracks, Disc Number, or Disc Count / Total Discs is non-zero. Running the scan changes nothing.
Clear Selected Track / Disc Numbering... Explicit Music.app metadata change	Targets only the four numbering fields shown above. It does not intentionally change unrelated metadata or the physical audio file.
Undo Last Change In-session recovery	Successfully cleared rows can be restored with Undo Last Change during the same LMO session when an undo record is available.

22.1 Safety contract for Clear Numbering

Critical confirmation before writing

LMO shows a critical confirmation that names the four fields being cleared and the protected values that must remain unchanged. No numbering is intentionally changed if you cancel.

Before and after each selected record, LMO live-verifies the Music.app record. Title, Artist, Album, Album Artist, Composer, Genre, Year, Comments, location, Persistent ID, and artwork count are protected by the cleanup contract.

If Music.app temporarily changes editable protected metadata while numbering is written, LMO restores the exact pre-cleanup values and verifies the final protected fingerprint. Persistent ID, location, and artwork count are fail-closed boundaries: an unexpected change stops the batch rather than being silently repaired.

If numbering or protected-metadata restoration cannot be fully verified, LMO attempts to restore the row's original numbering and protected Music.app metadata before stopping. Whitespace is not silently normalized by this cleanup.

Physical files are outside this action

Clear Numbering does not rename, move, delete, replace, convert, or rewrite the physical audio file. Naming and whitespace cleanup are separate tools.

23. Batch Folder Relink

Batch Folder Relink is the Health Check workflow for a set of existing Music.app records whose saved files were moved together to another folder. It updates existing Music.app file links; it is not a library import and it does not move the audio files.

23.1 Mapping and preflight

Old root Derived from the reviewed missing record	LMO uses the missing track's saved parent folder as the old root.
Replacement folder Explicit macOS folder selection	You choose the folder that now contains the same relative folder/file structure.
Exact relative-path mapping Fail-closed mapping rule	Only files whose exact relative path exists below the replacement folder are candidates. Targets must be readable supported audio files.
Available technical preflight Additional rejection checks	When saved facts are available, LMO rejects a mapped target whose byte size differs or whose duration differs beyond the allowed tolerance. Duplicate mapped target paths are not applied.

23.2 What is preserved and verified

Existing records only

The confirmation states the exact candidate count and preview. LMO preserves each existing Music.app Persistent ID and playlist memberships. It does not intentionally create new Music.app records, and it does not move, copy, rename, rewrite, or delete physical audio files.

For each candidate, LMO re-reads the live record, blocks a target path that already belongs to a different Music.app Persistent ID, updates the location serially, and requires the same Persistent ID to be visible at the exact mapped target path. Protected Music.app metadata is compared before and after; if editable protected metadata changes, LMO attempts to restore it exactly before accepting the row.

A post-relink problem is not blindly rolled back to an old path that may no longer exist. If Music.app already accepted and verified the same Persistent ID at the target, LMO keeps that verified live state, records a warning, and pauses the batch for safety review.

23.3 Final Library Index checkpoint

After successful live relinks, LMO updates the verified per-record Library Index rows. When a complete saved Library Index is available, it also saves a typed Music Library snapshot, reapplies the verified relinked Persistent IDs after earlier queued SQLite writes have drained, and reads the saved exact paths back from SQLite. A clean finish requires the successful target paths to be observable in the saved index; otherwise the result reports a checkpoint warning instead of pretending the saved state is sealed.

Recommended verification after the batch

Run Health Check once immediately after Batch Folder Relink to verify the batch checkpoint by itself. Then run Refresh Music Changes and Health Check again to confirm that the saved and live states remain stable.